Cluster Editing above a P_3 packing: W[1]-hardness in the excess and the treewidth of the intersection graph

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Abstract

Cluster Editing asks for the fewest edge insertions and deletions that turn a graph into a disjoint union of cliques. Li, Pilipczuk and Sorge parameterize it by the excess ℓ of the solution size over a modification-disjoint packing H of induced P_3's, and leave open for which packings the problem is fixed-parameter tractable. We use the treewidth w of the intersection graph $I(H)$ of the packing as the structural measure. Our main result is a negative answer to the question whether $\ell + w$ suffices: the problem is W[1]-hard parameterized by $\ell + w$, and also by $\ell + \mathrm{pw}(I(H))$ and by $\ell + c$, where c is the maximum number of packed P_3's through a vertex, even for pair-maximal packings. Hence there is no algorithm in time $f(\ell, w)n^{O(1)}$ unless FPT = W[1], and under ETH none in time $f(\ell, w)n^{o(\sqrt{\ell})}$. The reduction is from Grid Tiling with $\leq$; it uses a selection chain of bounded width whose position is released by cutting one of many "tunnels", and is computer-assisted at four finite steps, which we list. The matching upper bounds come from a CSP view: for $\ell = 0$ the problem is a constraint satisfaction problem with domain size 3 whose constraint scopes are cliques of $I(H)$; for general ℓ, guessing at most ℓ edits gives such a problem with domain size 5. This yields algorithms in time $3^{w+1}n^{O(1)}$ and $5^{w+1}n^{2\ell+O(1)}$, and $n^{\ell}f(\ell, w)\,5^{w+\ell+1}n^{O(1)}$ after a locality argument. Under ETH there is no $2^{o(w)}n^{O(1)}$ algorithm even for $\ell = 0$, and neither w nor ℓ alone suffices. If the components of $I(H)$ have bounded size, the problem is fixed-parameter tractable in ℓ and has a kernel. We isolate a configuration, three pairs of a vertex triple packed by three distinct members (a *soft triangle*), as the source of difficulty: without soft triangles the set of zero-excess solutions is closed under every majority operation, so $\ell = 0$ is polynomial for every c and w, lock cores have at most two anchors, a far-lift model of the excess is fixed-parameter tractable, and on forests the problem is fixed-parameter tractable in ℓ conditionally on a bounded-certificate hypothesis; with soft triangles minimal certificates of the natural branching scheme have unbounded size, and the hardness reduction shows that soft triangles suffice. For pair-maximal packings we give a normal form of optimal solutions, a dichotomy of inserted pairs into local and non-local ones, and an instance family in which an optimum must delete an edge between members at arbitrary distance in $I(H)$. The parameters belong to the packing, not to the graph. Exchanging one packed P_3 for an induced P_3 keeps the size and raises w by at most one; we use such exchanges to analyse two obstructions, and we show that locating a good repacking cannot be done in fixed-parameter time unless FPT = W[1]. Still open are hardness for bounded c (our reduction has $c = O(\ell)$), the case $c = 2$, and the exact exponent of n under ETH ($n^{o(\sqrt{\ell})}$ is excluded, n^{ℓ} is achieved).

1 Introduction

Cluster Editing asks for the fewest edge insertions and deletions that turn a graph G into a disjoint union of cliques. Every induced P_3 must be hit by an edit, so a packing H of induced P_3's that pairwise share no vertex pair (a *modification-disjoint* packing) gives the lower bound $|H|$ on the solution size. Li, Pilipczuk and Sorge [15] parameterize by the *excess* ℓ over this bound (problem CEaMP): is there a solution of size at most $|H| + \ell$? They show NP-hardness already for $\ell = 0$, with every vertex in at most 23 packed P_3's, and an $n^{2\ell+O(1)}$ algorithm when every vertex lies in at most two packed P_3's; whether the latter case is fixed-parameter tractable is open [15]. For vertex-disjoint packings the problem is fixed-parameter tractable in ℓ [17].

The question of this paper is which measurable *interaction structure* of H makes the local repair choices (one edit per packed P_3, in three ways) combine into a global solution in $f(\ell, w)\, n^{O(1)}$ time. We take $w = \mathrm{tw}(I(H))$, where the intersection graph $I(H)$ has the packed P_3's as nodes and joins two of them when they share a vertex. The answer is negative: the problem is W[1]-hard parameterized by $\ell + w$ (Theorem 5.1), already for $\ell + \mathrm{pw}(I(H))$ and for $\ell + c$, where $c \le w + 1$ is the maximum number of packed P_3's through a vertex. The matching upper bounds are XP in ℓ with single-exponential dependence on w.

1.1 Results

Our first theorem collects the main results of Sections 3–5 and 10.

Theorem 1.1 (Main theorem)**.** *Let (G, H, ℓ) be an instance of CEaMP, $w = \mathrm{tw}(I(H))$, and assume a tree decomposition of $I(H)$ of width w is given.*

(a) W[1]-hardness. *CEaMP is W[1]-hard parameterized by $\ell+w$, by $\ell+\mathrm{pw}(I(H))$ and by $\ell+c$, even for pair-maximal packings; the hard instances have $\ell = k^2$, $c \le 4p+2$ and $\mathrm{pw}(I(H)) \le 24kp^2$ with $p = O(k^2)$. Hence there is no algorithm in time $f(\ell, w)n^{O(1)}$ unless* FPT = W[1]*, and, unless ETH fails, none in time $f(\ell, w)N^{o(\sqrt{\ell})}$ on instances with N vertices. So the XP algorithms in (c) are optimal up to the exponent of n.*

(b) CSP representation. *$(G, H, 0)$ is a YES instance iff a constraint satisfaction problem with one variable of domain $\{1, 2, 3\}$ per packed P_3 is satisfiable; every constraint has at most three variables, and its scope is a clique of $I(H)$. For $\ell > 0$, for each of the at most $n^{2\ell}$ sets U of at most ℓ hard pairs there is such a problem with domain size 5 (the five edit sets of a packed P_3 that leave it a cluster graph, Lemma 3.3) and unary costs, and (G, H, ℓ) is a YES instance iff for some U its minimum cost is at most $|H| + \ell - |U|$.*

(c) Algorithms. *CEaMP is solvable in time $3^{w+1}n^{O(1)}$ for $\ell = 0$ and in time $5^{w+1}n^{2\ell+O(1)}$ for general ℓ. The latter is XP in ℓ with single-exponential dependence on w; by (a) it is not a fixed-parameter algorithm and cannot be made one unless* FPT = W[1].

(d) ETH lower bound in the exponent of 2. *Unless ETH fails, there is no $2^{o(w)}n^{O(1)}$ algorithm for $\ell = 0$, even if every vertex lies in at most 23 packed P_3's.*

(e) Neither parameter suffices alone. *CEaMP is NP-hard when $w = 0$ (with ℓ part of the input) and, by [15], when $\ell = 0$.*

(f) Soft triangles. *Call a vertex triple a* soft triangle *if its three pairs are packed by three distinct members. If (G, H) has no soft triangle (in particular if $I(H)$ is triangle-free, e.g. a forest), then the set of zero-excess state vectors is closed under every majority operation; consequently*

$(G, H, 0)$ *is decidable in polynomial time for every* c *and* w*, and every lock core has at most two anchors.*

Part (a) is Theorem 5.1 and Corollary 5.2 (Section 5; the reduction and its proof are in Appendix I, and parts of the proof are computer-assisted, as listed in Section 5.3); parts (b) and (c) are Theorems 3.5 and 3.6 (and Theorem 8.2); (d) and (e) are Theorem 4.1 and Proposition 4.3; (f) is Theorem 10.4 with Corollaries 10.5 and 10.7 (Section 10). The CSP view and the dynamic program are standard once the interaction structure is identified (Observation 3.2); (d) is a consequence of the reduction of [15]. The hardness reduction is from Grid Tiling with $\leq$; it combines a bounded-width selection chain, blown-up vertices that cannot be split, guests that forbid positions, and hard edges (tunnels) between far vertices whose cut releases two chains at once.

The remaining results concern the structure of optimal solutions and the question whether (ℓ, w) suffices.

- *Bounded components* (Section 6). If s is the maximum number of vertices spanned by a connected component of $I(H)$, CEaMP is solvable in time $1.62^{(\binom{s}{2}+1)\ell} n^{O(1)} + n3^s$ and has a kernel with at most $(s^2 - s + 2)\ell$ vertices; in particular $I(H)$ a perfect matching is fixed-parameter tractable in ℓ (Theorem 6.1). This is the reduction rule of van Bevern, Froese and Komusiewicz [17] applied to components of $I(H)$; we verify its hypotheses and the counting at the level of components, and cite the soundness of the rule [17, Lemma 6.3], the algorithm of Böcker [2] and the $2k$ kernel of Chen and Meng [4]. A variant for components of arbitrary size leaves at most 2ℓ components (Theorem 6.10).

- *Normal form and an* n^ℓ *algorithm* (Sections 7 and 8). For a pair-maximal packing (no induced P_3 has all three pairs hard), in an optimal solution of size at most $|H| + \ell$, every cluster that contains an end of a hard edit is connected, of size at most $2D + \ell + 1$ and of bounded hard degree, and one end of a hard edit determines at most Z^2 candidates for the other, where $Z = O(w+\ell)$ (Corollary 7.7). This gives an algorithm in time $(n+1)^\ell (6w+9\ell+7)^{2\ell} 5^{w+\ell+1} n^{O(1)}$ for every packing (Theorem 8.2), halving the exponent of Theorem 3.6, and a fixed-parameter algorithm for a stronger parameter ω (Theorem 8.3). The parameter ω is not bounded by any function of (ℓ, w) (Proposition 8.4).

- *Locality of inserted pairs* (Section 9). Every inserted hard pair of an optimal solution of a pair-maximal packing has a middle vertex and is either of type T1 (two soft steps in distinct packed P_3's; local in a decomposition of $I(H)$) or of type T2 (one kept hard step). Merges of two pieces joined by no soft pair are of type T1 only (Corollary 9.3); merges joined by a soft pair can be of type T2 (examples with $n = 7$ and $n = 8$, Proposition 9.7). For every $m \geq 3$ there is an instance with $w = 1$, $c = 2$, $\ell = 1$ in which every optimum deletes a hard edge whose ends lie in packed P_3's at distance $m - 1$ in $I(H)$ (Proposition 9.8).

- *Soft triangles are the source of hardness* (Section 10). Besides Theorem 1.1(f): for soft-triangle-free packings a far-lift model of the excess (any set of at most ℓ far hard edits and no other deviation) is solvable in time $6^\ell n^{O(1)}$ for every c and w (Theorem 10.14), via the fact that unit lists have Helly number two (Lemma 10.12); on forests the full problem is fixed-parameter tractable in ℓ *conditionally* on a bounded-certificate hypothesis, which affects only the running time (Theorem 10.18); the natural abstract hitting model of the problem on forests is fixed-parameter tractable but not faithful (Propositions 10.16 and 10.17). A certificate-branching scheme has a fixed-parameter leaf test (Theorem 10.21) and bounded minimal certificates at $\ell = 0$ without soft triangles (Theorem 10.22), but with soft triangles minimal certificates of

unbounded size exist already at $c = 2$, $w = 2$, $\ell = 0$ (Theorem 10.23). An abstract linked-pair path-hitting problem is W[1]-hard (Theorem 10.27); without soft triangles tunnel releases cannot realise it (Theorem 10.28), and with soft triangles they can: this is the reduction of Theorem 5.1, so soft triangles are both necessary (along this route) and sufficient for hardness.

- *Repacking* (Section 11). The parameters belong to the packing: replacing a packed P_3 by an induced P_3 keeps $|H|$ and raises w by at most one, with a pair-maximal repair costing at most ℓ more (Lemma 11.1); every hard edit of an optimum lies in an induced P_3 and is either bridged (local in every decomposition) or absorbable by one exchange (Lemma 11.5); no hard edit has two uncovered ends. The far-deletion family of Proposition 9.8 has intrinsic excess one, but after one exchange $I(H^*)$ is a cycle and its only hard edit is at an uncovered vertex; the spine family has excess zero and $\mathrm{tw}(I(H^*)) \leq 2$ after one exchange and one augmentation (Propositions 11.10 and 11.11). A one-sided algorithm in time $5^{w+1}(Z^2+1)^{3\ell(w+1)}n^{O(1)}$ is exact whenever some optimum has only hard edits that are co-bagged or at local hubs (Theorem 11.13); its promise cannot be tested. No polynomial procedure outputs a repacking with excess and width bounded in ℓ alone unless P = NP (Theorem 11.15), bounded-radius augmentation cannot find the good packing (Proposition 11.17), and the naive normal form "hard edits have an uncovered end" is false (Proposition 11.16); a corrected normal form is a conjecture (Conjecture 11.18). Locating the repacking in fixed-parameter time is impossible unless FPT = W[1] (Open Problem 1 with Theorem 5.1).

Appendices A–J contain further proved statements that are not used in the main text: twins, the structure of the vertices outside the packing and a shrink rule, counting lemmas, the dynamic program for ω, conflict weights and a pruned variant, a family on which reduction on unions of packed P_3's fails, and tunnel gadgets. Appendix G contains the proofs for Section 10, Appendix H those for Section 11, and Appendix I the reduction and the proofs for Section 5.

What is open. The main question is settled: CEaMP is W[1]-hard in ℓ+tw$(I(H))$ (Theorem 5.1), so no $f(\ell, w)n^{O(1)}$ algorithm exists unless FPT = W[1]. What remains open is the fine structure. (1) **OPEN:** hardness for *bounded* c: the reduction has $c = 4p + 2$ with $p \geq k^2 + 5$ and $\ell = k^2$, because the clique blow-up that prevents splitting needs $p > \ell$; the cases $c = 2$ and $c \leq 22$ are open (Open Problems 2 and 7). (2) **OPEN:** the exact exponent: the reduction excludes $f(\ell, w)n^{o(\sqrt{\ell})}$ under ETH, while the algorithms have exponent ℓ resp. 2ℓ (Open Problem 3). (3) **OPEN:** forests $I(H)$ with large trees and soft-triangle-free packings, where the reduction does not apply (it needs soft triangles, Theorems 10.28 and 10.23); there the question reduces to the bounded-certificate hypothesis BC (Theorem 10.18). Section 11 shows that the far-deletion family and the spine family are artifacts of the chosen packing, and that *locating* the repacking is impossible in fixed-parameter time unless FPT = W[1], since it would give fixed-parameter tractability in ℓ+w (Proposition 11.19); the normal form of Section 7 applies to a pair-maximal extension of the given packing, whose intersection graph may have larger treewidth. Section 12 lists the open problems and summarises, in one table, which result holds for the input packing and which for a pair-maximal extension, and which steps are computer-assisted.

1.2 Related work

van Bevern, Froese and Komusiewicz [17] parameterize F-free editing above packings; for Cluster Editing they obtain fixed-parameter algorithms for vertex-disjoint P_3 packings and leave open the modification-disjoint case, which Li, Pilipczuk and Sorge [15] settle negatively for $\ell = 0$ and study for $1/c$-integral packings. Modification-disjoint P_3 packings are also used as a lower-bound tool for

cluster vertex splitting [8]; for broader context see the survey [5]. To our knowledge, the treewidth of the intersection graph of the packing has not been used as a parameter before. Komusiewicz and Uhlmann [13] show that Cluster Editing is NP-hard on graphs of maximum degree 6 and has no $2^{o(k)}n^{O(1)}$ algorithm under ETH. The critical-clique (twin) property used in Appendix A goes back to [10, 7]; the half-degree property of Lemma 7.2 is the same single-vertex-move argument. Cubic expanders of linear treewidth are cited from Grohe and Marx [9].

2 Preliminaries

Graphs and solutions. Graphs are simple and undirected, $n = |V(G)|$. A *pair* is a 2-subset of $V(G)$; for a set S of pairs, $G \triangle S$ flips the adjacency of each pair in S. A graph is a *cluster graph* iff it has no induced P_3, i.e. no vertex triple with exactly two edges. A *solution* is a set S of pairs with $G \triangle S$ a cluster graph; $\mathrm{OPT}(G)$ is the minimum size of a solution. Solutions correspond to partitions of V into *clusters* (the cliques of $G \triangle S$); the *cost* of a partition is the number of pairs ab with $[ab \in E] \neq [a, b \text{ in one cluster}]$, and an *optimum* is a partition of minimum cost. We write $\alpha(\cdot)$ for the independence number and $\nu(\cdot)$ for the matching number.

Packings, soft and hard pairs. A *packing* $H = \{P_1, \ldots, P_p\}$ is a set of induced P_3's $P_i = (a_i, b_i, c_i)$ with centre b_i ($a_i b_i, b_i c_i \in E$, $a_i c_i \notin E$); its elements are the *members* or *packed P_3's*. Write $\mathrm{pairs}(P_i)$ for the three pairs inside $V(P_i)$. H is *modification-disjoint* if the sets $\mathrm{pairs}(P_i)$ are pairwise disjoint; this is equivalent to: any two members share at most one vertex. All packings below are modification-disjoint. A pair is *soft* if it lies in $\bigcup_i \mathrm{pairs}(P_i)$ (its *owner* is then unique) and *hard* otherwise; a *hard edge* is a hard pair in E and a *hard non-edge* a hard pair outside E. For a vertex v, $Q(v)$ is the set of soft partners of v, $N_h(v)$ the set of hard neighbours and $d_h(v) = |N_h(v)|$. G_{soft} is the *soft graph* on V whose edges are the soft pairs.

Problem CEaMP. Given G, a modification-disjoint packing H and $\ell \geq 0$, decide whether there is a solution S with $|S| \leq |H| + \ell$ (the convention of [15]; ℓ is the *excess*). We call (G, H, ℓ) a *YES instance* if $\mathrm{OPT}(G) \leq |H| + \ell$.

The intersection graph. The *intersection graph* $I(H)$ has node set H and an edge PP' iff $V(P) \cap V(P') \neq \emptyset$. The members through one vertex form a clique of $I(H)$; hence (Helly property of tree decompositions) they lie in a common bag. Put $c = c(H) = \max_v |\{P \ni v\}|$ and $w = \mathrm{tw}(I(H))$ (with $w = 0$ if $I(H)$ has no edge), so $c \leq w + 1$. Tree decompositions of width w are assumed *given* in the algorithmic statements; one of width w is computed in time $2^{O(w^3)}n$ [3, 6], and one of width at most $2w + 1$ in time $2^{O(w)}n$ [14], which would change the bases below (e.g. 7^{w+1} to 7^{2w+2}).

Lemma 2.1 (Degrees). *For every vertex v, $|Q(v)| = 2\,|\{P \in H : v \in V(P)\}|$. Hence $D(H) := \max_v |Q(v)| = 2c \leq 2(w + 1)$.*

Proof. Each member $P \ni v$ contributes the two soft pairs of P at v, i.e. two partners. Two members through v cannot have a common partner $x \neq v$, since they would then share the two vertices v, x, hence a pair. So the partner sets are disjoint and $|Q(v)| = 2|\{P \ni v\}|$. □

Below, D denotes a *number* with $D \geq \max_v |Q(v)|$ (for instance $D = D(H) = 2c$), and $D(H)$ the function value.

Lemma 2.2 (Solutions decompose)**.** *Let S be any set of pairs such that $G \triangle S$ is a cluster graph, $x_i = S \cap \text{pairs}(P_i)$ and $U = S \setminus \bigcup_i \text{pairs}(P_i)$ (the* hard edits*). Then every $x_i \neq \emptyset$, $|S| = \sum_i |x_i| + |U|$, and $|U| \leq |S| - |H|$.*

Proof. If $x_i = \emptyset$ the triple $V(P_i)$ keeps its adjacencies in $G \triangle S$ and is an induced P_3. The pair sets are disjoint, so $|S| = \sum |x_i| + |U| \geq |H| + |U|$. □

Consequently the *excess* of a solution is $\sum_i (|x_i| - 1) + |U|$, and every solution of size at most $|H| + \ell$ has $|U| \leq \ell$.

2.1 Maximalisation and pair-maximal packings

A family Φ of induced P_3's is *hard over H* if the sets $\text{pairs}(P)$, $P \in \Phi$, are pairwise disjoint and disjoint from the soft pairs of H; then $H \cup \Phi$ is a packing with $|H| + |\Phi|$ members. H is *pair-maximal* if no induced P_3 of G has all three pairs hard, i.e. if H is maximal among modification-disjoint packings. This is strictly stronger than maximality among vertex-disjoint packings (Proposition 4.5).

Lemma 2.3 (Hard families)**.** *Let (G, H, ℓ) be a YES instance and Φ a family hard over H. Then $|\Phi| \leq \ell$. In general* $\text{OPT} \geq |H| + |\Phi|$.

Proof. $H \cup \Phi$ is a modification-disjoint packing of $|H| + |\Phi|$ induced P_3's. Lemma 2.2, applied to $H \cup \Phi$ and an optimum solution, gives $\text{OPT} \geq |H| + |\Phi|$. In a YES instance $\text{OPT} \leq |H| + \ell$. □

Lemma 2.4 (Maximalisation)**.** *Let H be any packing and M a maximal family hard over H (inclusion-maximal; found greedily in polynomial time). Put $H' = H \cup M$, $\ell' = \ell - |M|$. Then:*

(a) $\text{OPT} \geq |H'|$. *In particular $|M| > \ell$ implies that (G, H, ℓ) is a NO instance.*

(b) $\text{OPT} \leq |H| + \ell$ *iff* ($|M| \leq \ell$ *and* $\text{OPT} \leq |H'| + \ell'$). *So (G, H, ℓ) and (G, H', ℓ') are equivalent whenever $\ell' \geq 0$.*

(c) H' is a pair-maximal packing ("hard" now means hard with respect to H'; the pairs of M have become soft).

(d) $\text{tw}(I(H')) \leq \text{tw}(I(H)) + |M|$, *hence $w' \leq w + |M| \leq w + \ell$.*

(e) $c' \leq \min\{c + |M|,\ w' + 1\}$ *and* $D(H') = 2c' \leq D(H) + 2|M|$.

A tree decomposition of $I(H')$ of width at most $w + |M|$ is obtained in polynomial time from one of $I(H)$.

Proof. (a) is Lemma 2.3 (second sentence: $\text{OPT} \geq |H| + |M|$). (b) If $\text{OPT} \leq |H| + \ell$ then $|M| \leq \ell$ by (a), and $|H'| + \ell' = |H| + \ell$; the converse is the same identity.

(c) The members of M have pairwise disjoint pair sets, disjoint from those of H, so H' is a packing. Suppose an induced P_3 Q has all its pairs hard with respect to H'. Then $M \cup \{Q\}$ is a family hard over H ($Q \notin M$ because the pairs of Q are not in $\text{pairs}(M)$), contradicting maximality of M.

(d) Let $(\mathcal{T}, \{\beta_t\})$ be a tree decomposition of $I(H)$ of width w (if $H = \emptyset$ take a single empty bag). $I(H')[H] = I(H)$. Put $\beta'_t = \beta_t \cup M$. Every node of H still occupies a connected subtree. Every node of M occupies all of $\mathcal{T}$. An edge of $I(H')$ with both ends in H is covered as before; an edge with an end in M is covered by any bag containing its other end (if both ends are in M every bag covers it). Bags have at most $w + 1 + |M|$ nodes.

(e) At most $|M|$ members of M pass through a vertex, so $c' \leq c + |M|$; $c' \leq w' + 1$ is the inequality $c \leq w + 1$ for H'. By Lemma 2.1, $D(H') = 2c' \leq D(H) + 2|M|$. □

Example 2.5 (Tightness of Lemma 2.4(d))**.** Let G be the star with centre v and leaves $p, q, a_1, b_1, \ldots, a_m, b_m$, $H = \{(p, v, q)\}$ and $M = \{(a_i, v, b_i) : 1 \le i \le m\}$. M is a maximal family hard over H: every induced P_3 of a star is (x, v, y) with leaves x, y, and every pair xv is already soft. $I(H)$ is one node, $\mathrm{tw} = 0$; $I(H \cup M) = K_{m+1}$, $\mathrm{tw} = m = 0 + |M|$.

Convention: pair-maximal packings and primed parameters. Sections 7–9 and the appendices state their results for a *pair-maximal* H; where a result needs a YES instance or an optimal solution, it says so. For an arbitrary input packing H let M be as in Lemma 2.4 and read the statements for the *primed instance* (G, H', ℓ') with parameters $w' \le w+|M|$, $c' \le c+|M|$, $D' \le D+2|M|$, $\ell' = \ell - |M|$. The family M is not unique (it depends on the greedy order), so quantities defined from H' (such as ω in Section 8) are parameters of the primed instance, not functions of the input (G, H, ℓ). Only Theorems 3.5, 3.6, 6.1, 6.10 and 8.2 are statements about the input H in terms of w, ℓ and n; the table in Section 12 lists the scope of every result.

Notation. $P_i = (a_i, b_i, c_i)$ with centre b_i; c is the maximum number of members through a vertex (unsubscripted), and $\kappa_i \ge 1$ the number of edited pairs among $\mathrm{pairs}(P_i)$. $U = U^- \cup U^+$ is the set of hard edits, U^- the deleted hard edges and U^+ the inserted hard non-edges, $u = |U|$. $\ell' = \ell - |M|$ is the primed excess and $\bar{\ell}$ the excess left after component accounting (Section 6). $\mathcal{C}_j$ are the components of $I(H)$ and W_j the vertex sets they span. $R = V \setminus V(H)$ (Appendix A). $B = 2D + 3\ell$, $Z = B + D + 1$ and BIG are defined in Section 7. Hard components K and the cliques $\mathcal{K}_v$ of the constructions in Section 8 are distinct objects.

3 The interaction CSP and its dynamic programs

Observation 3.1. *Let S be a solution with $|S| \le |H|$. Then S has exactly one pair in each* $\mathrm{pairs}(P_i)$ *and no other pair, and $G \triangle S$ is a cluster graph iff no vertex triple is an induced P_3 of $G \triangle S$. Consequently $(G, H, 0)$ is a YES instance iff the following CSP is satisfiable: variables $x_i \in \{1, 2, 3\}$ (which pair of P_i is flipped), and for every vertex triple t the constraint "t is not an induced P_3 after flipping", whose truth depends on the owners' variables of the soft pairs of t only (hard pairs keep their status in G).*

Proof. By Lemma 2.2, $|x_i| \ge 1$ for all i and $|S| = \sum |x_i| + |U| \le |H|$ force $|x_i| = 1$ and $U = \emptyset$. The rest is the definition of a cluster graph. □

The CSP has $\Theta(n^3)$ constraints, not $O(|H|)$.

Observation 3.2. *The set of owners of the soft pairs of any triple is a clique of $I(H)$ (at most three nodes). Hence the primal graph of the CSP is a subgraph of $I(H)$ (not every edge of $I(H)$ is witnessed by a triple), and a tree decomposition of $I(H)$ is one of the primal graph.*

Proof. Two pairs of one triple share a vertex v; their owners both contain v, so they coincide or are adjacent in $I(H)$. □

Lemma 3.3 (Five states)**.** *Let $P = (a, b, c)$ be a member, with edges ab, bc and non-edge ac. Of the seven non-empty subsets $x \subseteq \mathrm{pairs}(P)$, exactly five leave no induced P_3 on $\{a, b, c\}$ after editing: $\{ab\}$, $\{bc\}$, $\{ac\}$ (cost 1, states 1, 2, 3), $\{ab, bc\}$ (cost 2, state 4) and $\{ab, bc, ac\}$ (cost 3, state 5). The sets $\{ab, ac\}$ and $\{bc, ac\}$ leave an induced P_3.*

Proof. After editing $\{ab, ac\}$ the edges on $\{a, b, c\}$ are bc, ac, after $\{bc, ac\}$ they are ab, ac: two edges, an induced P_3. The other five sets leave the edge sets $\{bc\}$, $\{ab\}$, $\{ab, bc, ac\}$, $\emptyset$ and $\{ac\}$, none of which has exactly two edges. □

Hence in every solution each x_i is one of these five sets (its triple $V(P_i)$ must be a cluster graph), and the variables of the dynamic programs below may range over five values; this changes no solution and no cost.

The natural parameter of the CSP is therefore the treewidth of its primal graph, which is at most w; the ETH bound of Section 4 is stated for $I(H)$.

Lemma 3.4 (Dynamic programming over $I(H)$)**.** *Let there be one variable x_i per member, with a domain $\mathcal{D}_i$ of size at most d and a unary cost $\gamma_i : \mathcal{D}_i \to \mathbb{N}$, and let $\mathcal{C}$ be a set of constraints, each given by a test of its satisfaction on $\prod \mathcal{D}_i$ in polynomial time, whose scope is a clique of $I(H)$. Given a tree decomposition of $I(H)$ of width w, a satisfying assignment of minimum total cost is found in time $d^{w+1}(|H| + |\mathcal{C}|)^{O(1)}$.*

Proof. Let $(\mathcal{T}, \{\beta_s\})$ be the tree decomposition. By the Helly property each clique lies in a bag; assign each constraint to one such bag. For a node s and an assignment α of the variables of β_s, let $f_s(\alpha)$ be the minimum cost over the variables in the subtree of s, among assignments extending α and satisfying all constraints assigned in the subtree. The standard recurrences (introduce, forget, join; see [6]) compute all f_s; each table has at most d^{w+1} entries. □

Theorem 3.5. *Given (G, H) and a tree decomposition of $I(H)$ of width w, CEaMP with $\ell = 0$ is solvable in $3^{w+1}n^{O(1)}$ time.*

Proof. Observations 3.1 and 3.2 give a CSP with domains of size 3, all unary costs 1 and $O(n^3)$ constraints with clique scopes; apply Lemma 3.4. □

Theorem 3.6 (CSP algorithm)**.** *Given (G, H, ℓ) and a tree decomposition of $I(H)$ of width w, CEaMP is solvable in $5^{w+1}n^{2\ell+O(1)}$ time.*

Proof. Let $\mathcal{D}_i$ be the 5 subsets of pairs(P_i) of Lemma 3.3. By Lemma 2.2, a solution of size $\leq |H|+\ell$ has the form $S = U \cup \bigcup x_i$ with $x_i \in \mathcal{D}_i$ and U a set of at most ℓ hard pairs; there are at most $\sum_{j \leq \ell} \binom{n^2/2}{j} \leq n^{2\ell}$ candidates for U.

Fix U and put $G_U = G \triangle U$. Since U avoids soft pairs, every P_i is still an induced P_3 of G_U with the same pairs, so H is still a modification-disjoint packing. For a triple t, the status of each hard pair of t in $G_U \triangle x$ is fixed, and each soft pair is determined by G_U and the variable of its owner. Hence t gives a constraint C_t ("t is not an induced P_3 in $G_U \triangle x$") whose scope is the owner set of t, a clique of $I(H)$ (the proof of Observation 3.2 does not use G itself). Triples with empty scope are tested directly, in $O(n^3)$ time. Seek $x \in \prod \mathcal{D}_i$ satisfying all C_t and minimizing $\sum_i |x_i|$; by Lemma 3.4 this takes $5^{w+1}n^{O(1)}$ time. Accept iff for some U the minimum m_U satisfies $m_U + |U| \leq |H| + \ell$. Soundness: any feasible (U, x) gives the solution $U \cup \bigcup x_i$. Completeness: Lemmas 2.2 and 3.3. □

Theorem 3.5 is the case $\ell = 0$ of Theorem 3.6 with the smaller base 3. Together with Observations 3.1 and 3.2 and Lemma 3.4 these are parts (b) and (c) of Theorem 1.1. The algorithm of Theorem 3.6 is not fixed-parameter tractable: ℓ is part of the input and appears in the exponent of n, and Proposition 4.3 shows that w alone does not suffice. By Theorem 5.1 this cannot be repaired by a fixed-parameter algorithm in $\ell + w$ unless FPT = W[1].

Remark 3.7 (Domain $\{1,2,3\}$ does not suffice for $\ell > 0$). In the encoding above, a variable ranges over the five states of Lemma 3.3. For $\ell > 0$ it cannot be restricted to the three single-pair values: an edited packed P_3 can receive two or three edits in every optimum (the states 4 and 5). For two: $n = 5$, $E = \{02, 04, 12, 23, 24\}$, $H = \{(1,2,3)\}$, OPT $= 2$, unique optimum $\{12, 23\}$. For three: $n = 6$, $E = \{01, 02, 03, 04, 12, 14, 15, 24, 25, 35, 45\}$, $H = \{(0,3,5)\}$, OPT $= 3$, unique optimum $\{03, 05, 35\}$ (both checked by enumerating all set partitions). We do not claim that 5 is the smallest possible domain size in other encodings. For $c = 2$, $I(H)$ is subcubic but may have unbounded treewidth, so Theorem 3.6 is a different tradeoff from the $n^{2\ell}$ algorithm of [15], not a recovery of it.

4 Limits of the parameterization

4.1 An ETH lower bound in the exponent

Theorem 4.1. *Assume ETH. There is no algorithm that decides CEaMP with $\ell = 0$ in time $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$. This holds even when every vertex lies in at most 23 packed P_3's (so $I(H)$ has maximum degree at most 66), and whether or not a tree decomposition is supplied.*

The theorem is a corollary of the reduction of Li, Pilipczuk and Sorge.

Proof. By the construction behind [15, Thm. 1] there is a polynomial reduction from 3-SAT to CEaMP with $\ell = 0$ in which every vertex lies in at most 23 packed P_3's and the graph has $|V| = O(N+m)$ vertices for a formula with N variables and m clauses (each variable occurrence and each clause is replaced by a gadget of constant size). By the Sparsification Lemma [11] (see also [6]) we may assume $m = O(N)$ at the cost of $2^{o(N)}$ extra time, so $|V| = O(N)$.

Every member has three vertices and every vertex lies in at most 23 members, hence $3|H| = \sum_v |\{P \ni v\}| \le 23|V|$, i.e. $|H| \le 23|V|/3 = O(N)$. A single bag containing all members is a tree decomposition of $I(H)$, so $\mathrm{tw}(I(H)) \le |H| - 1 = O(N)$, and such a decomposition can be supplied. An algorithm with running time $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$, which may compute its own decomposition, would therefore decide 3-SAT in time $2^{o(N)}$, contradicting ETH. The degree bound: a member meets at most $3 \cdot 22 = 66$ others. □

Remark 4.2. (1) The bound is on the exponent only: it does not show that the base 3 (or 5) is tight, and $2^{o(w)}$ excludes $3^{o(w)}$ and $5^{o(w)}$ alike. A SETH-type bound $(3-\varepsilon)^w$ is open. (2) On the instances of the theorem every vertex lies in at most 23 members, so $\mathrm{tw}(I(H)) \le |H| - 1 \le 23n/3$ (for general packings $|H|$ can exceed n); hence on these instances a $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$ algorithm is in particular a $2^{o(n)}$ algorithm, the constant 23/3 being absorbed in the $o(\cdot)$, so the theorem is no stronger than the statement that CEaMP at $\ell = 0$ has no $2^{o(n)}$ algorithm under ETH; it says that the exponent of Theorem 3.5 cannot be improved to $o(w)$ on instances with $|H| = \Theta(n)$.

4.2 Other parameters do not suffice

Proposition 4.3. *CEaMP is NP-hard when $I(H)$ has no edge ($w = 0$, vertex-disjoint packing), with ℓ part of the input. Hence w alone is not enough, and (by Theorem 1 of [15]) neither is $\ell = 0$ alone.*

Proof. Cluster Editing is NP-hard (e.g. already on graphs of maximum degree 6 [13]). Given (G,k) compute greedily a maximal set H of vertex-disjoint induced P_3's (if none, G is a cluster graph). Vertex-disjoint implies modification-disjoint and $I(H)$ is edgeless. Since OPT $\ge |H|$, (G,k) is a YES instance iff $(G, H, k - |H|)$ is (a NO instance if $k < |H|$). □

Proposition 4.4. *(a) For the star $G = K_{1,2m}$ ($m \geq 1$) every packing $H \neq \emptyset$ has $I(H) = K_{|H|}$ and every YES instance (G, H, ℓ) satisfies $\ell + \mathrm{tw}(I(H)) \geq 2m - 2$, while $\mathrm{tw}(G) = 1$. (b) For the $k \times k$ grid G and a family H of vertex-disjoint induced P_3's along rows, $\mathrm{tw}(I(H)) = 0$ and $\mathrm{tw}(G) = k$.*

Proof. (a) Leaves have degree one, so the centre of any induced P_3 is the hub v; all packed P_3's are (a, v, b) with distinct leaves (the pairs va are disjoint), pairwise sharing v, so $I(H) = K_p$, $p = |H| \leq m$, $\mathrm{tw} = p - 1$. If the hub's cluster contains j leaves, the cost is $(2m - j) + \binom{j}{2}$ (the other leaves are best left as singletons), which is minimized, with value $2m - 1$, at $j \in \{1, 2\}$; so $\mathrm{OPT} = 2m - 1$ and a YES instance has $\ell \geq 2m - 1 - p$, i.e. $\ell + \mathrm{tw}(I(H)) \geq 2m - 2$. (b) Rows of a grid are induced P_3's (no triangles) and $\mathrm{tw}(k \times k \text{ grid}) = k$ is standard. □

Proposition 4.4 concerns the numbers $\mathrm{tw}(G)$ and $\mathrm{tw}(I(H))$ only, which are incomparable. Part (a) is the only part that bears on the pair (ℓ, w): bounded $\mathrm{tw}(G)$ does not make $\ell + w$ small, because on a star small $\mathrm{tw}(I(H))$ forces large ℓ. Part (b) says nothing about ℓ.

4.3 Weaker maximality hypotheses fail

The following two instances show that the structure theory of Appendix A and Section 7 needs pair-maximality and that edits touching vertices outside $V(H)$ cannot be excluded.

Proposition 4.5 (Pair-maximality cannot be replaced by vertex-maximality)**.** *Let G have vertices $0, \ldots, 4$ and edges $03, 13, 14, 24, 34$, and $H = \{(0, 3, 1)\}$ (pairs $03, 13, 01$). Then H is maximal among vertex-disjoint packings (only the two vertices $2, 4$ remain), $\mathrm{OPT} = 2$ and the unique optimum is $\{03, 24\}$; it contains the edit 24 with both endpoints outside $V(H)$. Here $\{3, 4, 2\}$ is an induced P_3 all of whose pairs are hard, so H is not pair-maximal.*

Proof. The partition $\{\{0\}, \{1, 3, 4\}, \{2\}\}$ has the two edits $03, 24$. Conversely, $0\,3\,1$ and $3\,4\,2$ are induced P_3's with disjoint pair sets $\{03, 13, 01\}$ and $\{34, 24, 23\}$, so $\mathrm{OPT} \geq 2$. That this is the only optimum is checked by enumerating all 52 partitions of the five vertices. □

A second example: for the path $0\,1\,2\,3\,4$ and $H = \{(0, 1, 2)\}$ the remaining vertices $3, 4$ cannot form a P_3, so H is maximal among vertex-disjoint packings, but $(2, 3, 4)$ is an induced P_3 whose pairs $23, 34, 24$ are disjoint from $\mathrm{pairs}(H)$, so H is not pair-maximal; the pair-maximal extension is no longer vertex-disjoint.

Proposition 4.6 (Strong locality fails)**.** *Let $G = C_4$ with edges $01, 02, 13, 23$ and $H = \{(0, 1, 3)\}$ (pairs $01, 13, 03$; vertex $2 \notin V(H)$). Then H is pair-maximal, $\mathrm{OPT} = 2 = |H| + 1$, so $(G, H, 0)$ is a NO instance, and there are exactly three optima $\{03, 12\}$, $\{01, 23\}$, $\{02, 13\}$, each containing a hard pair at vertex 2.*

Proof. Every induced P_3 of C_4 contains one of the soft pairs $01, 13, 03$, so H is pair-maximal. $\mathrm{OPT} \geq |H| = 1$, and a single edit leaves an induced P_3 (checked on the six pairs), so $\mathrm{OPT} = 2$; the three optima are found by enumerating all 15 partitions. □

This refutes the claim that all hard edits of some optimum have both endpoints in $V(H)$; what remains true is that no edit has both endpoints outside $V(H)$ (Theorem A.5 in Appendix A).

5 W[1]-hardness

Theorems 3.5 and 3.6 solve CEaMP in time $5^{w+1}n^{2\ell+O(1)}$, and Theorem 8.2 halves the exponent of n; all of these are XP in ℓ. This section shows that the exponent of n cannot be removed: the main question of this paper has a negative answer.

Theorem 5.1 (W[1]-hardness)**.** *There is a polynomial-time reduction that maps an instance $(k, n, (S_{ij}))$ of Grid Tiling with $\leq$ to an instance (G, H, ℓ) of CEaMP with the following properties. Let p be the least even integer with $p \geq k^2 + 5$.*

(i) $\ell = k^2$*; H is a pair-maximal modification-disjoint packing;* $|V(G)| = O(k^2 n^2 p)$.

(ii) $c(H) \leq 4p + 2$*, and* $\mathrm{tw}(I(H)) \leq \mathrm{pw}(I(H)) \leq 24\,k\,p^2$*, independently of n; $I(H)$ has at most $2k$ components.*

(iii) (G, H, ℓ) is a YES instance iff the Grid Tiling instance has a solution; if it has one, then $\mathrm{OPT}(G) = |H| + k^2$.

Consequently CEaMP is W[1]-hard parameterized by $\ell + \mathrm{tw}(I(H))$, by $\ell + \mathrm{pw}(I(H))$ and by $\ell + c$, even for pair-maximal packings. Unless FPT = W[1] *there is no algorithm with running time $f(\ell, w)\, n^{O(1)}$, and unless ETH fails there is no algorithm with running time $f(\ell, w)\, N^{o(\sqrt{\ell})}$, where $N = |V(G)|$.*

Here $p \leq k^2 + 6$, so on the produced instances $\ell = k^2$, $c \leq 4k^2 + 26$ and $\mathrm{tw}(I(H)) \leq 24k(k^2+6)^2$ are all functions of k alone. Grid Tiling with $\leq$ is the following problem [6] (Chapter 14): given k, n and non-empty sets $S_{ij} \subseteq [n]^2$ $(1 \leq i, j \leq k)$, choose $s_{ij} = (a_{ij}, b_{ij}) \in S_{ij}$ with $a_{ij} \leq a_{i,j+1}$ and $b_{ij} \leq b_{i+1,j}$; it is W[1]-hard in k and has no $f(k)n^{o(k)}$ algorithm under ETH (Marx; see [6], Chapter 14). (We use the version of the cited chapter after exchanging the two coordinates.) The reduction, all lemmas and the proof of Theorem 5.1 are in Appendix I; this section states what is proved, outlines the construction, and says precisely which steps of the proof use a computer and what has been tested.

5.1 What the theorem says, and what it does not

Corollary 5.2 (The ETH bound that follows)**.** *Assume ETH. There are no computable f and no algorithm that decides CEaMP in time $f(\ell, \mathrm{tw}(I(H)))\, N^{o(\sqrt{\ell})}$ on instances with N vertices.*

Proof. Suppose such an algorithm runs in time $f(\ell, w)N^{h(\ell)}$ with $h(\ell) = o(\sqrt{\ell})$. Apply the reduction of Theorem 5.1 to a Grid Tiling instance with parameter k: then $\ell = k^2$, $w \leq 24k(k^2+6)^2 =: g(k)$ and $N \leq \mathrm{poly}(k)\, n^2$. The running time is at most $f(k^2, g(k))\,(\mathrm{poly}(k)n^2)^{h(k^2)}$, and $h(k^2) = o(k)$, so it is at most $f'(k)\, n^{o(k)}$ for a computable f' (the factor $\mathrm{poly}(k)^{h(k^2)} = 2^{o(k \log k)}$ is absorbed in f'). This solves Grid Tiling with $\leq$ in time $f'(k)n^{o(k)}$, contradicting ETH [6]. □

Nothing stronger follows from this reduction: since $\ell = k^2$, the lower bound is $n^{o(\sqrt{\ell})}$, while the upper bounds of Theorems 3.6 and 8.2 have exponent 2ℓ and ℓ. The gap between $\sqrt{\ell}$ and ℓ is open (Open Problem 3). Since w is a polynomial in k on the produced instances, no statement in terms of $\ell + w$ jointly is claimed beyond Corollary 5.2.

Matching the XP algorithm. Theorem 5.1 shows that the exponent of n in Theorems 3.6 and 8.2 cannot be made independent of ℓ, even though w and c are bounded by functions of ℓ alone on the hard instances. Up to the exact exponent of n ($\Omega(\sqrt{\ell})$ under ETH, against ℓ resp. 2ℓ proved), the complexity of the pair (ℓ, w) is therefore settled.

What is still open. (1) The reduction has $c = 4p + 2$ with $p \geq k^2 + 5$, so c grows with ℓ; whether CEaMP is W[1]-hard (or fixed-parameter tractable in ℓ) when c is bounded by a constant, in particular for $c = 2$, is open (Open Problems 2 and 7). The reason is local to the proof: p is the size of the cliques into which every chain vertex is blown up, and the argument that no clique is split (Lemma I.7) needs $p > \ell$. (2) The exact exponent under ETH (Open Problem 3). (3) Packings with $w \leq 1$ with tree components of unbounded size, and soft-triangle-free packings (Open Problems 6 and 8); the instances here have many soft triangles and large w (the width is $\Theta(p^2)$ per chain).

5.2 The construction in outline

The reduction follows the plan of the tunnel attempt of Appendix F, and realises the abstract problem of Theorem 10.27 with soft triangles; the obstruction of Theorem 10.28 is circumvented because the instances are not soft-triangle-free. We use the language of zero-excess (ZE) solutions: a member (a, b, c) with centre b is in state L, R or M if its single edit deletes ab, deletes bc, or inserts ac (states $1, 2, 3$ of Appendix F).

A selection chain. A *thin chain* OH_m ($m \geq 7$) is an explicit tight instance on the vertices $s_0, \ldots, s_{2m}, e_0, \ldots, e_m$ with spine members $P_j = (s_{2j}, s_{2j+1}, s_{2j+2})$, further interior members G_j, F_j that carry the vertices e_j, two caps at the ends and four hard edges (Appendix I). Its ZE partitions are exactly m *positions* $a \in \{0, \ldots, m-1\}$, in which the spine members are in states $R^a M L^{m-1-a}$ (Lemma I.1). This is a tight instance of bounded c and bounded width in which a "position" is selected, so that a cut at any one of arbitrarily many places repairs it (Remark I.13).

Thick chains. Every thin vertex v becomes a *super-vertex* $K_v = \{(v, i) : i \in \mathbb{Z}_p\}$, glued by hard edges, and every thin member (a, b, c) is replaced by the p^2 *Latin members* $((a, i), (b, j), (c, i + j))$. These cover every pair between K_a, K_b, between K_b, K_c and between K_a, K_c exactly once. All pairs of such a triple of super-vertices are soft; so if $i + j \neq k$, the triple $(a, i), (b, j), (c, k)$ is a soft triangle. The purpose is that a ZE partition cannot split a super-vertex (the glue is hard) and that, for an arbitrary solution of excess at most ℓ, splitting a super-vertex costs more than ℓ hard edits when $p \geq \ell + 5$ (Lemma I.7). Hence the thick chain behaves like the thin chain, while a vertex outside the chain can be attached to a super-vertex at a cost that is independent of the position.

Guests, locks and comparators. A *guest* is a single new vertex y with a set W_y of super-vertices to which it is soft (through $p/2$ *pad members* $(y, (v, i), (v, k))$ for a perfect matching of K_v); all other pairs of y are hard non-edges. A guest can join the cluster of a base vertex h only if the entire cluster of h lies in W_y; so the set W_y forbids exactly those positions of the chain at which the cluster of h contains a vertex outside W_y (Lemma I.2). Each chain is divided into $U = 1 + n(n+1)$ *groups* of two consecutive positions. A *permanent guest* sits in the cluster of a host vertex of each group and, for groups that carry no tile, forbids both positions of the group. For a tile $(a, b) \in S_{ij}$ the chain A_{ij} has a *lock guest* and an *anchor* at the slot group (a, b), and the chain B_{ij} at the slot group (b, a); the two anchors of one tile are joined by a hard edge, a *tunnel*. An uncut tunnel forces the lock guest into the host cluster, hence forbids the group; cutting it (one hard edit, cost one) frees the group. So a chain whose tunnels are all uncut has no position, and a chain with exactly one cut tunnel is in a position of the group of that tunnel; as the same hard edit releases the chains A_{ij} and B_{ij}, *it chooses the tile* s_{ij} *in both orders* (a, b) *and* (b, a) *at the price of one edit.* Finally, for consecutive chains of a row (of columns) and a separator group $\alpha < n$, a *comparator guest* y with a member (z, y, z') (z, z' the permanent guests of the two chains) must join exactly one of the two chains, and can do so at the first chain only if its position is before the separator and at the

second only if it is after; this is the clause $(\beta \le \alpha) \vee (\beta' > \alpha)$ on the blocks β, β' of the two chosen tiles, and the $n-1$ clauses together express $\beta \le \beta'$. This is Proposition I.5.

Soundness by counting. The completeness direction is Proposition I.5. For soundness, an arbitrary solution S of excess at most $\ell = k^2$ is analysed without assuming its form. By Lemma I.7 no super-vertex is split. Partition the pairs into regions Π_c, one for each of the $2k^2$ chains; every region needs at least one edit beyond one per member (Lemma I.8), because with all tunnels uncut the chain has no position. A pair of two regions is a *cross-anchor pair* (a pair between the anchor sets of the two chains of one cell); only these pairs can serve two regions, and they are hard. If q of them are edited, the excess is at least $\max(q, 2k^2 - q) \ge k^2$ (Theorem I.9); if it equals k^2, then S is a set of exactly one tunnel cut per cell plus one edit per member, and Proposition I.5 gives a tiling. Thus the instance has $\mathrm{OPT} \le |H| + k^2$ iff the Grid Tiling instance is solvable. In this argument the restricted model of Appendix F is not assumed; it is derived.

Where soft triangles are used. Without soft triangles every instance satisfies the conclusions of Theorem 10.4: lock cores have at most two anchors (Corollary 10.7) and tunnel releases cannot realise Theorem 10.27 (Theorem 10.28). A single chain with its anchors has a core of unbounded size (Remark I.13), so by Lemma F.6 its set of ZE state vectors is closed under no majority operation. The Latin members are the soft triangles: this answers the question of Section 10 whether soft triangles suffice for hardness.

5.3 Computer assistance and what has been tested

The proof of Theorem 5.1 is *computer-assisted*. Precisely, the following finite statements are established by computation and are used as lemmas; all the other steps (Lemmas I.4, I.7, I.8, Proposition I.5, Theorems I.9 and I.10, the pathwidth bound, the pair-maximality) are proved by hand from them.

(CA1) *Lemma I.1(a).* The explicit partitions P_a of the thin chain are ZE partitions for all $m \ge 7$ and a. Each constraint involves at most three consecutive cells; windows at the left end do not depend on m, windows at the right end depend only on the distance to the right end, and interior windows are isomorphic, so a check for $7 \le m \le 60$ and every a (no failure) covers all m.

(CA2) *Lemma I.1(b).* The chain has no other ZE partition. The number N_m of ZE partitions is computed by a transfer-matrix dynamic programme over the cells (six states per interior cell; an explicit integer matrix of size 6×6, identical for all interior cells, with characteristic polynomial $x^4(x-1)^2$); this makes N_m an affine function of m for large m, and a computation gives $N_m = m$ for $7 \le m \le 40$. The statement "the transfer matrix is the same at every interior cell of every m" rests on translation invariance of the member list and was checked at all cells $7, \ldots, 19$ of $m = 26$.

(CA3) *Lemma I.2.* The partner sets of the host vertices (the clusters of the hosts at the four classes of positions). The statement is read off the formula of the partitions P_a; it was in addition checked by computer for all even $m \in [8, 60]$ and all groups (1,836 cases), because the two cap groups have exceptional e-labels.

(CA4) *Counts.* The values $c \le 4p + 2$ and $u_{\max} \le 3$ (Lemma I.6) are obtained by a count that is easily checked by hand from the sets of Lemma I.2; the count was also carried out by computer for $k \le 4$.

Tests that are not part of the proof. (a) The chain lemma was re-verified by an independent implementation (own validity check and own SAT enumeration of all ZE partitions) for $m = 7, \ldots, 14$ and then for a sample up to $m = 100$: always exactly the m partitions P_a. (b) The *ZE semantics* of Proposition I.5: for random Grid Tiling instances and every choice of one tile per cell (a sample when there are many), ZE feasibility of the instance with the corresponding tunnels cut agrees with "the choice is a tiling" ($k = 2$, $n = 2, 3$; $k = 3$, $n = 2$; $k = 1$, $n = 4, 5$; thousands of cases in two independent implementations, no mismatch); for $k = n = 2$ all 386 cut sets of size at most 4 were tested, and exactly the 15 tilings (one tunnel per cell) are feasible. The comparator was tested on all 100 pairs of positions of two chains with $m = 10$, in the thin model and on the thick instance with $p = 8$ (515 vertices): feasible exactly when ($a \leq 3$ or $b \geq 6$) and neither a nor b is a permanently forbidden position. A thick instance with $k = 2$, $n = 2$, $p = 8$ (2,916 vertices, 23,176 members) agreed with the thin model on four choices. (c) *Exact optimum* (all pairs editable; SAT with a cardinality bound on the excess, an UNSAT answer is a proof) on small thick instances only: for $k = 1$ (two chains with one tile and its tunnel, $p = 8$, 428 vertices, $|H| = 3258$) the optimum exceeds $|H|$ and is at most $|H| + 1$ (a certified solution of cost $|H| + 1$); the same instance without the tile has OPT $\geq |H| + 2$; and for a comparator instance (two chains, $p = 6$, 401 vertices, $|H| = 2349$, $\ell = 2$) three YES choices are satisfiable at $|H| + 2$ and one NO choice is unsatisfiable at $|H| + 2$. *No end-to-end exact solve of an instance with $k \geq 2$ was possible* (about 3,000 vertices); for $k = 2, 3$ the evidence is the tests of the thin model above together with the proof of Theorem I.9. (d) Pair-maximality and the validity of the packing were checked on the instances with $(k, n, p) = (1, 2, 8)$, $(1, 2, 6)$ to $(1, 5, 6)$ and $(2, 2, 10)$ (no induced P_3 with three hard pairs). (e) The measured pathwidth upper bounds (an explicit vertex ordering) are 497 for $k = 1$ and 5,049 for $k = 2$ at $p = 6$ resp. 10 on all components together; they are consistent with Lemma I.11.

5.4 Consequences for the rest of the paper

Theorem 5.1 does not contradict any result below: Theorem 6.1 needs components of bounded size, but $I(H)$ here has components of size growing with n; Theorem 8.3 is fixed-parameter tractable in (ℓ, D, ω) where $D = 2c = O(k^2)$ is bounded by a function of k, so ω is unbounded on these instances (the tunnels are hard pairs between far vertices and make the graph Γ' of Section 8 have unbounded treewidth; compare Proposition 8.4); Theorem 10.4 is about soft-triangle-free packings. Two consequences follow from results proved earlier. By Proposition 11.19, the problem of locating a repacking (Open Problem 1) has no fixed-parameter algorithm unless FPT = W[1]; and a reduction lemma in the sense of Open Problem 5 cannot exist unless FPT = W[1], since by Theorem 8.3 it would give fixed-parameter tractability in $\ell + w$.

6 Components of $I(H)$

Let $\mathcal{C}_1, \ldots, \mathcal{C}_r$ be the connected components of $I(H)$, $p_j = |\mathcal{C}_j|$ the number of members in $\mathcal{C}_j$, and $W_j = \bigcup_{P \in \mathcal{C}_j} V(P)$. Put $s = \max_j |W_j|$, the largest number of *vertices* spanned by a component. If $s_P = \max_j p_j$ counts members instead, then $s \leq 2s_P + 1$: order the members of a component so that each shares a vertex with an earlier one; the first spans 3 vertices and each later one adds at most 2.

Theorem 6.1 (Component theorem)**.** *CEaMP can be solved in time* $1.62^{\binom{s}{2}+1)\ell}\, n^{O(1)} + n\, 3^s$ *and admits a kernel with at most* $(s^2 - s + 2)\,\ell$ *vertices, where s is the maximum number of vertices spanned by a connected component of $I(H)$. No maximality hypothesis is used. The kernel is an*

instance of CEaMP with empty packing. The proof uses the soundness of Reduction Rule 6.2 of [17] (their Lemma 6.3), the algorithm of Böcker [2] and the $2k$-vertex kernel of Chen and Meng [4].

Corollary 6.2. *If $I(H)$ is a perfect matching (so $w = 1$, $c = 2$), CEaMP is fixed-parameter tractable in ℓ: here $s = 5$, the time is $1.62^{11\ell} n^{O(1)}$ and the kernel has at most 22ℓ vertices. More generally components with at most s_P members give Theorem 6.1 with $s = 2s_P + 1$.*

Proof. Two adjacent nodes of $I(H)$ share exactly one vertex (H is modification-disjoint), so $|W_j| = 3 + 3 - 1 = 5$; $s^2 - s + 2 = 22$ and $\binom{5}{2} + 1 = 11$. □

What is cited and what is proved. We use the problem of van Bevern, Froese and Komusiewicz [17]: an instance $(G, \mathcal{G}, k)$ consists of a graph, a set $\mathcal{G}$ of *vertex-disjoint induced subgraphs* G_j with $1 \le \tau(G_j) \le t$, where τ is the cluster-editing optimum, and a budget k; its parameters are t and $k - \sum_j \tau(G_j)$ (Definition 1.2 and Problem 1.3 of [17]; connectivity of the G_j is not required, and their size enters only through t). We prove below that the components of $I(H)$ define such an instance with excess at most ℓ (Lemma 6.3), that the reduction rule can be applied at component level without disturbing the other components (Lemma 6.4), and the bound $r \le 2\bar{\ell}$ on the number of irreducible components (Lemma 6.6); the last re-derives, for components, the counting argument of [17, Lemmas 3.2 and 6.5]. The following are *cited, not reproved*: (i) soundness of their Reduction Rule 6.2 for an arbitrary induced subgraph $G[W]$ (their Lemma 6.3, whose proof uses only G, W and the optimal solution S of $G[W]$); (ii) the time to test and apply the rule (their Lemma 6.4); (iii) the algorithm of Böcker [2] running in $O(1.62^k + n + m)$ and the kernel with at most $2k$ vertices for Cluster Editing [4], both as used in [17].

6.1 Components as a packing in the sense of van Bevern et al.

Lemma 6.3 (Components form a packing in the sense of [17])**.** *Let $G_j = G[W_j]$, $\tau_j = \mathrm{OPT}(G_j)$, $k = |H| + \ell$ and $\bar{\ell} = k - \sum_j \tau_j$.*

1. *The sets W_j are pairwise disjoint and each G_j is an induced subgraph.*
2. *$p_j \le \tau_j \le \lfloor \binom{s}{2}/2 \rfloor =: t_s$ and $\tau_j \ge 1$.*
3. *$\bar{\ell} = \ell - \sum_j (\tau_j - p_j) \le \ell$.*
4. *For every solution S of G and every j, $S_j := S \cap \binom{W_j}{2}$ is a solution of G_j, so $|S_j| \ge \tau_j$; the sets $\binom{W_j}{2}$ are pairwise disjoint, so $|S| = \sum_j |S_j| + |S_{\mathrm{ext}}|$ where S_{ext} are the pairs of S inside no W_j. In particular $\mathrm{OPT}(G) \ge \sum_j \tau_j$, and if $\bar{\ell} < 0$ then (G, H, ℓ) is a NO instance.*

Proof. (1) A vertex in $W_i \cap W_j$, $i \ne j$, lies in a member of $\mathcal{C}_i$ and in a member of $\mathcal{C}_j$; these are adjacent in $I(H)$, contradicting that $\mathcal{C}_i, \mathcal{C}_j$ are different components. Induced by definition.

(2) The members of $\mathcal{C}_j$ are induced P_3's of G_j with pairwise disjoint pair sets inside $\binom{W_j}{2}$. A solution of G_j must edit a pair of each of them (Lemma 2.2); hence $\tau_j \ge p_j \ge 1$. For the upper bound, let m_j be the number of edges of G_j; deleting all edges, or inserting all non-edges, yields a cluster graph, so $\tau_j \le \min(m_j, \binom{|W_j|}{2} - m_j) \le \binom{|W_j|}{2}/2 \le \binom{s}{2}/2$.

(3) $|H| = \sum_j p_j$ because the components partition H.

(4) $(G \triangle S)[W_j] = G_j \triangle S_j$ is an induced subgraph of a cluster graph, hence a cluster graph. The rest is counting with disjoint pair sets. □

Thus (G, H, ℓ) is a YES instance iff $(G, \{G_j\}, k)$ is a YES instance of the problem of [17] with $t \le t_s$ and excess $\bar{\ell} \le \ell$ (answer NO if $\bar{\ell} < 0$). Components, not single members, are needed: a pair uv with $u \in W_j$ and $v \notin W_j$ lies in no W_i at all when v is outside $V(H)$ or in another component, hence is not counted in $\sum_j \tau_j$ and is an excess edit (Lemma 6.3(4)); for a packing whose members share vertices, the same pair may be a soft pair of another member and is then free, which is the obstruction of [15].

6.2 Reducible sets and the counting lemma

A set $\mathcal{F}$ of members is *pair-closed* if no member outside $\mathcal{F}$ has two vertices in $W_{\mathcal{F}} := \bigcup_{P \in \mathcal{F}} V(P)$; every component of $I(H)$ is pair-closed, and then no outside member meets $W_{\mathcal{F}}$ at all. An optimal solution $S_{\mathcal{F}}$ of $G[W_{\mathcal{F}}]$ satisfies the *boundary condition* if for every $y \notin W_{\mathcal{F}}$ the set $N(y) \cap W_{\mathcal{F}}$ is empty or exactly one cluster of $G[W_{\mathcal{F}}] \triangle S_{\mathcal{F}}$. $\mathcal{F}$ is *reducible* if some optimal $S_{\mathcal{F}}$ satisfies it, and *irreducible* otherwise. This is the condition of Reduction Rule 6.2 of [17] for the induced subgraph $G[W_{\mathcal{F}}]$ (stated there as two clauses: vertices in one cluster have the same neighbours outside $W_{\mathcal{F}}$; vertices in different clusters have no common neighbour outside $W_{\mathcal{F}}$; the two forms are equivalent). A component $\mathcal{C}_j$ is *tight* if $\tau_j = p_j$.

Lemma 6.4 (Rule 6.2 at component level)**.** *If component $\mathcal{C}_j$ is reducible with S_j, then $(G, \{G_i\}, k)$ is a YES instance iff $(G \triangle S_j, \{G_i\}_{i \neq j}, k - \tau_j)$ is. In the new instance the graphs G_i ($i \neq j$) are unchanged, still vertex-disjoint and induced, with the same τ_i; the excess $\bar{\ell}$ is unchanged; and for every $i \neq j$ the boundary condition, hence reducibility, is unchanged. So one pass over the components, in any order, reaches an instance in which no component is reducible.*

Proof. The equivalence is Lemma 6.3 of [17] (cited; sound for every induced subgraph). Both k and $\sum \tau$ drop by τ_j. The set S_j consists of pairs with both ends in W_j, so no pair incident to a vertex of W_i, $i \neq j$, changes: $G[W_i]$ and all neighbourhoods $N(u)$, $u \in W_i$, are the same before and after, and the boundary condition for i refers only to these. □

Lemma 6.5 (Boundary violations force an edited boundary pair)**.** *Let S be any solution, $\mathcal{F}$ pair-closed, $S_{\mathcal{F}} = S \cap \binom{W_{\mathcal{F}}}{2}$ and $y \notin W_{\mathcal{F}}$. If $N(y) \cap W_{\mathcal{F}}$ is neither empty nor a cluster of $G[W_{\mathcal{F}}] \triangle S_{\mathcal{F}}$, then S contains a pair $\{y, u\}$ with $u \in W_{\mathcal{F}}$.*

Proof. Suppose not. Then $N_{G \triangle S}(y) \cap W_{\mathcal{F}} = N_G(y) \cap W_{\mathcal{F}}$. In the cluster graph $G \triangle S$ this set is $\mathrm{cluster}(y) \cap W_{\mathcal{F}}$ (as $y \notin W_{\mathcal{F}}$), which is empty or a cluster of the induced subgraph $(G \triangle S)[W_{\mathcal{F}}] = G[W_{\mathcal{F}}] \triangle S_{\mathcal{F}}$. □

Lemma 6.6 (Counting irreducible components)**.** *Let S be a solution of G and J a set of irreducible components of $I(H)$. Then $|S| \ge \sum_j \tau_j + |J|/2$. In particular, if $|S| \le k = \sum_j \tau_j + \bar{\ell}$, then $|J| \le 2\bar{\ell}$.*

Proof. Claim: for every $j \in J$, (a) $|S_j| \ge \tau_j + 1$, or (b) S contains a pair with exactly one end in W_j (an *external* pair of j). Suppose (a) fails; then $|S_j| = \tau_j$ by Lemma 6.3(4), so S_j is an optimal solution of G_j, and since $\mathcal{C}_j$ is irreducible, S_j violates the boundary condition at some $y \notin W_j$. A component is pair-closed, so Lemma 6.5 gives a pair $\{y, u\} \in S$ with $u \in W_j$.

Let J_a be the indices of J satisfying (a) and J_b the others, which satisfy (b). An external pair of j has exactly one end in W_j; it lies in no $\binom{W_i}{2}$, so it belongs to S_{ext}, and it is external for at most two indices (the components containing its ends). Hence $|S_{\mathrm{ext}}| \ge |J_b|/2$. Since $|S| = \sum_j |S_j| + |S_{\mathrm{ext}}|$ and $|S_j| \ge \tau_j$ for all j, we get $|S| \ge \sum_j \tau_j + |J_a| + |J_b|/2 \ge \sum_j \tau_j + |J|/2$. □

The bound concerns irreducible components, not components with positive internal excess $\tau_j > p_j$: a reducible component may have large internal excess, and it is removed by Lemma 6.4 at no cost in $\bar{\ell}$.

Proof of Theorem 6.1. Algorithm. (1) Compute the components of $I(H)$, the optima τ_j by the subset dynamic programme over W_j ($O(3^{|W_j|})$), and $\bar{\ell}$; if $\bar{\ell} < 0$ answer NO (Lemma 6.3(4)). (2) For each component test reducibility and, if reducible, apply the rule (Lemma 6.4); the test of [17, Lemma 6.4] needs a constant number of exact cluster-editing computations on subgraphs of G_j plus polynomial time, which the subset programme provides in $O(3^{|W_j|})$ each. (3) Let r^* be the number of remaining components. If $r^* > 2\bar{\ell}$, answer NO (Lemma 6.6 with J the set of all remaining components). Otherwise the current instance has budget $k^* = \sum_{\text{remaining}} \tau_j + \bar{\ell} \le r^* t_s + \bar{\ell} \le (2t_s + 1)\bar{\ell}$. Forget the packing and solve plain Cluster Editing with budget k^* by Böcker's algorithm [2] in $O(1.62^{k^*} + n + m)$.

Correctness follows from Lemma 6.3 (the instance is of the required form and $\bar{\ell}$ is the excess), Lemma 6.4 (steps preserve the answer), and Lemma 6.6.

Time. The components have at least 3 vertices and are disjoint, so there are at most $n/3$ of them and steps (1), (2) take $O(n3^s) + n^{O(1)}$. Step (3) takes $1.62^{k^*} n^{O(1)}$ with $k^* \le (2t_s+1)\bar{\ell} \le (\binom{s}{2}+1)\ell$, since $2t_s \le \binom{s}{2}$ and $\bar{\ell} \le \ell$.

Kernel. After step (2) and the checks of step (3), apply the kernelization of Cluster Editing with at most $2k^*$ vertices [4] to (G, k^*) and add the empty packing: an equivalent CEaMP instance with empty packing, budget $\le k^*$ and at most $2k^* \le (4t_s+2)\ell \le (s^2 - s + 2)\ell$ vertices. □

Remark 6.7 (Scope). Theorem 6.1 is fixed-parameter tractable in (ℓ, s); it is not an FPT algorithm in $\ell + w$ (which does not exist unless FPT = W[1], Theorem 5.1), since s is not bounded by a function of w (a path or tree $I(H)$ has $w \le 1$ and arbitrarily large components), and the exponent $\binom{s}{2}\ell$ is of no practical interest. The argument uses neither pair-maximality nor Appendix A. For a perfect matching it covers the smallest non-trivial instance of the case $c = 2$ left open in [15] for fixed-parameter tractability in ℓ.

6.3 Components of arbitrary size

For components too large for the subset programme, tightness and reducibility can still be decided when w is small.

Proposition 6.8. *Given a tree decomposition of $I(H)$ of width w and a component $\mathcal{C}$ of $I(H)$, one decides in time $3^{w+1}n^{O(1)}$ whether $\mathcal{C}$ is tight; if it is tight, one decides in the same time whether it is reducible and finds a witnessing $S_{\mathcal{C}}$.*

Proof. A tight solution of $G[W_{\mathcal{C}}]$ consists of one edited pair per member and nothing else (each member needs at least one edited pair and the pair sets are disjoint), and every such solution is optimal since $\tau \ge p$; so for a tight $\mathcal{C}$ the optimal solutions are exactly the tight ones. Variables: the three choices of each member. Constraint: $G[W_{\mathcal{C}}] \triangle S$ has no induced P_3, i.e. every triple of $W_{\mathcal{C}}$ is fine. A triple with a soft pair involves at most three members (one per soft pair, pair sets being disjoint); each owns two of the three vertices of the triple, so any two of them share a vertex, they form a clique of $I(H)$ and lie in a common bag. A triple without soft pair is checked statically. So $\mathcal{C}$ is tight iff this CSP is satisfiable, and Lemma 3.4 decides this in $3^{w+1}n^{O(1)}$ time. The boundary condition at $y \notin W_{\mathcal{C}}$ with $A_y = N(y) \cap W_{\mathcal{C}} \neq \emptyset$ says: (i) every pair inside A_y is an edge of $G \triangle S$ and (ii) every pair between A_y and $W_{\mathcal{C}} \setminus A_y$ is a non-edge of $G \triangle S$ (given that $G[W_{\mathcal{C}}] \triangle S$ is a cluster graph this says A_y is a cluster). Each such pair is hard (static) or owned by one member (a unary constraint). Reducibility is therefore the same CSP with additional unary constraints. □

(For a non-tight component the optimal $S_{\mathcal{C}}$ has more than p edits and is not a 3^p assignment; the proposition says nothing about it.)

Lemma 6.9 (Reduction preserves pair-maximality)**.** *Let H be pair-maximal, $\mathcal{F} \subseteq H$ pair-closed and tight with $S_{\mathcal{F}}$ an optimal solution of $G[W_{\mathcal{F}}]$ satisfying the boundary condition, $G^\circ = G \triangle S_{\mathcal{F}}$ and $H^\circ = H \setminus \mathcal{F}$. Then H° is a modification-disjoint packing of G°, (G, H, ℓ) and (G°, H°, ℓ) are equivalent, H° is pair-maximal in G°, and $I(H^\circ)$ is an induced subgraph of $I(H)$, so* $\mathrm{tw}(I(H^\circ)) \leq \mathrm{tw}(I(H))$.

Proof. Members outside $\mathcal{F}$ have at most one vertex in $W_{\mathcal{F}}$, so none of their pairs lies in $W_{\mathcal{F}}$; $S_{\mathcal{F}}$ does not touch them, so they remain induced P_3's with the same pairs, and H° is a modification-disjoint packing of G°. Since $\mathcal{F}$ is tight, $|S_{\mathcal{F}}| = |\mathcal{F}|$, and Rule 6.2 (Lemma 6.3 of [17]) gives $\mathrm{OPT}(G) = \mathrm{OPT}(G^\circ) + |\mathcal{F}|$, hence the equivalence with unchanged ℓ.

Pair-maximality. Let Q be an induced P_3 of G°; we show that it has a soft pair with respect to H°. Let $q = |V(Q) \cap W_{\mathcal{F}}|$. If $q = 3$, Q is an induced P_3 of $G^\circ[W_{\mathcal{F}}] = G[W_{\mathcal{F}}] \triangle S_{\mathcal{F}}$, a cluster graph: impossible. If $q = 2$, $V(Q) = \{p, p', y\}$ with $p, p' \in W_{\mathcal{F}}$, $y \notin W_{\mathcal{F}}$. The pairs yp, yp' are unchanged, and by the boundary condition $N_{G^\circ}(y) \cap W_{\mathcal{F}}$ is empty or one cluster K of the cluster graph $G^\circ[W_{\mathcal{F}}]$. If it is empty, Q has at most one edge. If both $p, p' \in K$, they are adjacent and Q is a triangle. If exactly one is in K, the other is a non-neighbour of y and of the first (different clusters), so Q has one edge. If neither is in K, Q has at most one edge. In no case is Q an induced P_3. If $q \leq 1$, every pair of Q has at most one end in $W_{\mathcal{F}}$, so Q is an induced P_3 of G with the same pairs and, H being pair-maximal, has a pair in pairs(H); this pair is not in pairs$(\mathcal{F})$ (those lie inside $W_{\mathcal{F}}$), so it is soft with respect to H°. □

Theorem 6.10 (Component reduction)**.** *Let H be* any *packing, $\bar{\ell} = \ell - \sum_j (\tau_j - p_j)$, and apply Rule 6.2 to every reducible tight component (found by Proposition 6.8, which needs a tree decomposition of $I(H)$ of width w and costs $3^{w'+1} n^{O(1)}$ with $w' \leq w + \ell$ per test). Then, modulo the cited soundness of Rule 6.2 (Lemma 6.3 of [17]), the reduced instance is equivalent to (G, H, ℓ), and if it is a YES instance then $\bar{\ell} \geq 0$ and the number r of components of $I(H^o)$, H^o the remaining packing, satisfies $r \leq \ell + \bar{\ell} \leq 2\ell$. In particular* if more than 2ℓ components remain, the answer is NO. *No τ_j of a non-tight component is needed by the algorithm.*

Proof. The rule. The set S_j used for a reducible tight component $\mathcal{C}_j$ consists of pairs inside W_j. So $G[W_i]$ and the sets $N(y) \cap W_i$ $(i \neq j)$ do not change, τ_i, tightness and (ir)reducibility of the other components are unchanged, ℓ and $\bar{\ell}$ are unchanged (Lemma 6.4), and one pass reaches an instance with no reducible tight component.

Counting. Work in the reduced instance. Let $\mathcal{A}$ be the set of tight (irreducible) components and $\mathcal{B}$ the set of non-tight ones. Since $\sum_j \tau_j = |H| + \sum_j (\tau_j - p_j)$, we have $|H| + \ell = \sum_j \tau_j + \bar{\ell}$. If $\bar{\ell} < 0$ the instance is a NO instance (Lemma 6.3). Otherwise let S be an optimal solution with $|S| \leq \sum_j \tau_j + \bar{\ell}$. By Lemma 6.6 with $J = \mathcal{A}$, $|\mathcal{A}| \leq 2\bar{\ell}$. Each $j \in \mathcal{B}$ has $\tau_j - p_j \geq 1$, so $|\mathcal{B}| \leq \sum_j (\tau_j - p_j) = \ell - \bar{\ell}$. Therefore $r \leq 2\bar{\ell} + \ell - \bar{\ell} = \ell + \bar{\ell} \leq 2\ell$ (as $\bar{\ell} \leq \ell$). □

The theorem is phrased through the number of remaining components so that it does not need the τ_j of non-tight components, which are not computable in polynomial time for a large component.

Corollary 6.11. *(a) For $\ell = 0$ a YES instance is reduced to the empty packing; this agrees with Theorem 3.5. (b) For an arbitrary packing H, compute M and $H' = H \cup M$ (Lemma 2.4) and then reduce tight reducible components repeatedly. This ends, in time $3^{w+\ell+1} n^{O(1)}$ given a tree decomposition of $I(H)$ of width w, with a pair-maximal packing H^*, a budget $\ell^* = \ell - |M| \leq \ell$, $\mathrm{tw}(I(H^*)) \leq w + |M| \leq w + \ell$, and at most $2\ell^*$ components of the intersection graph, or with a correct NO answer.*

Proof. (a) $r \le 2\ell = 0$. (b) After the single maximalisation H' is pair-maximal and $\mathrm{tw}(I(H')) \le w + |M|$ (Lemma 2.4). Each reduction step removes the members of a tight reducible component, keeps the budget (Lemma 6.9), keeps the packing pair-maximal and does not increase the treewidth of the intersection graph (it is an induced subgraph of the previous one). The loop ends after at most $|H'|$ steps, when no tight component is reducible; Theorem 6.10 bounds the number of components by $2\ell^*$. □

What remains after the reduction is the case of *few but arbitrarily large* components of $I(H)$ (for $w = 1$, a single tree of members), which is Open Problem 6.

7 Optimal solutions of pair-maximal packings

Setting. Throughout this section H is pair-maximal (Section 2; for an arbitrary packing read the primed instance), $D \ge \max_v |Q(v)|$ is a number, and S is a solution with $|S| \le |H| + \ell$, so that its set U of hard edits has $u = |U| \le \ell$ (Lemma 2.2); U^- and U^+ are the deleted hard edges (hard edges across clusters) and the inserted hard non-edges (hard non-edges inside clusters), and $\mathrm{del}_x, \mathrm{ins}_x$ count the elements of U^- and U^+ at the vertex x. Solutions and partitions correspond (Section 2). A *hard component* is a connected component of the graph $(V, \text{hard edges of } G)$; it is *clean* if all its non-adjacent pairs are soft.

Lemma 7.1 (Cost formula)**.** *Let π be a partition of V and $f(C) = \binom{|C|}{2} - 2e(C)$, where $e(C)$ is the number of edges of G inside C. Then* $\mathrm{cost}(\pi) = |E| + \sum_{C \in \pi} f(C)$.

Proof. The cost counts non-edges inside blocks, $\sum_C [\binom{|C|}{2} - e(C)]$, plus edges across blocks, $|E| - \sum_C e(C)$. □

7.1 Clusters of optimal solutions: half-degree and diameter

Lemma 7.2 (Half-degree and diameter)**.** *Let S be an optimal solution, C a cluster of $G \triangle S$ and $v \in C$. Put $d_C(v) = |N_G(v) \cap C|$.*

(a) $d_C(v) \ge (|C| - 1)/2$.

(b) If $a, b \in C$ and $ab \notin E$, then some $z \in C \setminus \{a, b\}$ has $az, zb \in E$.

This is the single-vertex-move argument behind the critical-clique property of Cluster Editing [10, 7]; we include the short proof.

Proof. (a) Let $S' = S \triangle \{vx : x \in C \setminus \{v\}\}$. Then $G \triangle S'$ is $G \triangle S$ with v cut off from the rest of C, a cluster graph. For $x \in C \setminus \{v\}$ with $vx \in E$ the pair $vx \notin S$ and $vx \in S'$: this adds $d_C(v)$ pairs. For $x \in C \setminus \{v\}$ with $vx \notin E$ the pair is in S and not in S': this removes $|C| - 1 - d_C(v)$ pairs. All other pairs are unchanged. So $|S'| = |S| + 2d_C(v) - (|C| - 1)$, and $|S'| \ge |S|$ by optimality.

(b) $N_G(a) \cap C$ and $N_G(b) \cap C$ are subsets of $C \setminus \{a, b\}$ (as $a \notin N(a)$ and $ab \notin E$), a set with $|C| - 2$ elements, while $d_C(a) + d_C(b) \ge |C| - 1$ by (a). So the two sets intersect. □

Lemma 7.2 uses neither H nor ℓ; only local optimality under single-vertex moves.

7.2 Hard pairs

Lemma 7.3 (HE and H2)**.** *Let H be pair-maximal. (a) If ab, bc are hard edges (a, b, c distinct), then $ac \in E$ or ac is soft. (b) If uv is a hard non-edge, then $N_h(u) \cap N_h(v) = \emptyset$.*

Proof. (a) If $ac \notin E$ and ac were hard, then $a\,b\,c$ would be an induced P_3 with all three pairs hard, contradicting pair-maximality. (b) If $z \in N_h(u) \cap N_h(v)$, then $u\,z\,v$ is an induced P_3 ($uz, zv \in E$, $uv \notin E$) all of whose pairs are hard. □

No YES hypothesis is needed. Without pair-maximality the lemma fails ($H = \emptyset$ and any induced P_3).

Lemma 7.4 (EX: bounds at a hard edit)**.** *Let H be pair-maximal, S a solution with hard edits U, $u = |U|$, let $xy \in U$ and let C be the cluster of x. Then:*

(a) if xy is deleted, $d_h(x) \le D + \mathrm{del}_x + \mathrm{del}_y - 1$; if xy is inserted, $d_h(x) \le D + \mathrm{ins}_y - 1 + \mathrm{del}_x$. In both cases $d_h(x) \le D + u$;

(b) $|C| \le 1 + D + d_h(x) + \mathrm{ins}_x$, hence $|C| \le 1 + 2D + u$;

(c) every $z \in C$ has $d_h(z) \le |C| - 1 + \mathrm{del}_z \le 2D + 2u$.

Proof. (a) *Deleted:* $x \in C$, $y \notin C$. A hard neighbour z of x with xz not deleted ("kept") lies in C, and $z \neq y$. Apply Lemma 7.3(a) with centre x to the hard edges xz, xy: $zy \in E$ or zy is soft. If zy is a hard edge, it joins $z \in C$ to $y \notin C$, so it is a deleted hard edge at y other than xy (there are $\mathrm{del}_y - 1$ of these); otherwise $z \in Q(y)$, at most D vertices. So at most $D + \mathrm{del}_y - 1$ hard neighbours are kept, and $d_h(x) \le D + \mathrm{del}_y - 1 + \mathrm{del}_x$. *Inserted:* $x, y \in C$. A kept hard neighbour z of x lies in C and $z \neq y$ (as $xy \notin E$); zy is a pair inside C, hence an edge of $G \triangle S$. If zy were a hard edge, $z \in N_h(x) \cap N_h(y)$, contradicting Lemma 7.3. So zy is soft ($z \in Q(y)$, at most D choices) or a hard non-edge, inserted at y and different from xy ($\mathrm{ins}_y - 1$ choices). So $d_h(x) \le D + \mathrm{ins}_y - 1 + \mathrm{del}_x$. In the deleted case the pairs counted by $\mathrm{del}_x + \mathrm{del}_y - 1$ are distinct pairs of U, so the sum is $\le u$; in the inserted case $\mathrm{ins}_y - 1 + \mathrm{del}_x \le u - 1$ (an inserted pair at y other than xy and a deleted pair at x are distinct elements of $U \setminus \{xy\}$).

(b) For $z \in C \setminus \{x\}$ the pair xz is an edge of $G \triangle S$. Either xz is soft (at most D such z), or a kept hard edge (at most $d_h(x)$), or a hard non-edge, inserted at x (at most ins_x). With (a): in the deleted case $|C| \le 1 + D + (D + \mathrm{del}_x + \mathrm{del}_y - 1) + \mathrm{ins}_x \le 1 + 2D + u$ because the deleted pairs at x or y and the inserted pairs at x are distinct elements of U; in the inserted case $\mathrm{ins}_x + \mathrm{ins}_y - 1$ counts the inserted pairs at x or y and del_x is disjoint from them, so the total is again $\le u$.

(c) Hard neighbours of z inside C are at most $|C| - 1$; hard neighbours outside C are joined by deleted edges, at most $\mathrm{del}_z \le u$. Use (b). □

Lemma 7.5 (Connected clusters)**.** *In an optimal solution every cluster C induces a connected subgraph $G[C]$.*

Proof. If $C = C_1 \cup C_2$ with $C_1, C_2 \neq \emptyset$ and no edge of G between them, splitting C into C_1, C_2 turns the $|C_1||C_2| > 0$ inserted pairs between them into non-edges outside S and creates no deletion. □

7.3 Normal form and candidates

Fix ℓ and put

$$B = 2D + 3\ell, \qquad L_B = \{v : \ d_h(v) \le B\}, \qquad Z = B + D + 1 = 3D + 3\ell + 1.$$

In an optimal solution of size at most $|H| + \ell$ call a cluster *exceptional* if it contains an endpoint of a pair of U. Let BIG be the union of the hard components that contain a vertex v with $d_h(v) > B$.

Lemma 7.6 (NF: normal form)**.** *Let H be pair-maximal and S an optimal solution with $|S| \le |H|+\ell$, so $u \le \ell$.*

(α) There are at most $2u \le 2\ell$ exceptional clusters. Each exceptional cluster C has $|C| \le 1 + 2D + u \le 2D + \ell + 1$, lies in L_B, and $G[C]$ is connected.

(β) Let C be a non-exceptional cluster. Then no hard edge leaves C and no hard non-edge lies in C; so C is a union of whole hard components, each clean, and all pairs between different components inside C are soft. If C contains at least two components, $|C| \le 2D$.

(γ) Every hard component K of BIG *is a whole cluster of S, and this cluster is non-exceptional. Consequently every exceptional cluster is disjoint from* BIG*, and every component of* BIG *is clean.*

Proof. (α) Each pair of U has at most two clusters at its ends, which gives the count. The bound on $|C|$ is Lemma 7.4(b) (for an endpoint $x \in C$ of a pair in U), the inclusion $C \subseteq L_B$ is Lemma 7.4(c) together with $2D + 2u \le B$, and connectivity is Lemma 7.5.

(β) A hard edge leaving C would be deleted, with an endpoint in C; a hard non-edge inside C would be inserted. Both would make C exceptional. If K is a hard component meeting C, hard edges at a vertex of $K \cap C$ do not leave C, so $K \subseteq C$ by connectivity of K; K is clean since C contains no hard non-edge. Two vertices of different components inside C are not joined by a hard edge, and are not a hard non-edge, so their pair is soft. If C contains components $K_1 \neq K_2$, a vertex $x \in K_1$ has all vertices of $C \setminus K_1$ as soft partners, so $|C \setminus K_1| \le D$; likewise $|C \setminus K_2| \le D$; since $K_1 \subseteq C \setminus K_2$, $|C| = |K_1| + |C \setminus K_1| \le 2D$.

(γ) Let $d_h(v) > B$ and let $K = K(v)$ be its hard component. By Lemma 7.4(a) no endpoint of a pair of U has hard degree above $D + u \le B$, and by (α) no vertex of an exceptional cluster has hard degree above B. So the cluster C of v is non-exceptional, and by (β) $K \subseteq C$ and C is a union of components. Now $|K| \ge d_h(v) + 1 > B + 1 > 2D$, so by ($\beta$) C cannot contain two components; hence $C = K$. Every other vertex of K lies in the same cluster. Cleanness is (β). The disjointness statement follows since a component of BIG is a non-exceptional cluster. □

Corollary 7.7 (CAND: bounded candidate partners)**.** *Let H be pair-maximal, S optimal with $|S| \le |H| + \ell$, and $xy \in U$. Then $x, y \in L_B$ and:*

(i) if xy is deleted, $y \in N_h(x)$, and $|N_h(x)| \le D + \ell$;

(ii) if xy is inserted, x and y have a common neighbour in their cluster, so their distance in $G[L_B]$ is exactly 2;

(iii) in both cases y lies in the ball Ball$(x, 2)$ *of $G[L_B]$, which has at most Z^2 vertices.*

The same holds with the roles of x and y exchanged.

Proof. $x, y \in L_B$ by Lemma 7.4(a) ($d_h \le D + u \le B$). (i) is the definition together with Lemma 7.4(a). (ii) x, y lie in the exceptional cluster C and are non-adjacent, so Lemma 7.2(b) gives $z \in C$ adjacent to both; $C \subseteq L_B$ by Lemma 7.6(α), so the path $x\,z\,y$ lies in $G[L_B]$ and the distance is 2 (not 1 since $xy \notin E$). (iii) A vertex of $G[L_B]$ has at most B hard neighbours and at most D soft ones, so degree at most $B + D = Z - 1$, and a ball of radius 2 has at most $1 + (Z-1) + (Z-1)^2 \le Z^2$ vertices; deleted pairs have distance 1. □

Lemma 7.2(b) is what makes the radius 2 independent of D and ℓ: the candidates for the other end of a hard edit lie within distance 2 of one end in $G[L_B]$. Corollary 7.7 says that *one* endpoint of a hard edit determines a set of at most Z^2 candidates for the other, for every hard edit, including those of type T2 and deleted edges (Section 9). What it does not bound is the number of candidates for the first endpoint, which is n.

8 An n^ℓ algorithm and the limits of the cluster-identity route

This section uses Corollary 7.7 in two ways: guessing the positions of the hard edits (Theorem 8.2), and following the clusters directly (Theorem 8.3). The first halves the exponent of Theorem 3.6; the second is fixed-parameter tractable for a stronger parameter ω which, by Proposition 8.4, is not bounded by any function of (ℓ, w). **Whether CEaMP is fixed-parameter tractable in $\ell + w$ remains open.**

8.1 The n^ℓ algorithm

Remark 8.1 (Fixed set of hard edits). The proof of Theorem 3.6 shows: for a *fixed* set U' of hard pairs, $m_{U'} = \min \sum_i |x_i|$ over all $x \in \prod \mathcal{D}_i$ with $G \triangle (U' \cup \bigcup x_i)$ a cluster graph is computed in $5^{w+1} n^{O(1)}$ time from a tree decomposition of $I(H)$ of width w, and for *every* U' of hard pairs the value $m_{U'} + |U'|$ is the size of an actual solution.

Theorem 8.2 (n^ℓ algorithm)**.** *Given (G, H, ℓ) for an arbitrary packing H and a tree decomposition of $I(H)$ of width w, CEaMP is solvable in time*

$$(n+1)^\ell \, Z_*^{2\ell} \, 5^{w+\ell+1} \, n^{O(1)}, \qquad D_* = D(H) + 2\ell, \quad Z_* = 3D_* + 3\ell + 1 = 3D(H) + 9\ell + 1.$$

Since $D(H) \le 2(w+1)$, $Z_* \le 6w + 9\ell + 7$, so the factor $Z_*^{2\ell}$ is a function of (ℓ, w) only. This improves the exponent 2ℓ of Theorem 3.6 to ℓ; it is still XP in ℓ.

Proof. Compute M and $H' = H \cup M$ by Lemma 2.4; reject if $|M| > \ell$. Then H' is pair-maximal, $\ell' = \ell - |M| \le \ell$, $D(H') \le D(H) + 2|M| \le D_*$, a tree decomposition of $I(H')$ of width $w' \le w + |M| \le w + \ell$ is available, and the instance is equivalent to (G, H', ℓ'). All lemmas of Section 7 apply to H' with $D(H')$ and ℓ'; put B, L_B, Z accordingly. Then $Z = 3D(H') + 3\ell' + 1 \le Z_*$.

Algorithm. Compute L_B and, for $x \in L_B$, the set $P(x)$ of vertices $y \in \mathrm{Ball}_{G[L_B]}(x, 2) \setminus \{x\}$ with xy hard; $|P(x)| \le Z^2$. Enumerate all sequences of at most ℓ' pairs $(x_1, y_1), \ldots, (x_j, y_j)$ with $x_i \in L_B$ and $y_i \in P(x_i)$; there are at most $(1 + nZ^2)^{\ell'} \le (n+1)^{\ell'} Z^{2\ell'}$ of them. For each, let $U' = \{x_i y_i\}$ (a set of at most ℓ' hard pairs), compute $m_{U'}$ by Remark 8.1 for H' in $5^{w'+1} n^{O(1)}$ time, and accept iff $m_{U'} + |U'| \le |H'| + \ell'$.

Soundness. By Remark 8.1, $m_{U'} + |U'|$ is the size of an actual solution of G for every U' of hard pairs of H'; acceptance gives a solution of size $\le |H'| + \ell' = |H| + \ell$.

Completeness. Suppose the instance is a YES instance and let S be an optimal solution, $|S| \le |H'| + \ell'$, with hard edits U, $|U| \le \ell'$ (Lemma 2.2 for H'). For each pair of U take one endpoint x; it lies in L_B and the other endpoint lies in $P(x)$ by Corollary 7.7. So U is among the enumerated sets, and for it $m_U + |U| \le |S| \le |H'| + \ell'$ (the soft part of S is a feasible choice of x for U by Lemma 2.2).

Time. $(n+1)^{\ell'} Z^{2\ell'} 5^{w'+1} n^{O(1)} \le (n+1)^\ell Z_*^{2\ell} 5^{w+\ell+1} n^{O(1)}$. □

The exponent is halved because the candidate list is attached to *one* endpoint (Corollary 7.7); Theorem 3.6 guesses both. Note that ℓ bounds the *number* of hard edits but not the cost of locating them: the n^ℓ comes from the position of the first endpoint of each pair. A fixed-parameter algorithm along this route needs a bound on these positions as well.

8.2 A fixed-parameter algorithm for a stronger parameter

Let H be pair-maximal and $\ell, D, B, Z, L_B, \mathrm{BIG}$ as in Section 7. Let $V' = V \setminus \mathrm{BIG}$ and let Γ' be the graph on V' whose edges are all edges of G and all soft pairs with both ends in V'.

Theorem 8.3 (Fixed-parameter tractability in (ℓ, D, ω))**.** *Let $\omega = \mathrm{tw}(\Gamma')$ and assume a tree decomposition of Γ' of width ω is given. Then one can compute in time $(Z(\omega+1))^{O(\omega)}n + n^{O(1)}$ a number m^* such that*

$$\mathrm{OPT}(G) \le |H| + \ell \iff |E| + \sum_K f(K) + m^* \le |H| + \ell,$$

where K ranges over the hard components of BIG *and f is as in Lemma 7.1. The proof, with the dynamic program, is in Appendix C.*

All quantities in Theorem 8.3 refer to the pair-maximal instance (G, H', ℓ'); they are not functions of the input (G, H, ℓ) (Section 2).

8.3 Why ω is not bounded by (ℓ, w)

Proposition 8.4 (ω is unbounded at $\ell = 0$, $w = 0$)**.** *For every cubic graph F there is an instance $(G_F, H_F, 0)$ with H_F a vertex-disjoint pair-maximal packing ($w = 0$, $c = 1$, $D = 2$), $\mathrm{OPT}(G_F) = |H_F|$ (a YES instance with $\ell = 0$), $\mathrm{BIG} = \emptyset$, every hard component of size at most 6 and every hard degree at most 4, and $\mathrm{tw}(\Gamma') \ge \mathrm{tw}(F)$. By [9] (cited, not re-proved here) there are cubic graphs F (expanders) with $\mathrm{tw}(F) = \Omega(|V(F)|)$. Hence ω is not bounded by any function of (ℓ, w).*

Proof. Orient each edge of F arbitrarily; for each oriented edge $e = (p \to q)$ introduce three vertices a_e, b_e, c_e. Let $\mathcal{K}_v = \{a_e : e \text{ leaves } v\} \cup \{b_e, c_e : e \text{ enters } v\}$ and make each $\mathcal{K}_v$ a clique; add the cross edges $a_e b_e$ and nothing else. Then $|\mathcal{K}_v| = \mathrm{out}(v) + 2\,\mathrm{in}(v) = 3 + \mathrm{in}(v) \le 6$. $H_F = \{P_e = (a_e, b_e, c_e)\}$: $a_e b_e, b_e c_e \in E$ and $a_e c_e \notin E$ since $a_e \in \mathcal{K}_p$, $c_e \in \mathcal{K}_q$, $p \ne q$ and cross edges are of the form $a_{e'} b_{e'}$; so P_e is an induced P_3. The members are vertex-disjoint, so modification-disjoint, $c = 1$, $w = 0$ and $D = 2$.

Pair-maximality. In an induced P_3 $x\,y\,z$ not both edges can lie inside cliques, since then x, z would be in the clique of y (each vertex is in exactly one clique) and adjacent. So one of xy, yz is a cross edge $a_e b_e$, which is soft.

$\mathrm{OPT} = |H_F|$. The partition $\{\mathcal{K}_v\}$ has no inserted pair and deletes exactly the cross edges, $|H_F|$ of them; and $\mathrm{OPT} \ge |H_F|$ by Lemma 2.2.

Hard structure. The soft pairs inside $\mathcal{K}_v$ are $b_e c_e$ (e entering v); all other pairs inside $\mathcal{K}_v$ are hard edges. So each $\mathcal{K}_v$ is one hard component and it is clean. The hard degree of $a_e \in \mathcal{K}_p$ is $|\mathcal{K}_p| - 1 \le 4$ (as a_e is an out-edge, $\mathrm{in}(p) \le 2$, $|\mathcal{K}_p| \le 5$); that of b_e or $c_e \in \mathcal{K}_q$ is $|\mathcal{K}_q| - 2 \le 4$. So $d_h \le 4 = 2D + 3\ell$ at $\ell = 0$, hence no vertex has $d_h > B$ and $\mathrm{BIG} = \emptyset$.

Minor. $V' = V$, each $\mathcal{K}_v$ is connected in Γ' (a clique), and an edge or soft pair of Γ' between $\mathcal{K}_p \ne \mathcal{K}_q$ is one of $a_e b_e$, $a_e c_e$, which exist only for the oriented edge $e = (p \to q)$ of F. Contracting the $\mathcal{K}_v$ gives exactly F, so F is a minor of Γ' and $\mathrm{tw}(\Gamma') \ge \mathrm{tw}(F)$. □

Remark 8.5 (Scope of Proposition 8.4). The proposition refutes only the claim "$\omega \le f(\ell, w)$"; it does *not* refute fixed-parameter tractability. The family has $w = 0$ and $\ell = 0$; it is solved in polynomial time by Theorem 3.6 and has $s = 3$ in Theorem 6.1. The obstruction is to the representation by Γ' (and by any single decomposition used both for the soft pairs and for the clusters): hard edges between far vertices are free (not in the budget), arbitrary in number, and make footprints of clusters disconnected in any decomposition of $I(H)$ or of G_{soft}. At $\ell = 0$ they are harmless because

the CSP of Theorem 3.5 needs no identities; for $\ell > 0$ an exceptional cluster can be assembled from vertices that are far apart (a $c = 1$ instance with two vertex-disjoint packed P_3's joined by one hard edge that is deleted). So a dynamic program on one decomposition cannot be used for both purposes without a preprocessing step; a reduction lemma that would supply it is Open Problem 5.

9 Locality of inserted pairs and a non-local obstruction

Section 7 bounds the *number* of hard edits and the *candidates for the second endpoint* of a hard edit, but not the number of candidate *positions* of the first. This section studies the inserted hard pairs of an *optimal* solution. The main tool is the diameter bound of Lemma 7.2, which, combined with pair-maximality, classifies every inserted pair into one of two types (Theorem 9.2). Pairs of the first type are local in a decomposition of $I(H)$ (Corollary 9.3, Proposition 9.5); the second type, which includes the generic merge of two pieces, is *not* local in the same sense (Proposition 9.7). Finally, deleted hard edges can join packed P_3's at arbitrary distance in $I(H)$ (Proposition 9.8). **Nothing in this section gives a fixed-parameter algorithm in $\ell + w$ or a hardness result.**

Setting. H is pair-maximal (for an arbitrary packing read the primed instance), $D(H) = 2c$, S is an *optimal* solution with $|S| \le |H| + \ell$, C is a cluster of $G \triangle S$, and $U = U^- \cup U^+$, $u = |U| \le \ell$ are as in Section 7. An *inserted pair* is a pair of U^+: a hard non-edge of G inside one cluster. A hard edge of G inside a cluster is an edge of $G \triangle S$ and not in S, hence *kept*. The *pieces* of S are the connected components of the graph (V, E_{keep}) where E_{keep} is the set of kept hard edges (isolated vertices are singleton pieces); every piece lies inside one cluster. Pieces depend on S, not only on (G, H).

Lemma 9.1 (Cross pairs)**.** *If $A \neq B$ are pieces contained in a cluster C, then every pair ab ($a \in A, b \in B$) is soft or inserted. This holds for every solution S; neither optimality nor pair-maximality is used.*

Proof. Since $a, b \in C$, ab is an edge of $G \triangle S$, i.e. $ab \in E$ xor $ab \in S$. Suppose ab is hard. If $ab \in E$ then $ab \notin S$, so ab is a kept hard edge and a, b lie in one piece, a contradiction. So $ab \notin E$, hence $ab \in S$, and as ab is hard, $ab \in U^+$. □

(Appendix B bounds the sizes of pieces that share a cluster.)

9.1 The T1/T2 dichotomy

For an inserted pair ab with $a, b \in C$, a *witness* is a vertex $z \in C \setminus \{a, b\}$ with $az, zb \in E$.

Theorem 9.2 (T1/T2)**.** *Let H be pair-maximal, S optimal and ab an inserted pair, $a, b \in C$. Then ab has a witness, and for every witness z at least one of az, zb is soft. Moreover:*

(T1) If az and zb are both soft, they are owned by two distinct *members $P_a \neq P_b$ of H, and $z \in V(P_a) \cap V(P_b)$.*

(T2) If az is hard (the case of zb hard is symmetric), then az is a kept hard edge, so a and z lie in one piece, and zb is soft.

We say that ab is of type T1 *if some witness has both steps soft, and of* type T2 *otherwise; then every witness has exactly one hard step.*

The type is a property of the pair ab (not of a chosen witness): it is T1 iff a witness with two soft steps exists.

Proof. A witness exists by Lemma 7.2(b). If az and zb were both hard for a witness z, then $a\,z\,b$ would be an induced P_3 with all three pairs hard (ab is hard by assumption), contradicting pair-maximality. If both are soft, let P_a, P_b own az, zb. If $P_a = P_b$ then $a, b \in V(P_a)$ and ab is soft, contradicting that ab is hard; so $P_a \neq P_b$, and both contain z. If az is hard, it is a hard edge inside C, hence kept. □

Corollary 9.3 (Locality of soft-free merges)**.** *Let H be pair-maximal, S optimal, and let $A \neq B$ be pieces inside one cluster C such that* no soft pair joins A and B. *Then for every $(a, b) \in A \times B$:*

(i) ab is an inserted pair, of type T1;

(ii) there is $z \in C \setminus (A \cup B)$ with az, zb soft, owned by distinct members $P_a \neq P_b$ with $z \in V(P_a) \cap V(P_b)$;

(iii) given a, the number of possible partners b in any such pair is at most $D(H)(D(H) - 2) = 4c(c - 1)$.

In particular a and b are at distance exactly 2 in the soft graph G_{soft}, and $A \cup B$ lies in the ball of radius 4 of G_{soft} around any of its vertices.

Proof. (i) The cross pair ab is not soft by hypothesis, so it is hard, hence inserted by Lemma 9.1; its type is T1 by (ii). (ii) Apply Theorem 9.2 to ab with a witness z. Suppose az is hard (the other case is symmetric); then z lies in the piece A of a. The pair zb is an edge of G and joins A to B: if it were hard it would be kept and z, b would share a piece, contradicting $A \neq B$; if soft it contradicts the hypothesis. So az and zb are both soft, which is (T1). If $z \in A$, then zb (soft) is a soft pair between A and B; similarly for $z \in B$ and az. So $z \notin A \cup B$. (iii) Fix a. By (ii) $z \in Q(a)$: at most $D(H)$ choices. For such z, let P_a own az. The vertex b satisfies bz soft owned by $P_b \neq P_a$, and $b \notin V(P_a)$ (otherwise $ab \in \text{pairs}(P_a)$ is soft). Now $Q(z) \cap V(P_a) = V(P_a) \setminus \{z\}$ has two elements, so $b \in Q(z) \setminus V(P_a)$, a set of at most $|Q(z)| - 2 \leq D(H) - 2$ vertices. Altogether at most $D(H)(D(H) - 2) = 2c(2c - 2) = 4c(c - 1)$ partners. (For $c = 1$ the count is 0: (T1) needs two members through z.)

The distance statement: ab is not soft and a, z, b is a soft path; so $d_{G_{\text{soft}}}(a, b) = 2$. Every $a' \in A$ has $d_{G_{\text{soft}}}(a', b) = 2$ for a fixed $b \in B$, and every $b' \in B$ has $d_{G_{\text{soft}}}(b', a') = 2$; hence all of $A \cup B$ is within distance 4 of b (and of any other vertex of $A \cup B$, by symmetry). □

Remark 9.4 (Scope). If no soft pair joins A and B then all $|A||B|$ cross pairs are inserted (Lemma 9.1), so $|A||B| \leq u \leq \ell$. The corollary covers *only* merges without a soft pair between the pieces. Merges in which some cross pairs are soft (the generic case when the pieces are neighbours in the soft graph) are not covered, and Proposition 9.7 shows they can genuinely be of type T2.

Proposition 9.5 (Bag locality of type T1 pairs)**.** *Let $(\mathcal{T}, \{\beta_t\})$ be a tree decomposition of $I(H)$ of width w, let $c = c(H)$, and for a set Y of nodes of $I(H)$ let $N[Y]$ be the set of nodes at distance at most 1 from Y (a set of* members of H*).*

(a) If ab is an inserted pair of type T1 with a witness z and members P_a, P_b as in Theorem 9.2, then P_a, P_b are adjacent in $I(H)$, so some bag β_t contains both and $a, b \in \bigcup_{P \in \beta_t} V(P)$, a set of at most $3(w + 1)$ vertices. Moreover the member through z and every member through a or b lies in $N[\beta_t]$.

(b) $(\mathcal{T}, \{N[\beta_t]\})$ *is a tree decomposition of the square* $I(H)^2$ *of width at most* $(w+1)(3c-2)-1 \le (w+1)(3w+1)-1$.

Proof. (a) P_a, P_b share z, so they are adjacent in $I(H)$ and, the edge being covered, lie in a common bag. Every member through a contains $a \in V(P_a)$, hence meets $P_a \in \beta_t$ and lies in $N[\beta_t]$; similarly for b and z.

(b) *Size.* Each member has 3 vertices, each lying in at most c members, so $I(H)$ has maximum degree at most $3(c-1)$, $|N[\beta_t]| \le (w+1)(1+3(c-1)) = (w+1)(3c-2)$, and $c \le w+1$. *Connectivity.* For a node v let $\mathcal{T}_v = \{t : v \in \beta_t\}$ (a subtree). The nodes t with $v \in N[\beta_t]$ form $\bigcup_{y \in N[v]} \mathcal{T}_y$. Each $\mathcal{T}_y$, $y \in N(v)$, meets $\mathcal{T}_v$ because the edge yv lies in some bag; a union of subtrees that all meet the connected set $\mathcal{T}_v$ is connected. *Edges.* Let yv be an edge of $I(H)^2$. If y, v are adjacent in $I(H)$, any $t \in \mathcal{T}_y$ has $y \in \beta_t$ and $v \in N[y]$. If they have a common neighbour m, any $t \in \mathcal{T}_m$ has $y, v \in N[m] \subseteq N[\beta_t]$. □

A type T1 pair is therefore a pair of vertices inside one bag of a decomposition of bounded width, and, given one end, it has at most $4c(c-1)$ candidate partners in the soft-free case. In particular each *cross pair* (a, b) of a soft-free merge lies in a bag; this does not put the pieces A, B themselves in one bag.

Lemma 9.6 (Near-twin agreement)**.** *Let S be any solution (optimal or not), $u = |U|$, and let $z \ne b$ lie in one cluster of $G \triangle S$. Then*

$$A(z,b) = \big\{a \notin \{z,b\} : \ az, ab \text{ both hard}, \ [az \in E] \ne [ab \in E]\big\}$$

has at most u elements. In particular, if z, b lie in one cluster of $G \triangle S$ and zb is soft, the set $\Theta(z,b) = N_h(z) \setminus (N[b] \cup Q(b))$ satisfies $|\Theta(z,b)| \le u \le |S| - |H|$ and is computable from G and H alone.

Proof. Let $a \in A(z,b)$. Exactly one of az, ab is a hard edge and the other a hard non-edge. If a lies in the cluster of z, b, the hard non-edge is a hard non-edge inside a cluster, so it is in U. If a lies in another cluster, the hard edge joins two different clusters, so it is deleted and in U. Assign to a this pair. Its ends are a and one of z, b; since $a \notin \{z, b\}$, the map $a \mapsto$ pair is injective. So $|A(z,b)| \le u$. For the second statement, $a \in \Theta(z,b)$ means $a \ne z$, az a hard edge, $a \notin N[b]$ (so $a \ne b$ and $ab \notin E$) and ab not soft. Thus $a \in A(z,b)$. □

In a type T2 pair with az hard and zb soft, $a \in \Theta(z,b)$: the far end a is one of at most ℓ vertices determined by the soft pair (z,b). The lemma does not say where a is relative to z in $I(H)$: a is a hard neighbour of z.

9.2 Merge pointers need not be bag-local

Proposition 9.7 (Examples of type T2 merges)**.** *Each of the following instances (graph G on vertices $0, \dots, n-1$ given by its edge list, H given by triples (x, y, z) with centre y) has a pair-maximal modification-disjoint packing H and a unique optimal solution, found by enumerating all set partitions of the vertex set.*

(a) $n = 7$, $E = \{03, 04, 12, 13, 14, 15, 23, 35, 36\}$, $H = \{(4,1,5), (0,3,2), (1,3,6)\}$. OPT $= 4 = |H| + 1$*; the clusters are* $\{0,4\}, \{1,2,3,5\}, \{6\}$*. The pieces of* $\{1,2,3,5\}$ *are* $\{1,2\}$ *and* $\{3,5\}$ *(kept hard edges* 12 *and* 35*); the cross pairs* 13, 15, 23 *are soft and* 25 *is inserted.* $Q(2) = \{0,3\}$ *and* $Q(5) = \{4,1\}$ *are disjoint, so the pair* 25 *has no witness with two soft steps: it is of type T2.*

(b) $n = 8$, $E = \{02, 06, 14, 16, 17, 23, 25, 37, 45, 46, 47, 57\}$, $H = \{(0,6,4), (3,7,4), (2,5,4), (0,2,3), (6,1,7)\}$. $\mathrm{OPT} = 6 = |H| + 1$*; the clusters are* $\{0,6\}, \{1,4,5,7\}, \{2,3\}$*; the only hard edit of the solution is the inserted pair* 15*; the pieces are* $\{1,4\}$ *and* $\{5,7\}$*, the cross pairs* $17, 45, 47$ *are soft,* $Q(1) = \{6,7\}$*,* $Q(5) = \{2,4\}$*, so* 15 *has no witness with two soft steps and is of type T2.*

(c) $n = 6$*,* $E = \{02, 03, 04, 05, 12, 13, 15, 25, 35, 45\}$*,* $H = \{(3,5,4), (0,2,1)\}$*. The unique optimum has clusters* $\{0,1,2,3,5\}, \{4\}$ *and* $\mathrm{OPT} = 4$*; it contains the inserted pair* 23 *(type T2:* 30 *hard and kept,* 02 *soft), but* 2 *and* 3 *lie in the* same *piece (the kept hard edges* $03, 05, 13, 15, 25$ *connect* $\{0,1,2,3,5\}$*).*

Consequently "every inserted pair with both ends in $V(H)$ *has a common soft neighbour" is false, and Corollary 9.3 cannot be extended to merges of two pieces that are joined by a soft pair: (a) and (b) are such merges, with inserted pairs of type T2. Example (c) is T2 but intra-piece.*

Proof. Direct enumeration (all 877, 4140 and 203 set partitions of 7, 8 and 6 vertices respectively) of the cost $|\{ab : (ab \in E) \neq (a, b \text{ in one block})\}|$, together with a check that each triple of H is an induced P_3, that the triples have pairwise disjoint pair sets and that no induced P_3 of G has all pairs hard. The structural claims (pieces, soft pairs, Q) are read off from the lists above. In (a) and (b), every witness z of the inserted pair has az or zb hard because $Q(a) \cap Q(b) = \emptyset$, so the pair is not of type T1. □

9.3 An obstruction: deleted hard edges between far members

An optimum may be forced to delete a hard edge whose ends lie in members that are far apart in $I(H)$.

Proposition 9.8 (FAR)**.** *For* $m \geq 3$ *let* G_m *be the path* $0, 1, \ldots, 2m$ *with the extra edge* $e^* = \{1, 2m-1\}$ *and* $H_m = \{P_i = (2i, 2i+1, 2i+2) : 0 \leq i < m\}$ *(centres* $2i+1$*). Then:*

(i) H_m *is a pair-maximal modification-disjoint packing,* $I(H_m)$ *is a path on* m *nodes (*$c = 2$*,* $w = 1$*);*

(ii) $\mathrm{OPT}(G_m) = m + 1 = |H_m| + 1$*;*

(iii) in every *optimum the hard edge* e^* *is deleted. Its ends lie in* P_0 *and* P_{m-1}*, at distance* $m - 1$ *in* $I(H_m)$*.*

Proof. (i) The chord joins 1 and $2m-1$, which are $2m-2 \geq 4$ apart, so no triple P_i contains both of them and each P_i is an induced P_3; the pair sets are disjoint. Every path edge lies in some P_i and is soft, so e^* is the only hard edge. An induced P_3 has two edges, at most one of which is e^*, so it contains a soft pair: H_m is pair-maximal.

(ii) $|E| = 2m + 1$ and G_m is connected with $2m + 1$ vertices, hence unicyclic (the unique cycle is on $1, \ldots, 2m-1$, of length ≥ 5). An induced subgraph on s vertices has $e \leq s - 1$ edges unless it contains the cycle, in which case $s \geq 5$ and $e \leq s$. With $f(C) = \binom{|C|}{2} - 2e(C)$ this gives $f(C) \geq (s-1)(s-4)/2 \geq -1$ in the first case, with equality iff $s \in \{2,3\}$ and $G[C]$ is connected, and $f(C) \geq s(s-5)/2 \geq 0$ in the second. Clusters with $f = -1$ have ≥ 2 vertices, so there are at most m of them, and by Lemma 7.1 cost $\geq (2m+1) - m = m + 1$. The partition $\{0,1\}, \{2,3\}, \ldots, \{2m-2, 2m-1\}, \{2m\}$ has cost $m + 1$.

(iii) In an optimum $\sum f = -m$, so exactly m clusters have $f = -1$ (each connected, of size 2 or 3) and all others have $f = 0$; at most one vertex lies outside the $f = -1$ clusters. A cluster

with $f = -1$ that contains the pendant vertex 0 contains its only neighbour 1; likewise for $2m$ and $2m-1$. Suppose $1, 2m-1$ lie in one cluster C^*. If $f(C^*) = -1$ and $|C^*| = 2$, then $C^* = \{1, 2m-1\}$ contains no pendant, and both pendants are outside the $f = -1$ clusters: two vertices outside, a contradiction. If $f(C^*) = -1$ and $|C^*| = 3$, it contains at most one pendant; any pendant not in C^* has its only neighbour in C^* and lies in no other $f = -1$ cluster, so exactly one pendant is outside and the remaining $2m-3$ vertices must form $m-1$ clusters of size ≥ 2, impossible. If $f(C^*) \geq 0$, at most $2m-1$ vertices remain for m clusters of size ≥ 2. So 1 and $2m-1$ are in different clusters, i.e. e^* is deleted. □

(After one exchange of a member this obstruction disappears: Proposition 11.10 shows that it is an artifact of the packing H_m, while the excess 1 is intrinsic.) So the pointer realised by the deleted edge e^* joins members at distance $m-1$ in $I(H_m)$: its two ends are not adjacent in any $I(H)^r$ with $r < m-1$, and a decomposition of $I(H)^r$ has no reason to put them in one bag. Corollary 7.7 bounds the candidates for the *other* end of e^* (at most $D + \ell$ hard neighbours), not the position of the first. For $m = 3, 4, 5$ a computer enumeration gives OPT $= m + 1$ with $7, 9, 11$ optima, each separating the ends of e^*. Appendix D shows that the idea of putting "locally bad" members into every bag does not repair this for distances 2 and 3 (Proposition D.10); for unbounded distance this is open.

10 Soft triangles are the source of hardness

The remaining sections locate where the difficulty of CEaMP sits. For zero-excess solutions the interaction between packed P_3's is as simple as a constraint system can be unless the packing contains one specific configuration, which we call a *soft triangle*. Without it, zero excess is polynomial for every c and w (Corollary 10.5), every lock core has at most two anchors (Corollary 10.7), and a far-lift model of excess is fixed-parameter tractable (Theorem 10.14); on forests $I(H)$ the full problem is fixed-parameter tractable in ℓ conditionally on one bounded-certificate hypothesis (Theorem 10.18). With soft triangles, the natural branching scheme provably has unbounded branching (Theorem 10.23). Section 5 shows that CEaMP is W[1]-hard in $\ell + w$, and the instances of that reduction are full of soft triangles; the present section shows that soft triangles are necessary for the routes that we can analyse and that without them the problem is much easier. The status is summarised in Section 10.6.

Conventions and labels. States are as in Appendix F: a member (a, b, c) with centre b is in state $1, 2, 3$ if its single edited pair is ab, bc, ac. Put $t_{ab} = (0, 1, 1)$, $t_{bc} = (1, 0, 1)$, $t_{ac} = (0, 0, 1)$ for the states $(1, 2, 3)$: $t_e(s) = 1$ iff the pair e of the member is inside one cluster after the edit. $\Sigma = \Sigma(G, H) \subseteq \{1, 2, 3\}^H$ is the set of zero-excess (ZE) state vectors; by Lemma 2.2, $s \in \Sigma$ iff no vertex triple is an induced P_3 of G after the edits s. A *majority operation* is $m : \{1, 2, 3\}^3 \to \{1, 2, 3\}$ with $m(a, a, b) = m(a, b, a) = m(b, a, a) = a$; applied to vectors it acts coordinatewise. Every statement is marked: *proved* (a proof is given, in Appendix G when long), *by computer* (a finite case check that is part of the proof, or of an example, and that was carried out by two independent implementations), or *evidence* (computer experiments that are not part of any proof). The statements of this section were also re-derived independently (new code, exhaustive enumeration of all set partitions for $n \leq 11$); this is evidence and not a substitute for the proofs.

10.1 Soft triangles and the structure of zero-excess solutions

Definition 10.1 (Soft triangle)**.** A vertex triple $\{x, y, z\}$ is a *soft triangle* of (G, H) if its three pairs are soft and are owned by three pairwise distinct members. (G, H) is *soft-triangle-free* if there is none. (This depends on H only, not on the edges of G.)

Lemma 10.2 (Soft triangles and $I(H)$)**.** *(a) The owners of a soft triangle form a triangle of $I(H)$. (b) If $c \leq 2$, every triangle of $I(H)$ consists of the owners of a soft triangle. (c) If $I(H)$ is triangle-free, in particular if $I(H)$ is a forest ($w \leq 1$), there is no soft triangle. For $c \geq 3$ a triangle of $I(H)$ need not come from a soft triangle (three members through one vertex).*

Proof. (a) Two pairs of a triple share a vertex, so their owners intersect. (b) Let A, B, C be pairwise adjacent. If $A \cap B$, $B \cap C$ and $A \cap C$ shared a vertex, it would lie in three members, contradicting $c \leq 2$; so the three intersection vertices $x \in A \cap B$, $y \in B \cap C$, $z \in A \cap C$ are distinct, $\{x, y\} \subseteq V(B)$, $\{y, z\} \subseteq V(C)$, $\{x, z\} \subseteq V(A)$ are soft pairs with three distinct owners. (c) follows from (a). □

Lemma 10.3 (Constraint structure)**.** *Let H be any packing and classify a vertex triple τ by the owners of its soft pairs.*

(i) No soft pair: a static *condition on G (if τ is an induced P_3 of G, then $\Sigma = \emptyset$).*

(ii) One soft pair, owner A: a unary *constraint on s_A.*

(iii) Two soft pairs, from distinct members A, B: these share a vertex $q \in V(A) \cap V(B)$, $\tau = \{p, q, r\}$ with $p \in V(A) \setminus \{q\}$, $r \in V(B) \setminus \{q\}$, and pr is hard; a binary *constraint on (s_A, s_B), namely $t_{pq}(s_A) + t_{qr}(s_B) + [pr \in E] \neq 2$.*

(iv) All soft pairs from one member: $\tau = V(A)$, no restriction on Σ beyond the states.

(v) Three soft pairs from three distinct members: a soft triangle, a ternary *constraint.*

Hence if (G, H) is soft-triangle-free, Σ is empty or the set of vectors satisfying unary lists $\Lambda_A \subseteq \{1, 2, 3\}$ and, for every pair A, B of adjacent members sharing q, the binary relation $R_{AB} \subseteq \{1, 2, 3\}^2$ that is the conjunction of the four triples $\{p, q, r\}$.

Proof. Two soft pairs of one triple share a vertex. If their owners coincide, the triple is $V(A)$, which is case (iv). If they differ, the shared vertex lies in both members, so it is the unique common vertex q; the third pair pr is not soft, since an owner A or B would then contain two vertices of the other member (forcing $|V(A) \cap V(B)| \geq 2$) and a third owner gives (v). The constraint of (iii) is "not exactly two edges". The final sentence is the classification, as R_{AB} depends only on A, B and the bits $[pr \in E]$ (the cross pairs). □

Theorem 10.4 (Majority closure)**.** *If (G, H) is soft-triangle-free, then $\Sigma(G, H)$ is closed under* every *majority operation m applied coordinatewise, with the same m at all coordinates: for $u, v, w \in \Sigma$, $m(u, v, w) \in \Sigma$. No pair-maximality is needed, and c and w are arbitrary.*

Proved; the classification of the binary relations R_{AB} is by computer (Appendix G.1). The relation R_{AB} depends on the roles of q in A and in B and on the four cross bits: 144 configurations, giving exactly 40 distinct relations, all of the shapes empty, singleton, 2-element set, a full row or column through $\{1, 2\}$, a 3-element subset of $\{1, 2\}^2$, or a "cross" (row ∪ column through $\{1, 2\}$); each shape is closed under every majority by a short argument. The classification was reproduced by an independent construction from 576 real graphs, and all 729 majority operations were checked against all relations. The point of the theorem is that single triples are not closed under any uniform majority; only the conjunction of the four triples at a shared vertex is.

Corollary 10.5 (Zero excess without soft triangles is polynomial)**.** *For soft-triangle-free (G, H), whether $(G, H, 0)$ is a YES instance is decidable in polynomial time. No bound on c or w and no tree decomposition is needed.*

Proof. By Lemma 10.3, $(G, H, 0)$ is YES iff there is no static violation and the binary CSP with $|H|$ variables of domain $\{1, 2, 3\}$, unary lists Λ_A and relations R_{AB} is satisfiable. By Theorem 10.4 every constraint is closed under a majority operation. By the theorem of Jeavons, Cohen and Cooper [12], a set of constraints closed under a near-unanimity operation (here a majority, of arity 3) is decided by enforcing strong consistency of a fixed level, which for binary constraints and a majority is path consistency; this takes time polynomial in $|H|$. Static triples are tested in $O(n^3)$. □

Evidence: an independent path-consistency solver built from the vertex triples agreed with exhaustive enumeration on 20,920 random soft-triangle-free instances ($n \leq 10$, $c \leq 4$, feasible and infeasible) and 3,400 tree-like ones, and every one of them had Σ closed under all 729 majority operations; on instances *with* soft triangles the same code finds 451 failures of closure.

Remark 10.6 (No conflict with the hardness of $\ell = 0$). Corollary 10.5 does not contradict the NP-hardness of $\ell = 0$ for $c \leq 23$ [15]: the variable gadget of [15, Sect. 4.1.1] (three cliques of five vertices over $\mathbb{F}_5$, one member for each arithmetic progression, 25 members, $c = 5$) packs every pair between consecutive cliques and a computer count finds 100 soft triangles in it (evidence; any triple with one vertex in each clique that is not an arithmetic progression is one). The hardness proof therefore lives on ternary constraints. Likewise Example F.11 ($c = 2$) has four soft triangles and is not closed under the operation m_3, and the flip gadget of Example F.10 has soft triangles. At $c \leq 2$ a soft triangle is a triangle of $I(H)$ (Lemma 10.2), hence $w \geq 2$, so for $w \leq 1$ the hardness of $\ell = 0$ cannot occur (this is also clear from Theorem 3.5).

Corollary 10.7 (Lock cores have at most two anchors)**.** *If (G, H) is soft-triangle-free, every core (minimal non-face of anchors, Appendix F) has at most two anchors. In particular this holds for every H with $c \leq 2$ and $I(H)$ triangle-free, whatever* $\mathrm{tw}(I(H))$.

Proof. Theorem 10.4 gives closure under a majority operation, and Lemma F.6 applies. □

This settles, for these instances, the question whether the hypothesis of Lemma F.6 holds: it holds exactly when there is no soft triangle in the sense that matters. The statement that the specific operation m_3 closes Σ for $c \leq 2$ is false with triangles (Example F.11) and true without soft triangles.

Corollary 10.8 (Tunnel releases are fixed-parameter tractable without soft triangles)**.** *In the tunnel-release model of Appendix F, for a soft-triangle-free packing the minimum number of cuts is computable in time $2^{\ell} n^{O(1)}$, with no dependence on c or w.*

Proof. A set S of tunnels is lockable iff the anchors at the ends of its tunnels in $V(H)$ form a face of G with all tunnels removed (Lemma F.3). By Corollary 10.7 a non-face contains a non-face of at most two anchors, hence of at most two tunnels, so every core has at most two tunnels. Proposition F.9(iv) with $d = 2$ applies, with the $\ell = 0$ test of Corollary 10.5. □

Remark 10.9 (With triangles at $c \leq 2$: evidence only). For $c \leq 2$ with triangles in $I(H)$ the set Σ can fail to be closed under a given majority (Example F.11). Whether it is always 2-*decomposable* (equal to the set of vectors whose binary projections lie in the binary projections of Σ), which is what Lemma F.6 actually needs, is open. Evidence: all tight triangle blocks (three members on six vertices, all role orders and hard bits; 3,712 tight configurations, 163 distinct Σ), all tight

K_4 gadgets (521 distinct Σ), 3,000 random tight instances on the triangular prism and random instances with $c \leq 2$ all gave 2-decomposable Σ and no core with three anchors, although 24 of the 163 sets Σ of the blocks and 56 of the 521 of the K_4 gadgets are not closed under all majorities. In an independent re-derivation, among random instances with soft triangles the non-2-decomposable ones (34 in the runs that recorded c) all had $c = 3$, and instances with $c = 2$ that are not closed under all majorities did occur (33).

10.2 Far hard pairs: what they can and cannot do

Call a hard pair $\{p, r\}$ a *cross pair* if $p \in V(A) \setminus \{q\}$ and $r \in V(B) \setminus \{q\}$ for two members $A \neq B$ with $V(A) \cap V(B) = \{q\}$ (these are the cross bits of Lemma 10.3); every other hard pair is *far*. A far pair never lies in a binary triple.

Lemma 10.10 (U-only identity and locality of hard edits)**.** *Call a solution* U-only *if every member receives exactly one edit, so that its other edits form a set U of hard pairs. For every packing, the least size of a U-only solution is $|H| + \min\{|U| : \Sigma(G \triangle U, H) \neq \emptyset\}$. For soft-triangle-free (G, H), $\Sigma(G \triangle U, H)$ differs from $\Sigma(G, H)$ only through (a) the static triples (these gate non-emptiness: an all-hard induced P_3 created or destroyed by U empties or un-empties Σ globally), (b) the lists Λ_A of the members A that contain an end of a pair of U, and (c) the relations R_{AB} of adjacent members A, B for which U contains a cross pair of (A, B). In particular a far pair changes only static triples and the lists at the members through its two ends.*

Proof. U avoids soft pairs, so every member is still an induced P_3 of $G \triangle U$, with the same pairs, and a vector is the single-edit part of a solution of G with hard edits U iff it lies in $\Sigma(G \triangle U, H)$. A hard pair e enters only the triples containing it; by Lemma 10.3 these are static, unary on a member through an end of e, or (if e is a cross pair of (A, B)) in a binary triple of R_{AB} or in a list of A or B. □

The model is a sub-model: one edit per member is not always optimal, as in Remark 3.7; for instance $n = 5$, $E = \{02, 03, 12, 23, 24\}$, $H = \{(1, 2, 4)\}$ has OPT $= |H| + 1$ but least U-only value $|H| + 2$ (by enumeration of all partitions).

Lemma 10.11 (Strictness)**.** (by computer; finite check) *Let A, B be adjacent members of a packing, R the relation R_{AB}, and Λ_A, Λ_B the lists that the cross bits of (A, B) impose on A and on B. For $X \subseteq \{1, 2, 3\}$ put* $\mathrm{img}(X) = \{s_A : \exists s_B \in X \cap \Lambda_B,\ (s_A, s_B) \in R\} \cap \Lambda_A$. *If $X \subseteq Y \subseteq \{1, 2, 3\}$ and* $\mathrm{img}(X) \neq \mathrm{img}(Y)$, *then* $|\mathrm{img}(X)| \leq 1$.

Across an edge of $I(H)$, a restriction of the neighbour's list that really changes the image leaves at most one state; locks propagate as unit propagation along edges of $I(H)$. This is a one-edge statement, and was checked for 144 configurations and 27 pairs $X \subseteq Y$ (803 strict cases, no violation); an independent check on 576 real graphs agrees. It is used only through Lemma 10.12.

Lemma 10.12 (Helly number two for unit lists)**.** *Let $\Sigma \subseteq D^V$ be closed under some majority operation (for any finite domain D) and let J be a family of unit constraints $x_{v_j} = s_j$. If Σ and J are jointly infeasible, then Σ together with some at most two members of J is infeasible.*

Proof. Take a minimal infeasible J with $|J| = k \geq 3$ and, for each j, $t_j \in \Sigma$ satisfying $J \setminus \{j\}$. For three indices a, b, c put $u = m(t_a, t_b, t_c) \in \Sigma$. For $j \notin \{a, b, c\}$ all three tuples have value s_j at v_j; at v_a the tuples t_b, t_c both have value s_a, so u has s_a there (and symmetrically for b, c). Hence u satisfies all of J, a contradiction. (If two constraints concern one coordinate with different values, the same argument applies.) □

Observation 10.13 (Lists of an outside vertex)**.** *Let $A = (a, b, c)$ and let $t \notin V(A)$ be such that the three pairs $\{x, t\}$, $x \in V(A)$, are hard; put $S = N(t) \cap V(A)$. The three triples $\{a, b, t\}, \{b, c, t\}, \{a, c, t\}$ allow exactly the following states of A:*

S	$\emptyset$	$\{a\}$	$\{b\}$	$\{c\}$	$\{a, b\}$	$\{a, c\}$	$\{b, c\}$	$\{a, b, c\}$
allowed in $\{1, 2, 3\}$	$\{1, 2, 3\}$	$\{1\}$	$\emptyset$	$\{2\}$	$\{2\}$	$\emptyset$	$\{1\}$	$\{3\}$
allowed in $\{1, \ldots, 5\}$	$\{1, \ldots, 5\}$	$\{1, 4\}$	$\{4, 5\}$	$\{2, 4\}$	$\{2\}$	$\{5\}$	$\{1\}$	$\{3\}$

(Direct check; the second row is for the five states of Lemma 3.3: S must be empty or a single part of the cluster partition of $\{a, b, c\}$ after the edit.) So a zero-excess list of an outside vertex is trivial, empty, or a unit *list; there are no two-element lists. This is why forked locks never occur for zero excess.*

Theorem 10.14 (Far-lift zero-excess model)**.** *Let (G, H) be soft-triangle-free, with c and w arbitrary. It is decidable in time $6^\ell n^{O(1)}$ whether there is a set W of at most ℓ far pairs with $\Sigma(G \triangle W, H) \neq \emptyset$ (the* far-lift model*). A YES answer gives a solution of CEaMP of size $|H| + |W|$.*

Proved (Appendix G.2). Algorithm: (1) if $G' = G \triangle W$ has an all-hard induced P_3, branch on its far pairs (none: NO); (2) otherwise accept if $\Sigma(G') \neq \emptyset$ (Corollary 10.5); (3) delete all *liftable* unit lists (those of Observation 10.13 for which at least one of the three pairs is far); if the rest is already infeasible, answer NO; (4) otherwise find a minimal infeasible family of liftable lists, which has at most two members by Lemma 10.12 (the rest of Σ is closed under all majorities by Theorem 10.4), and branch on the at most 3 far pairs of each: at most 6 branches, depth at most ℓ. A far pair changes only liftable lists (Lemma 10.10), so some far pair of the family must be edited. *Evidence*: the algorithm agreed with exhaustive search on 1,420 instances (and, in an independent re-implementation, on 1,050 more; the largest minimal family had two lists).

The model is restricted (no double edits, no edits of cross pairs): it is an upper bound on the excess, exact when some optimum uses neither. It contains the tunnel-release model of Corollary 10.8 and removes the need for tunnels: *arbitrary* far hard edits, including those that close static violations, are handled. It also shows what the cores of far edits look like: a pair of unit lists, a vertex-cover-like structure whose branching factor depends neither on n, nor c, nor w.

10.3 Five states, an exact model, and the abstract models

By Lemma 3.3 the variables of the dynamic programs of Section 3 range over five states: besides $1, 2, 3$, the state 4 (edits ab, bc; extra cost 1) and 5 (edits ab, bc, ac; extra cost 2). For a graph G' with the same packing let $\mathrm{Ex}(G')$ be the least value of $|F| + \sum_i \mathrm{extra}(x_i)$ over sets F of cross pairs and state vectors $x \in \{1, \ldots, 5\}^H$ such that $G' \triangle F \triangle$(soft edits of x) is a cluster graph (∞ if none); extra $= 0, 0, 0, 1, 2$ for the states $1, \ldots, 5$.

Lemma 10.15 (Exactness)**.** *For every packing,* $\mathrm{OPT}(G) = |H| + \min_W \big(|W| + \mathrm{Ex}(G \triangle W)\big)$*, where W ranges over sets of far pairs.*

Proof. Split a solution as in Lemma 2.2 into its soft edits (one non-empty edit set per member, which by Lemma 3.3 is a state; its cost is $1 +$ extra), its edits of cross pairs F, and its edits of far pairs W. Conversely every (W, F, x) is a solution of that size. □

Hence the *far* edits are the only edits whose position is not controlled by the dynamic program, which is what Section 8 found.

Breaks. Let RM2 be the following abstract model on a forest: a set of *conflicts* (pairs of nodes, with the path joining them) and two kinds of operations, a *break* at a node (cost 1; it hits every conflict whose path contains the node) and a *tunnel*, a link in a graph of maximum degree d (cost 1; it hits every conflict that has an end at one of its two nodes). We want at most L operations hitting all conflicts.

Proposition 10.16 (RM2 is fixed-parameter tractable)**.** *RM2 is solvable in time* $(2d+1)^L n^{O(1)}$ *by a search tree.*

Proof. Root every tree and let Q be an unhit conflict whose top node v is deepest. If an optimal solution hits Q by a break at a node $w \in Q$, replace it by a break at v: every other unhit conflict R through w has its top at depth at most $\mathrm{depth}(v)$ and contains w, hence contains the whole path from w to its top, which passes through v. If Q is hit by a tunnel, the tunnel has an end at an end of Q ($2d$ choices). So the search tree branches $2d+1$ ways to depth L. □

(Evidence: agreement with exhaustive search on 5,000 + 2,100 random instances in two independent implementations.) RM2 was a plausible abstraction of CEaMP on forests; it is not a faithful one.

Proposition 10.17 (RM2 is not faithful)**.** *Let* $n = 11$ *(vertices* $0, \ldots, 10$*), let* G *have the twelve edges*

$$01,\ 02,\ 04,\ 05,\ 06,\ 13,\ 28,\ \{2,10\},\ 34,\ 37,\ 38,\ 78$$

and let $H = \{M_0, M_1, M_2, M_3\}$ *with* $M_0 = (2,0,1)$*,* $M_1 = (4,3,1)$*,* $M_2 = (5,0,6)$*,* $M_3 = (2,8,7)$*. This instance is valid, pair-maximal,* $I(H)$ *is a star with centre* M_0*,* $c = 2$ *and there is no soft triangle. Then* $\mathrm{OPT} = |H| + 3$ *(enumeration of all set partitions). The members* M_1 *and* M_2 *cannot be in any of the states* $1, 2, 3$ *in a solution without hard edits (an outside vertex gives them the list* $\{4,5\}$ *of the second row of Observation 10.13), while* M_0, M_3 *can; each of* M_1, M_2 *is repaired alone by one break (state* 4*), so the break model predicts excess* 2*, but the true excess is* 3*: two breaks next to a common member put the cluster-views* $(0,0)$ *at two vertices of* M_0*, which then cannot be in a state of* $\{1,2,3\}$ *either. The cheapest vectors without hard edits are* $(1,5,4,1)$ *and* $(4,4,4,1)$*, both of cost* 7*.*

The proposition (checked by exhaustive enumeration, and by an independent subset dynamic program) says that a break is not a node-hit: it locks its neighbours. Consequently the abstract model of "hitting conflict paths by breaks and tunnels" cannot be used to decide CEaMP on forests, and the difficulty of the interior of a chain, where breaks are placed, stays inside the dynamic program (as the Core-Bound analysis below also shows).

The full problem on forests, conditionally. Let $I(H)$ be a forest ($w \le 1$, so $c \le 2$, $D = 2c \le 4$, no soft triangle). The *sites* of (G, H) are the members and the edges of $I(H)$; a triple that contains a far pair has as sites the owners of its soft pairs and the edges at which its cross pairs lie. For a set C of sites let $\mathrm{Ex}_C(G')$ be $\mathrm{Ex}(G')$ computed after deleting every triple that contains a far pair and has no site in C; Ex_C is monotone in C. Let $V(C)$ be the set of vertices of the members in C and of the members incident to an edge in C.

Hypothesis BC(t) (bounded certificates; open). There is $K(t, D)$ such that for every G' (with the same forest H, and no static violation) with $\mathrm{Ex}(G') > t$, every inclusion-minimal set C of sites with $\mathrm{Ex}_C(G') > t$ has $|C| \le K(t, D)$.

Assumption (O). $\mathrm{Ex}_C(G')$ can be computed in time $f(w)n^{O(1)}$, by a dynamic program over a tree decomposition whose bags are enlarged by the members at distance one. (Every triple with a

soft pair or a cross pair has its vertices in one such bag; the argument is a sketch and is not proved here.)

Theorem 10.18 (Forests, conditional)**.** *Let $I(H)$ be a forest. Algorithm $\mathcal{A}$ of Appendix G.3 decides CEaMP; it is correct unconditionally (given the oracle). Its running time is $(6KZ^2)^\ell \cdot T_{\mathrm{O}} \cdot n^{O(1)}$, where $Z = 3D + 3\ell + 1$, T_{O} is the time of the oracle (O) and K is the largest minimal set C met. Hence, if BC(t) holds for $t < \ell$ and (O) holds, CEaMP on forests is fixed-parameter tractable in ℓ.*

Proved implication; BC and (O) are not proved. The hypothesis BC affects the *running time only*, not correctness. BC(0) holds with $K = 2$ (Lemma 10.12). The case $\ell \le 1$ is already polynomial by Theorem 3.6 and is not progress. Evidence for BC: minimal certificates had at most $t+2$ sites in every adversarial search ($1, 3, 4$ for $t = 0, 1, 2$). A proof would need a weighted Helly statement for the five-state problem; breaks interact through locks (Proposition 10.17), so an exchange argument in the style of Proposition 10.16 fails.

10.4 Certificates for far edits: a branching scheme and its unbounded branching

We now use a different split of the hard pairs, tuned to a branching algorithm. For a vertex v let $\mathcal{M}(v)$ be the set of members through v. A pair $\{u, v\}$ is *local* if $u, v \in V(H)$ and some member through u equals or shares a vertex with some member through v, and *remote* otherwise; soft pairs are local, every pair with an end outside $V(H)$ is remote. Let $\mathcal{N}$ be the set of remote hard pairs. For $T \subseteq \mathcal{N}$ let $F'(T)$ be the least number of edits in an assignment of a status to every soft pair (each member gets a non-empty subset of its pairs) and to every local hard pair, such that every vertex triple *all of whose remote pairs lie in T* is not an induced P_3 (remote pairs of T keep their status in G; triples with a remote pair outside T are dropped).

Lemma 10.19 (Certificates)**.** *(a) F' is monotone: $T \subseteq T'$ implies $F'(T) \le F'(T')$. (b) If S is a solution with $S \cap T = \emptyset$, then $F'(T) \le |S|$. (c) $F'(\mathcal{N})$ is the least size of a solution with no remote edit. Hence if $F'(T) > |H| + \ell$ every solution of size at most $|H| + \ell$ contains a pair of T (such a T is a* certificate*).*

Theorem 10.20 (Certificate branching)**.** *Algorithm: if $F'(\mathcal{N}) \le |H| + \ell$, accept; if $\ell = 0$, reject; otherwise delete pairs from $\mathcal{N}$ one by one while $F' > |H| + \ell$ stays true, and for every pair e of the resulting minimal certificate T recurse on $(G \triangle e, \ell - 1)$. The algorithm is correct for every packing; depth at most ℓ and branching factor $d = \max |T|$ over the certificates met, so $d^\ell(|\mathcal{N}| + 2)$ oracle calls.*

Proof. (a) fewer constraints; (b) S restricted to soft and local pairs is an assignment, and every triple with its remote pairs in T has them unedited; (c) with $T = \mathcal{N}$ nothing is dropped. Soundness: a solution of $G \triangle e$ of size $\le |H| + \ell - 1$ gives $S \triangle \{e\}$, and H is still a packing since e is hard. Completeness: if no solution of size $\le |H| + \ell$ avoids all remote pairs then $F'(\mathcal{N}) > |H| + \ell$, a minimal certificate T exists, and every solution contains some $e \in T$. □

So the question is how large minimal certificates are, and how F' is computed. For the second, the former difficulty that a triple with two local pairs and one remote pair need not lie in a bag of a decomposition of $I(H)$ disappears for this definition of *local*.

Theorem 10.21 (The leaf test is fixed-parameter tractable)**.** *Given a tree decomposition of $I(H)$ of width w, $F'(T)$ for any $T \subseteq \mathcal{N}$, in particular $F'(\mathcal{N})$, is computable in time $f(w)n^{O(1)}$.*

Proved (Appendix G.4; radius 3 in $I(H)$ is attained and suffices). Remark: this is true for the *adjacency-defined* notion of local used here; for locality defined through the bags of a decomposition of $I(H)$ the corresponding statement remains unproved, so the earlier gap is bypassed by changing $\mathcal{N}$, not closed. Together, Theorems 10.20 and 10.21 give an algorithm in time $d^\ell f(w)n^{O(1)}$; the question is the size of d.

Theorem 10.22 (Core bound at $\ell = 0$ without soft triangles)**.** *Let H have no soft triangle and $\ell = 0$. Every minimal $T \subseteq \mathcal{N}$ with $F'(T) > |H|$ has $|T| \leq 12$ (≤ 3 if it is a static violation), for every c and w.*

Proved (Appendix G.5; the proof uses the far pairs as *guards*, unit or two-element lists on one member, together with Theorem 10.4 and a Helly lemma for 2-element lists on a 3-element domain). The bound 12 is not claimed tight. At $\ell = 0$ the algorithm of Theorem 10.20 never branches, so Theorem 10.22 is a statement about the structure of $\ell = 0$ certificates; its algorithmic use needs an $\ell \geq 1$ version, which is open (Erdős–Pósa-type argument sketched, not proved). *Evidence*: in 786 random soft-triangle-free instances the largest minimal inconsistent family of guards had two members, and minimal certificates found by deletion had at most 4 pairs.

With soft triangles the corresponding statement is false, already at $c = 2$ and $w = 2$.

Theorem 10.23 (Unbounded minimal certificates)**.** *For every $N \geq 1$ there is a graph G_N with a pair-maximal packing H_N of $3N$ members ($c = 2$, $\mathrm{tw}(I(H_N)) = 2$, N soft triangles, $6N+3$ vertices) and a set T_N of $2N + 4$ remote pairs with $F'(T_N) \geq |H_N| + 1$ and $F'(T_N \setminus \{e\}) = |H_N|$ for every $e \in T_N$. So T_N is a minimal certificate for $\ell = 0$ of unbounded size, and the minimal-certificate branching of Theorem 10.20 has unbounded branching factor.*

Proved (Appendix G.6; the block structure of the set of ZE vectors of G_N is by computer, for the first blocks and by isomorphism of interior blocks). Two comments. (1) The proof gives $F'(T_N) \geq |H_N| + 1$; the equality $F'(T_N) = |H_N| + 1$ was verified by computer ($N \leq 10$) and is not needed. (2) At $\ell = 0$ the set T_N is never branched on. For $\ell = 1$: the union of two disjoint copies, $T = T_N \cup T'_N$, is a minimal certificate of size $2(2N + 4)$ because F' is additive over disjoint unions (every triple meeting two copies has a remote pair outside T and is dropped); and for *connected* instances, a chain G_{2N} with one further pendant w_* adjacent to the glue vertex b_N and the remote pairs $w_*b_N, w_*r_N, w_*p_{N+1}$ added to T_{2N} gives $|T| = 4N + 7$ with $F'(T) - |H| = 2 > 1$ and $F'(T \setminus \{e\}) - |H| = 1$ for every e (evidence: $N = 1, \ldots, 4$, exact solvers). So the branching factor of Theorem 10.20 with an arbitrary minimal certificate is unbounded already at $\ell = 1$.

The mechanism is a chain of soft triangles that carries one bit through the glue vertices; a 2-element guard $\{1, 2\}$ ("the pair ac is apart", obtained from a pendant vertex adjacent to one end of the pair) switches off the escape state 3 of its block. This is exactly the *multi-valued unary guard* that cannot occur without a soft triangle (Theorem 10.22).

Remark 10.24 (Not a hardness result; minimum certificates). In G_N the real optimum is $|H| + N$: every pendant edge forces its anchor to be a singleton, impossible for a centre, so all N stubs are cut; the persistent bit exists only in the relaxation. Moreover there is always a small certificate there: if a far hard edge wx ends at the centre x of a member (y, x, y') and wy, wy' are far hard non-edges, then $\{wx, wy, wy'\}$ is a certificate at $\ell = 0$ (both pairs at x would have to be apart, two edits in one member). Every G_N has such a pendant at each centre of A_j. So Theorem 10.23 refutes "every minimal certificate is bounded", not "some certificate is bounded"; this is a property of the instance design, and computing a minimum certificate is itself not known to be fixed-parameter tractable. Whether a certificate of bounded size can always be chosen is open.

Lemma 10.25 (No conservative near-unanimity polymorphism with soft triangles, $c \geq 3$)**.** *Let $R_k = \{111, 113, 122, 131, 212, 221, 222, 232, 311, 331\}$ (states written as three-digit words in $\{1, 2, 3\}$). Any constraint language that contains R_k and all unary relations has no conservative near-unanimity polymorphism of any arity $r \geq 3$.*

Proved (Appendix G.7). The relation R_k is the conjunction of all triple constraints whose scope lies in $\{A, B, C\}$ for three pairwise adjacent members A, B, C on six vertices, in a packing with $c = 3$ that has four further members, each owning one of the four pairs of these six vertices that are neither in A, B, C nor hard non-edges (an explicit instance, found and checked by computer). It requires $c \geq 3$; for $c = 2$, where Theorem 10.23 lives, the lemma says nothing. What explains Theorem 10.23 at $c = 2$ is that a near-unanimity operation of arity r bounds the cores of unit lists but not of 2-element lists, since it only controls coordinates where the other $r - 1$ values coincide. The lemma shows that the route through a uniform near-unanimity polymorphism is closed for the full language with soft triangles; it is not a hardness proof.

10.5 Abstract hardness and the obstruction to realising it

W[1]-hardness through hard edges between far vertices needs an operation of cost one that repairs an obstruction at one of many interior positions. We separate the abstract problem from its realisation: the abstract problem is W[1]-hard (Theorem 10.27); tunnel releases cannot realise it without soft triangles (Theorem 10.28); and the reduction of Section 5 realises it with soft triangles.

Definition 10.26 (LPPH)**.** *Linked-pair path hitting*: a forest, a family of *obstruction paths* in it, a set of *links* (pairs of nodes) and a number L. An operation is a single node or a link; it hits every path containing one of its nodes. Can L operations hit all paths?

Theorem 10.27 (Abstract hardness)**.** *LPPH is W[1]-hard parameterized by L, already for a forest of $2k$ paths and a set of links that is a perfect matching (every node in at most one link), with $L = k^2$.*

Proved (Appendix G.8), by a reduction from Grid Tiling with $\leq$ [6]; cross-checked by brute force on 1,710 instances (two independent solvers, no mismatch). It is the abstract form of the mechanism "a far two-sided edit shares cost between two places"; an operation can hit an interior node of an obstruction, which is the feature the reduction uses.

Theorem 10.28 (Tunnel releases cannot realise Theorem 10.27 without soft triangles)**.** *Let (G, H) be soft-triangle-free and consider the tunnel-release model (Appendix F).*

(i) Let the conflict graph $\mathcal{C}$ *have as vertices the anchors that carry a tunnel or a stub, an edge xy if $\{x, y\}$ is not a face, and a loop at x if $\{x\}$ is not a face. Releasing a set R of these anchors (cutting their tunnels) gives a feasible instance iff R is a vertex cover of $\mathcal{C}$ (loops included); a cut at x repairs exactly the conflicts at x, and every conflict has at most two ends.*

(ii) Let X be a set of anchors and R_0 a set of released anchors disjoint from X such that the instance with release set R_0 is infeasible but $R_0 \cup \{x\}$ is feasible for every $x \in X$. Then $|X| \leq 2$. In particular an obstruction that must be repaired by cutting a tunnel at any *one of $N \geq 3$ anchors (as Theorem 10.27 needs, with $N = |S_{ij}|$ positions) cannot occur.*

Proof. (i) A set of locked anchors is feasible iff it is a face, iff (Corollary 10.7, using that a non-face contains a non-face of size at most two) it contains no loop and no edge of $\mathcal{C}$, i.e. iff the released set covers $\mathcal{C}$. (ii) R_0 is not a vertex cover, so some edge or loop e of $\mathcal{C}$ is uncovered; for each $x \in X$ the set $R_0 \cup \{x\}$ covers e, so x is an end of e, and e has at most two ends. □

Remarks (heuristic; not part of Theorem 10.28). (iv) *Breaks.* A double-edited member is an unpaired single-node operation, so the tile coupling between two arms is lost; the compatibility of a break with its neighbours is controlled by the same cross bits, so a break is a phase flip whose position can be restricted by the design but not a decoupler. We found no way to couple two break positions on different tracks at total cost 2. In the five-state model no list of an outside vertex excludes the state 4 (the break) without also excluding state 1 or state 2 (a finite check on the second row of the table of Observation 10.13, and on all intersections of such lists), so no hard vertex makes the break of a chain member dearer than both of its single-edit states. (v) *Other operations.* Inserted far pairs and cuts inside hard components of size at least three also act only through unary changes at their two ends (Lemma 10.10), and isolating a vertex of a hard K_s costs $s-1$ cuts. We have no proof that no combination of operations simulates interior hits.

10.6 What is known: the status

1. *The main question is answered*: CEaMP is W[1]-hard in $\ell + w$ (Theorem 5.1), so it is not fixed-parameter tractable in $\ell + w$ unless FPT = W[1].

2. *Soft-triangle-free instances reduce to one hypothesis.* Without soft triangles, zero excess is polynomial (Corollary 10.5), lock cores have at most two anchors (Corollary 10.7), the far-lift model and the tunnel-release model are fixed-parameter tractable (Theorem 10.14, Corollary 10.8), the leaf test of the branching scheme is fixed-parameter tractable (Theorem 10.21) and minimal certificates at $\ell = 0$ have bounded size (Theorem 10.22). For forests the full problem is fixed-parameter tractable in ℓ conditionally on BC (Theorem 10.18); BC and the $\ell \geq 1$ version of the core bound are open.

3. *Hardness needs soft triangles along the routes we can analyse, and soft triangles suffice.* The interior-hitting mechanism that Theorem 10.27 needs is impossible with tunnel releases when there is no soft triangle (Theorem 10.28), and the far-acting objects in the soft-triangle-free case are unit lists with cores of at most two (Lemma 10.12). With soft triangles the mechanism exists: the Latin members of the blown-up chains in Section 5 are soft triangles, and a chain with tunnels has cores of unbounded size (Remark I.13); this is a W[1]-hardness result with $c = O(\ell)$. Other operations in the soft-triangle-free case are excluded only heuristically, remarks (iv) and (v) above are not part of Theorem 10.28, and Theorem 10.23 alone is not a hardness result.

4. *With soft triangles at $c \leq 2$* (triangles of $I(H)$, $w \geq 2$): nothing is proved beyond Theorem 10.23; Remark 10.9 gives evidence that the lock structure is still 2-decomposable. The hardness reduction has $c \geq 26$; hardness for bounded c (Open Problem 2) and for $c = 2$ (Open Problem 7) is open.

11 Repacking: the parameters belong to the packing

The pair (ℓ, w) is a property of the packing H, not of the graph G. If H^* is another modification-disjoint packing with $|H^*| \geq |H|$, then (G, H^*, ℓ^*) with $\ell^* = |H| + \ell - |H^*| \leq \ell$ is an equivalent instance (the case $H^* = H \cup M$ is Lemma 2.4), with a new excess and a new width $\mathrm{tw}(I(H^*))$. The *intrinsic excess* $\mathrm{OPT}(G) - \nu_{\mathrm{mod}}(G)$, where $\nu_{\mathrm{mod}}(G)$ is the maximum size of a modification-disjoint packing, is a lower bound for ℓ^* whatever H^* is. This section asks which of the obstructions of Sections 9 and 10 survive when H may be replaced. The answer is a precise map of the remaining difficulty, not an algorithm: both known obstruction families to locality are artifacts of the choice

of H (Propositions 11.10 and 11.11), a one-sided fixed-parameter algorithm exists for a class of instances that contains them after repacking (Theorem 11.13), no polynomial repacking procedure can have bounds in ℓ alone (Theorem 11.15), and what remains is to *locate* the repacking (Open Problem 1). Fixed-parameter tractability in $\ell + w$ stays open. All proofs are in Appendix H.

Conventions. A vertex is *covered* if it lies in $V(H)$ and *uncovered* otherwise (the set R of Appendix A). H is pair-maximal unless said otherwise, S is an optimal solution with $|S| \le |H| + \ell$, and U is its set of hard edits, so $|U| \le \ell$ (Lemma 2.2). Ball(x) is the ball of radius 2 around x in $G[L_B]$ and D, B, Z, L_B are as in Section 7. For a tree decomposition $(\mathcal{T}, \{\beta_t\})$ of $I(H)$ put $X_t = \bigcup_{P \in \beta_t} V(P)$ and $\mathcal{T}_v = \{t : v \in X_t\}$ for $v \in V(H)$. The members through v pairwise intersect, so their subtrees pairwise meet and $\mathcal{T}_v$ is a subtree (Helly). Two covered vertices a, b are *co-bagged* if $\mathcal{T}_a \cap \mathcal{T}_b \neq \emptyset$. Statements labelled "computer" are finite checks that support, and do not prove, the claim; they are listed in Appendix H.

11.1 An exchange calculus

Lemma 11.1 (Exchange). *Let H be a packing, $Q \in H$, and T an induced P_3 of G whose three pairs are not soft pairs of $H - Q$. Then $H' = (H - Q) + T$ is a packing with $|H'| = |H|$, and $I(H') - T = I(H) - Q$; hence $\mathrm{tw}(I(H')) \le \mathrm{tw}(I(H)) + 1$. If $Q_1, \dots, Q_j \in H$ are distinct and $T_1, \dots, T_j$ are induced P_3's that are pairwise pair-disjoint and whose pairs are not soft pairs of $H - \{Q_1, \dots, Q_j\}$, the exchanged packing has size $|H|$ and $\mathrm{tw} \le \mathrm{tw}(I(H)) + j$. The packing H' need not be pair-maximal. If (G, H, ℓ) is a YES instance, a maximal family M hard over H' has $|M| \le \ell$, and $H'' = H' \cup M$ is a pair-maximal packing with $\mathrm{tw}(I(H'')) \le \mathrm{tw}(I(H)) + 1 + |M| \le w + 1 + \ell$ and $(G, H'', \ell - |M|)$ equivalent to (G, H, ℓ).*

Simultaneous exchanges need the T_i to be pairwise pair-disjoint; sequential exchanges, each applied to the current packing, need no such condition. In a computer test (random graphs with $6 \le n \le 10$, 27,177 single exchanges $H - Q + T$ with T a witness of a hard edit of an optimum) the exchanged packing was a packing of size $|H|$ with $\mathrm{tw}(I(H')) \le \mathrm{tw}(I(H)) + 1$ in all cases, and it was pair-maximal in 59% of them only: the two other pairs of Q become hard and can complete an all-hard induced P_3.

Lemma 11.2 (Closed twins). *Let S be an optimal solution and $N[a] = N[b]$ in G. Then a and b lie in one cluster of $G \triangle S$.*

Corollary 11.3 (Hard edits lie in induced P_3's). *Every edit of an optimal solution S lies in an induced P_3 of G. For an inserted pair ab one is (a, z, b) with z as in Lemma 7.2(b); for a deleted pair ab one is (x, a, b) or (x, b, a) with x adjacent to exactly one of a, b.*

This is the standard fact that optimal solutions of Cluster Editing edit only pairs lying in an induced P_3; we need the form with a witness.

Definition 11.4 (Bridged pairs). A hard pair $\{a, b\}$ with $a, b \in V(H)$ is *bridged* if a and b have a common neighbour z in the soft graph G_{soft}. Equivalently, there are members $P \ni a$, $P' \ni b$ with $P \neq P'$ and $V(P) \cap V(P') \neq \emptyset$. Then P, P' are adjacent in $I(H)$, so a, b are co-bagged in *every* tree decomposition of $I(H)$. Inserted pairs of type T1 (Theorem 9.2) are bridged.

Lemma 11.5 (Witness case split). *Let H be pair-maximal, S optimal and $e = \{a, b\} \in U$. Then e lies in an induced P_3 T of G (a* witness*), and every witness has exactly one or two soft pairs. Let z be the vertex of T opposite to e; it is the centre of T if e is inserted and an end of T if e is deleted.*

(a) If some witness has two soft pairs, they are $\{a, z\}$ and $\{b, z\}$, owned by two distinct members $P \ni a$ and $P' \ni b$ with $z \in V(P) \cap V(P')$; so e is bridged.

(b) Otherwise every witness has exactly one soft pair s, owned by some Q, and $(H - Q) + T$ is a packing of the same size in which e is soft (Lemma 11.1); we call e absorbable.

The two cases are a case split, not a dichotomy: an edit can have witnesses of both kinds (in a computer test, 2,699 of 6,595 hard edits). The shared vertex z is the centre of T only for inserted edits; for a deleted edit $e = ab$ with witness (x, a, b) the two soft pairs xa, xb share the end x. The conclusion of (a) is the same in both cases.

Corollary 11.6 (Non-bridged edits are absorbable)**.** *Let H be pair-maximal, S optimal and $e \in U$ not bridged (this includes every edit with an uncovered end). Then there is $Q \in H$ and an induced P_3 $T \supset e$ such that $H' = (H - Q) + T$ is a packing of size $|H|$ in which e is soft, with* $\mathrm{tw}(I(H')) \leq \mathrm{tw}(I(H)) + 1$. *The price is that the pairs of Q other than s become hard and may carry edits of S (Lemma 11.8).*

Lemma 11.7 (Uncovered ends)**.** *Let H be pair-maximal and S optimal. No hard edit of S has both ends uncovered.*

So every hard edit of an optimum has at least one covered end; with Corollary 7.7, its other end is covered or is an uncovered vertex within Ball-distance 2 of it.

Lemma 11.8 (Displacement, single-edit case)**.** *Let H be pair-maximal, S optimal, $e = ab \in U$ inserted, z a witness vertex in the cluster of a, b (Lemma 7.2(b)) such that $T = (a, z, b)$ has exactly one soft pair az, owned by $Q = (a, z, r)$ (as a vertex set $\{a, z, r\}$). Then S contains exactly one pair f of* $\mathrm{pairs}(Q)$*, namely one of ar, zr. After the exchange of Lemma 11.1, the set of hard edits relative to $(H - Q) + T$ is $(U - e) + f$. The pair f has the ends r and one vertex of T. If r lies in no member of $H - Q$, then r is uncovered after the exchange; otherwise f may again be non-bridged ("the defect moves").*

In a computer test of 5,027 inserted edits of this form, r became uncovered in 1,650 cases and stayed covered in 3,377 (67%). So iteration of Lemma 11.8 is the typical case. Iterating gives an alternating chain of exchanges; whether it terminates at an uncovered end or a bridged pair is **open** (Conjecture 11.18).

Lemma 11.9 (Hard components of uncovered vertices)**.** *Let H be pair-maximal and r uncovered. The hard component of r is $N[r]$, with $N_h(r) = N(r)$, and every pair inside $N[r]$ is a hard edge or a soft pair. In particular two vertices of $N[r]$ that are not co-bagged are joined by a hard edge.*

A computer test of 6,200 uncovered vertices of random pair-maximal packings found no violation.

11.2 The two known obstructions are artifacts of the packing

Proposition 11.10 (FAR after one exchange)**.** *Let G_m, H_m be as in Proposition 9.8, $m \geq 3$.*

(a) $\nu_{\mathrm{mod}}(G_m) = m$ and $\mathrm{OPT}(G_m) = m + 1$. So the excess 1 is intrinsic: $\ell^* \geq 1$ for every packing.

(b) $H^* = \{P_0, \ldots, P_{m-2}, T\}$ with $T = (1, 2m-1, 2m-2)$ (centre $2m-1$) is a pair-maximal packing of size m, equal to $H_m - P_{m-1} + T$, and $I(H^*)$ is a cycle on m nodes, so $\mathrm{tw}(I(H^*)) = 2$.

(c) The optimum $\{0,1\},\{2,3\},\dots,\{2m-2,2m-1\},\{2m\}$ *has cost* $m+1$*, and relative to* H^* *its only hard edit is the deleted pair* $\{2m-1,2m\}$*, whose end* $2m$ *is uncovered. The chord* e^* *is a soft pair of* T*. The vertex* $2m$ *is a local hub in every decomposition, since* $\mathrm{Ball}(2m)\cap V(H^*)\subseteq V(T)$*.*

So the far pointer of Proposition 9.8 becomes a hub edit at one uncovered vertex. By Proposition 9.8(iii) the original packing H_m has an optimum-forced deletion at distance $m-1$ in $I(H_m)$; the point of (a) is that the repacking does not remove the excess, only the non-locality. Computer: for $m=3,\dots,9$ ($n=7,\dots,19$) validity, pair-maximality, $\nu_{\mathrm{mod}}=m$ and $\mathrm{OPT}=m+1$ (MILP; enumeration for $m\le 5$), the cycle and the hard edit were confirmed.

Proposition 11.11 (The spine family after one exchange and one augmentation)**.** *Let* $\mathrm{Sp}(F)$*,* H*,* $\ell=1$ *be as in Proposition E.1 and put* $Y=(y,u_0,b_0)$*,* $Z=(b_0,u_1,c_1)$*,* $H'=(H-X_0)+Y+Z$*. Then* H' *is a packing with* $|H'|=|H|+1$*, hence* $\mathrm{OPT}=|H|+1=|H'|$*: the excess is* 0 *after repacking. Moreover* $\mathrm{tw}(I(H'))\le 2$*. Also, the original packing* H *lies in the class* $\mathcal{S}$ *of Definition 11.12 (with* $w=1$*,* $\ell=1$*): the unique hard edit* $\{y,b_0\}$ *of the optimum of Proposition E.1(S2) is a hub edit at the local hub* y*.*

This gives a two-line proof of $\mathrm{OPT}\ge|H|+1$ for $\mathrm{Sp}(F)$: H' is a packing of size $|H|+1$, and Lemma 2.2 applies; it replaces the case analysis (S6) for this purpose (S6 itself, irreducibility of pair-closed unions, is a different statement and is not affected). Computer: validity, pair-maximality of H, $|H'|=|H|+1$ and $\mathrm{tw}(I(H'))=2$ were checked for F the triangle ($n=18$), K_4 ($n=30$) and $K_{3,3}$ ($n=42$); $\mathrm{OPT}=|H|+1$ by MILP for the triangle and K_4 (for $K_{3,3}$ the upper bound is the explicit partition and the lower bound is H').

11.3 A one-sided algorithm for local instances

Definition 11.12 (Local hubs and the class $\mathcal{S}$)**.** Let H be pair-maximal, $\ell\ge 0$, $(\mathcal{T},\{\beta_t\})$ a tree decomposition of $I(H)$. An uncovered vertex $r\in L_B$ is a *local hub* if the vertices of $V(H)\cap\mathrm{Ball}(r)$ are pairwise co-bagged. A hard pair is *eligible* if it is (i) $\{a,b\}$ with $a,b\in V(H)\cap L_B$, $b\in\mathrm{Ball}(a)$, co-bagged, or (ii) $\{r,v\}$ with r a local hub and $v\in V(H)\cap\mathrm{Ball}(r)$. Let $\mathcal{S}(H,\mathcal{T},\beta)$ be the set of solutions S all of whose hard edits are eligible and such that every covered vertex lies in at most ℓ of them.

By Corollary 7.7 and Lemma 11.7, every hard edit of an optimal solution of size at most $|H|+\ell$ is a pair $\{x,y\}$ with $x,y\in L_B$, $y\in\mathrm{Ball}(x)$, at least one end covered; the class asks that such pairs are co-bagged (covered ends) or hub-local (one uncovered end). Bridged pairs are co-bagged in every decomposition (Definition 11.4).

Theorem 11.13 (One-sided local algorithm)**.** *Let* H *be pair-maximal and* $(\mathcal{T},\{\beta_t\})$ *a tree decomposition of* $I(H)$ *of width* w*. There is an algorithm of running time* $5^{w+1}(Z^2+1)^{3\ell(w+1)}n^{O(1)}$*, where* $Z=3D+3\ell+1\le 6w+3\ell+7$*, that outputs a number* $m^*=\min\{|S|:S\in\mathcal{S}(H,\mathcal{T},\beta)\}\ge\mathrm{OPT}(G)$*. If* $\mathrm{OPT}\le|H|+\ell$ *and some optimal solution lies in* $\mathcal{S}$*, then* $m^*=\mathrm{OPT}$*. Consequently, accepting iff* $m^*\le|H|+\ell$ *never accepts a NO instance and accepts every YES instance with an optimum in* $\mathcal{S}$*.*

This is a one-sided algorithm. Its promise (some optimum lies in $\mathcal{S}$) cannot be tested: $m^*>|H|+\ell$ does not certify NO. For FAR_m with the path tiling the YES instance $(G_m,H_m,1)$ is rejected, with $m^*=m+2=\mathrm{OPT}+1$; with H^* of Proposition 11.10 it is accepted. The algorithm is therefore not comparable with the unconditional Theorem 8.3: for the family $\mathrm{Sp}(F)$ of Proposition E.1 the instance $(G,H,1)$ is in the class with $w=1$ whereas $\omega=\mathrm{tw}(\Gamma')\ge\mathrm{tw}(F)$ is unbounded; for FAR_m

with the path tiling Γ' has treewidth at most 3 (it is G_m plus the soft pairs $\{2i, 2i+2\}$), but the instance is outside the class.

Where the proof needs care (Appendix H): the label of a covered vertex must be restricted to *co-bagged covered* partners and to *local hubs*, otherwise the symmetry check between the two ends of a pair cannot be enforced across a separator and the algorithm is unsound (computer: with that restriction lifted, 17 of 177 test instances gave a value below OPT); there is no edit between two uncovered vertices (Lemma 11.7); and every triple must be visible in one bag. The running time uses the five states of Lemma 3.3, $|X_t| \le 3(w+1)$ and $|\mathrm{Ball}| \le Z^2$.

Corollary 11.14 (Static form). *Let $\Lambda(H)$ be the set of pairs $\{a,b\} \subseteq V(H) \cap L_B$ with $a, b \in \mathrm{Ball}(r)$ for some $r \in L_B$. Let w_Λ be the minimum width of a tree decomposition of $I(H)$ that co-bags every pair of $\Lambda(H)$; equivalently, the minimum over the choices, for every $\{a,b\} \in \Lambda(H)$, of a member through a and a member through b, of the treewidth of $I(H)$ plus these links. Then CEaMP is solvable in time $5^{w_\Lambda+1}(Z^2+1)^{3\ell(w_\Lambda+1)}n^{O(1)}$, and the promise holds for every instance.*

Corollary 11.14 is only as strong as w_Λ, which is large on tunnel-rich instances; by Theorem 5.1 it cannot be bounded by a function of (ℓ, w) unless FPT = W[1].

11.4 Limits of repacking

Theorem 11.15 (No repacking with bounds in ℓ alone). *(a) If there is a polynomial-time algorithm that maps every YES instance $(G, H, 0)$ to a packing H^* with excess $\ell^* \le s(0)$ and $\mathrm{tw}(I(H^*)) \le f(0)$, for constants $s(0), f(0)$, then* P = NP. *The statement is stated for functions s, f of ℓ but only $\ell = 0$ is used.*

(b) Under ETH there is no polynomial-time algorithm that maps every YES instance $(G, H, 0)$ to a packing H^* with $|H^*| = |H|$ and $\mathrm{tw}(I(H^*)) \le |H|^{1-\varepsilon}$, $\varepsilon > 0$.

The guarantees hold only on YES inputs, so the proof validates the output: it tests that H^* is a modification-disjoint packing of induced P_3's of G, computes a tree decomposition of the stated width or rejects, and tests the excess before running the algorithm of the CSP theorem (Theorem 3.6). The theorem used is the *CSP algorithm* of Section 3, valid for every packing; it is not the majority-closure theorem (Theorem 10.4), which concerns soft-triangle-free packings. The reductions of [15] contain soft triangles, which is consistent: the NP-hardness at $\ell = 0$ is not contradicted by Theorem 10.4, and Theorem 11.15 does not use it. Hence any usable repacking must have $\mathrm{tw}(I(H^*)) \le w + g(\ell)$ *and* a locality property; with $H^* = H$ the width bound alone is trivial. The hope "if $\mathrm{tw}(I(H))$ is large relative to ℓ then a better packing exists, hence reject" is false for the same reason: every tight instance ($\ell = 0$, $\nu_{\mathrm{mod}} = \mathrm{OPT} = |H|$) has no better packing, and tight instances have unbounded treewidth under ETH.

Proposition 11.16 (The naive normal form is false). *Let $G = K_2 \vee 3K_1$ (hubs $3, 4$ adjacent, leaves $0, 1, 2$ each adjacent to both hubs; $n = 5$). Then $\nu_{\mathrm{mod}} = 2$,* OPT = 3 *with four optima, every maximum packing covers all five vertices, and for every maximum packing every optimum has a hard edit with both ends covered. Hence "every hard edit has an uncovered end for some maximum packing and some optimum" is false. The number of connected graphs with at least one induced P_3 on which it fails is* 3 *of* 20 *($n = 5$),* 2 *of* 111 *($n = 6$),* 3 *of* 852 *($n = 7$) and* 305 *of* 11,116 *($n = 8$) (computer, exhaustive). The surviving hard edit in the example is bridged.*

Proposition 11.17 (Bounded-radius augmentation is insufficient). *Let P be the path $0, \ldots, 2m+2$ and $H_0 = \{(2i+1, 2i+2, 2i+3) : 0 \le i < m\}$. Then H_0 is a pair-maximal packing, $\nu_{\mathrm{mod}} =$*

$\mathrm{OPT} = m + 1$, $\ell = 1$, $w = 1$, *and the unique packing of size* $m + 1$ *shares no member with* H_0. *So every improving exchange (remove* r *members, add* $r + 1$ *pair-disjoint ones) has* $r = m$ *(computer:* $m = 2, \ldots, 6$*).*

The scope is exactly bounded-r exchange search; the instance is a path. Moreover, in FAR_m the original tiling is already a maximum packing, so no improving exchange exists, and the helpful move of Proposition 11.10 is *neutral*: its benefit is invisible to any objective computed from (G, H), because it is chosen for the chord that the optimum cuts.

11.5 What remains: locating the repacking

Conjecture 11.18 (C1, normal form NF*)**.** *For every graph* G *there are a maximum modification-disjoint packing* H^* *and an optimal solution each of whose hard edits (relative to* H^**) has an uncovered end or is bridged.*

Status: open. Evidence (computer, independent implementation): no counterexample among all connected graphs with $n \leq 8$, among 1,500 random connected graphs with $n = 9$, and among 300 random connected graphs with $n = 10$. By Proposition 11.16 the clause "bridged" is necessary. C1 alone is not enough for Theorem 11.13, which also needs the hub clause of Definition 11.12 (the evidence above is for the covered-covered clause only), and it is an existence statement; it does not say how to find H^*.

Open Problem 1 (LOC: locating the repacking; **no, unless** FPT = W[1])**.** Is there an algorithm that, given (G, H, ℓ) and a decomposition of $I(H)$ of width w, runs in time $f(\ell, w)n^{O(1)}$ and outputs a pair-maximal packing H^* with $|H^*| \geq |H|$ and $\mathrm{tw}(I(H^*)) \leq g(\ell, w)$, together with a decomposition of that width, such that on every YES instance some optimal solution lies in $\mathcal{S}(H^*, \mathcal{T}^*, \beta^*)$ for $\ell^* = |H| + \ell - |H^*|$?

Proposition 11.19. *A positive answer to Open Problem 1 implies that CEaMP is fixed-parameter tractable in* $\ell + w$.

Corollary 11.20. *Unless* FPT = W[1]*, the answer to Open Problem 1 is negative: no algorithm locates a repacking of bounded width and excess in time* $f(\ell, w)n^{O(1)}$ *on every YES instance.*

Proof. Proposition 11.19 and Theorem 5.1. □

By Theorem 11.15 the width bound g cannot depend on ℓ alone, so g must involve w.

Why this is a different problem (heuristic). If S is an optimum with far hard edits $e_1, \ldots, e_k$, $k \leq \ell$, and H^* absorbs them (Corollary 11.6), then $H^* - H$ consists of witnesses T_i of the e_i, and conversely the e_i are read off from H^* as the hard pairs inside the new members. So computing H^* is computing the far edits: repacking moves the information problem (which of about n^2 candidate pairs is cut) into the packing, and does not remove it. A dynamic program that must carry for each pending far edit the identity of its partner across a separator would need more than $f(\ell, w)$ states; colour coding repairs completeness but not soundness. This paragraph is not a proof.

Precise status. (1) The obstructions of Proposition 9.8 (deleted hard edges between far members) and of Proposition E.1 (spine family, unbounded $\mathrm{tw}(\Gamma')$) are artifacts of the chosen packing: after one exchange, respectively one exchange and one augmentation, they have $\mathrm{tw}(I(H^*)) \leq 2$ and a single hard edit at an uncovered vertex, respectively excess 0. (2) The intrinsic excess and, for tight instances, the unbounded width are not artifacts (Proposition 11.10(a), Theorem 11.15). (3) Locating the repacking in fixed-parameter time is impossible unless FPT = W[1] (Corollary 11.20); the existence of a good repacking (Conjecture 11.18) is not affected by this.

12 Open problems and summary of results

Notation as above; $w = \mathrm{tw}(I(H))$.

The main question: answered

The main question of this paper was whether CEaMP is fixed-parameter tractable parameterized by $\ell + w$. **Answer: no, unless** FPT = W[1] (Theorem 5.1): the problem is W[1]-hard in $\ell + w$, in $\ell + \mathrm{pw}(I(H))$ and in $\ell + c$ (recall $c \leq w + 1$), even for pair-maximal packings, and under ETH there is no algorithm in time $f(\ell, w)N^{o(\sqrt{\ell})}$ (Corollary 5.2). The XP algorithms of Theorems 3.6 and 8.2, with exponent 2ℓ resp. ℓ and single-exponential dependence on w, are therefore optimal up to the exact exponent of n.
Positive results that remain: fixed-parameter tractable in ℓ when $w = 0$ [17], and in (ℓ, s) when the components of $I(H)$ span at most s vertices (Theorem 6.1); fixed-parameter tractable in (ℓ, D, ω) (Theorem 8.3), where ω is not bounded by (ℓ, w) (Proposition 8.4) and is unbounded on the hard instances; the case $H = \emptyset$ (plain Cluster Editing, $\ell = k$) is fixed-parameter tractable. *Precise status (Sections 10, 5 and 11).* (1) Without soft triangles (Definition 10.1) the set of zero-excess states is majority-closed, $\ell = 0$ is polynomial for every c and w, lock cores have at most two anchors, and the far-lift model is fixed-parameter tractable; on forests ($w \leq 1$) the full problem is fixed-parameter tractable in ℓ if the bounded-certificate hypothesis BC holds (Theorem 10.18), so for soft-triangle-free instances the question reduces to BC (Open Problem 8). (2) With soft triangles the problem is W[1]-hard (Theorem 5.1); the interior-hitting mechanism of the abstract problem of Theorem 10.27 cannot be realised by tunnel releases without soft triangles (Theorem 10.28), and is realised with them. The hard instances have $c = 4p + 2 \geq 26$ and $w = \Theta(p^2)$. (3) *Repacking.* The two known obstructions to locality, the far-deletion family FAR (Proposition 9.8) and the spine family $\mathrm{Sp}(F)$ (Proposition E.1), are repacking artifacts (Propositions 11.10 and 11.11), a one-sided algorithm is exact on the local class (Theorem 11.13), but locating the repacking in fixed-parameter time is impossible unless FPT = W[1] (Corollary 11.20). A positive answer to the old question would have needed a bound $g(\ell, w)$ on the number of candidate positions of the first endpoint of each hard edit; by Theorem 5.1 no such bound exists in general (given one endpoint, the other has at most Z^2 candidates, Corollary 7.7, so it is the placement of at most ℓ bounded objects that is free, and the reduction exploits exactly this freedom).

Open Problem 2 (Hardness for bounded c)**.** Is CEaMP W[1]-hard parameterized by ℓ when c is bounded by a constant (for instance $c \leq 34$)? The reduction of Theorem 5.1 has $c = 4p + 2$ with $p \geq k^2 + 5 = \ell + 5$ because a blown-up clique of size p cannot be split by an edit set of excess ℓ only if $p > \ell$ (Lemma I.7); the zero-excess semantics (Proposition I.5) hold for every even $p \geq 8$. What is missing is a gadget that holds a vertex of the chain together at bounded c against ℓ edits.

Open Problem 3 (ETH-tight lower bound)**.** Under ETH, Corollary 5.2 excludes $f(\ell, w)N^{o(\sqrt{\ell})}$ (from $\ell = k^2$ and the bound $n^{o(k)}$ for Grid Tiling), while the algorithms of Theorems 3.6 and 8.2 run in time $N^{2\ell+O(1)}$ resp. $N^{\ell+O(1)}$ times $f(\ell, w)$. Is there an algorithm with exponent $O(\sqrt{\ell})$, or is $N^{o(\ell)}$ excluded? A reduction with $\ell = O(k)$ would be needed for the latter.

Open Problem 4 (Pointers of type T2 and deleted hard edges)**.** Can the far ends of inserted pairs of type T2 (a kept hard edge az with a far from z in $I(H)$, and a soft pair zb; by Lemma 9.6 the end a is one of at most ℓ vertices fixed by (z, b)), and of deleted hard edges between far members, be handled in a dynamic program of width bounded by $g(\ell, w)$ *on a restricted class*? For the full problem the answer is no unless FPT = W[1] (Theorem 5.1), since such a program would give

a fixed-parameter algorithm; the question is open for soft-triangle-free packings and bounded c. Proposition 9.8 shows that an optimum can be forced to delete a hard edge between members at any distance in $I(H)$ (with $w = 1$, $c = 2$, $\ell = 1$); whether a merge pointer between two pieces joined by a soft pair can be forced at unbounded distance is open. For the deleted edge of Proposition 9.8 the far pointer disappears after one exchange (Proposition 11.10); finding such an exchange in fixed-parameter time is impossible in general unless FPT = W[1] (Corollary 11.20).

Open Problem 1 (LOC, locating the repacking) is stated in Section 11; by Proposition 11.19 a positive answer implies fixed-parameter tractability in $\ell + w$, so by Theorem 5.1 the answer is negative unless FPT = W[1] (Corollary 11.20), and Conjecture 11.18 (normal form NF*) is its existence counterpart.

Open Problem 5 (Reduction lemma; negative unless FPT = W[1])**.** Is there a polynomial-time algorithm that, given (G, H, ℓ) with H pair-maximal, either outputs NO correctly or outputs an equivalent instance (G^*, H^*, ℓ^*) with H^* pair-maximal, $\ell^* \leq \ell$, $w(H^*) \leq f_1(\ell, w)$ and $\mathrm{tw}(\Gamma'(G^*, H^*)) \leq f_2(\ell, w)$? A positive answer would give fixed-parameter tractability in $\ell + w$ by Theorem 8.3 (with $D^* \leq 2(w(H^*) + 1)$), so by Theorem 5.1 it is impossible unless FPT = W[1]. Proposition 8.4 shows that some reduction is necessary. For $c = 1$ the rule of [17] is such a reduction; Theorem 6.10 leaves at most 2ℓ components of $I(H)$; Appendix E shows that the same rule applied to pair-closed unions of members removes nothing on a family with $w = 1$, $\ell = 1$ and unbounded $\mathrm{tw}(\Gamma')$.

Open Problem 6 ($w = 1$ with unbounded tree components)**.** If $I(H)$ is a forest ($w \leq 1$, hence $c \leq 2$) with components of unbounded size, is CEaMP fixed-parameter tractable in ℓ? Theorem 6.1 settles bounded components, in particular perfect matchings (Corollary 6.2), but its counting argument is at the level of whole components and gives nothing for a single large tree component. For forests there is no soft triangle: $\ell = 0$ is polynomial, the far-lift model is fixed-parameter tractable (Theorem 10.14), and the full problem is fixed-parameter tractable in ℓ conditionally on BC and the oracle (O) (Theorem 10.18); the question is thus reduced to Open Problem 8.

Open Problem 7 (Packings with $c \leq 22$)**.** Is CEaMP with $c = 2$ fixed-parameter tractable in ℓ? What is the complexity of $\ell = 0$ for $3 \leq c \leq 22$? Both are open in [15], whose authors say a W[1]-hardness result for $c = 2$ "would also not be surprising". Partial answer: for soft-triangle-free packings (for $c \leq 2$: $I(H)$ triangle-free) $\ell = 0$ is polynomial for every c (Corollary 10.5), so a hardness proof for $\ell = 0$ at $c = 2$ needs triangles in $I(H)$, hence $w \geq 2$. The case $c = 2$ with triangles and arbitrary ℓ is open; the reduction of Theorem 5.1 has $c \geq 26$ (Open Problem 2).

Open Problem 8 (Bounded certificates BC; **OPEN**)**.** For forests $I(H)$ (more generally soft-triangle-free packings), is there $K(t, D)$ such that every inclusion-minimal set of sites C with $\mathrm{Ex}_C(G') > t$ has at most $K(t, D)$ elements (Section 10.3)? $t = 0$ holds with $K = 2$ (Lemma 10.12); $t \geq 1$ is open, as is the $\ell \geq 1$ version of Theorem 10.22 (Erdős–Pósa-type argument sketched, not proved). A proof needs a weighted Helly statement for five-state problems; breaks interact through locks (Proposition 10.17).

Open Problem 9 (Soft triangles at $c \leq 2$)**.** For $c \leq 2$ with triangles in $I(H)$, is Σ always 2-decomposable (equivalently, do lock cores have at most two anchors)? Evidence: yes for all triangle blocks, K_4 gadgets and random instances tried (Remark 10.9); Example F.11 shows that closure under the specific operation m_3 fails.

Open Problem 10 (Small certificates)**.** Can a certificate of bounded size always be chosen (or found in fixed-parameter time) when minimal certificates are unbounded (Theorem 10.23, Remark 10.24)?

Open Problem 11 (Base of the exponent)**.** For $\ell = 0$, does a $(3-\varepsilon)^w n^{O(1)}$ algorithm exist, or is there a SETH-type lower bound? Theorem 4.1 gives only the exponent.

Summary of results and their scope

The following table is the only status summary of the paper. "Input H" means that the statement is in terms of w, ℓ and n of the input packing; "pair-maximal H'" means that it is stated for a pair-maximal packing, and for an arbitrary input applies to a pair-maximal extension H' with the primed parameters of Section 2. "YES" and "optimal" mean that the statement needs a YES instance, respectively an optimal solution.

Result	Applies to	Needs	Status
Thm 3.5, 3.6 (CSP, 3^{w+1}, $5^{w+1}n^{2\ell}$)	input H	–	proved
Thm 10.4, Cor. 10.5, 10.7, 10.8 (no soft triangle: majority closure, $\ell = 0$ polynomial, cores ≤ 2)	input H, soft-triangle-free	–	proved; 40-relation classification by computer; cites [12]
Lemmas 10.10–10.12, Thm 10.14 (far-lift model, 6^ℓ), Lemma 10.15	input H (soft-triangle-free for Thm 10.14)	–	proved; restricted model
Prop. 10.16, 10.17 (RM2 FPT; not faithful)	model; explicit instance	–	proved (hub: enumeration)
Lemma 10.19, Thm 10.20, 10.21 (certificate branching, leaf test)	input H (adjacency-local)	–	proved
Thm 10.22 (certificates ≤ 12 at $\ell = 0$)	soft-triangle-free	$\ell = 0$	proved
Thm 10.23 (unbounded minimal certificates, $c = 2$)	explicit family	–	proved; block structure by computer; $= \lvert H\rvert + 1$ and connected $\ell = 1$ family: evidence
Lemma 10.25 (no conservative NU)	$c \geq 3$ language	–	proved
Thm 10.27, 10.28 (abstract W[1]-hardness; tunnels cannot realise it without soft triangles)	abstract; soft-triangle-free tunnel model	–	proved; remarks (iv), (v) heuristic
Thm 5.1 (W[1]-hardness in $\ell + w$, $\ell + \mathrm{pw}$, $\ell + c$; no $f(\ell, w)n^{o(\sqrt{\ell})}$ under ETH, Cor. 5.2)	pair-maximal H, $c \leq 4p + 2$, $\ell = k^2$	–	**proved, computer-assisted**: Lemma I.1 (transfer matrix, $m \leq 60$ checks), Lemma I.2, counts of Lemma I.6; the rest by hand; exact solves of small instances only ($k = 1$ and a comparator instance), thin model tested for $k \leq 3$
Remark I.13 (chain with unbounded core, bounded c and width)	single chain, $p \geq 8$	–	follows from the proof of Prop. I.5; not run separately
Thm 10.18 (forests, conditional)	forest $I(H)$	BC, oracle (O)	implication proved; BC, (O) open
Thm 4.1, Prop. 4.3 (ETH; w alone)	input H	–	proved from [15]
Thm 6.1 (components, kernel)	input H	–	proved; cites [17] Lemma 6.3, [2], [4]
Thm 6.10 (components of any size)	input H	–	proved; cites [17] Lemma 6.3
Thm 8.2 (n^ℓ algorithm)	input H (via H')	–	proved

Result	Applies to	Needs	Status
Lemmas 7.2–7.6, Cor. 7.7	pair-maximal H'	optimal, YES	proved
Thm 9.2, Cor. 9.3, Prop. 9.5	pair-maximal H'	optimal, YES	proved
Prop. 9.7, 9.8	explicit instances	–	proved (cex: enumeration)
Thm 8.3, Prop. 8.4	pair-maximal H'	YES	proved; expander treewidth cited [9]
Appendices A–D	pair-maximal H' (App. B: any H)	mostly YES	proved
Appendix E, F	explicit families; tunnel model	–	proved
Lemma 11.1, 11.2, Cor. 11.3, Lemma 11.5, Cor. 11.6, Lemma 11.7, 11.9 (exchange, twins, witness case split, uncovered ends)	pair-maximal H (exchange: any H)	optimal (not Lemma 11.1, 11.9)	proved; random tests
Lemma 11.8 (displacement, single edit)	pair-maximal H	optimal	proved; iteration and termination open
Prop. 11.10, 11.11 (FAR and Sp(F) repack)	explicit instances	–	proved; computer for $m \le 9$, $F \in \{$triangle, K_4, $K_{3,3}\}$
Thm 11.13, Cor. 11.14 (one-sided local algorithm)	pair-maximal H	promise: optimum in $\mathcal{S}$	proved; one-sided, promise untestable; implemented
Thm 11.15 (no repacking bounded in ℓ)	any H, $\ell = 0$	P $\neq$ NP resp. ETH	proved from [15]
Prop. 11.16, 11.17 (naive normal form false; augmentation depth m)	explicit instances	–	proved; counts by computer
FPT in $\ell + w$ (the main question)			**no unless** FPT = W[1] (Thm 5.1)
Hardness at bounded c (Open Problem 2); $c = 2$ (Open Problem 7)			**open**
Exact exponent under ETH (Open Problem 3): $N^{o(\sqrt{\ell})}$ excluded, N^{ℓ} achieved			**open**
Reduction lemma (Open Problem 5)			**no unless** FPT = W[1]
Positions of T2 / deleted pointers (Open Problem 4)			**open on restricted classes**; no in general unless FPT = W[1]
LOC (Open Problem 1); implies FPT (Prop. 11.19)			**no unless** FPT = W[1] (Cor. 11.20)
Conjecture C1 / NF* (Conj. 11.18)			**open**; no counterexample for all connected $n \le 8$, 1,500 random $n = 9$, 300 random $n = 10$
BC, small certificates, $c \le 2$ with triangles (Open Problems 8–10)			**open** (evidence only)

A Twins, vertices outside the packing, and a shrink rule

By Lemma 2.2 the whole difficulty of $\ell > 0$ is the set U of at most ℓ hard edits. This appendix proves what can be said about the vertices outside $V(H)$ for pair-maximal packings H. The results

are not used in the main text.

Conventions. Partitions, clusters, costs and optima are as in Section 2; $\Pi(v)$ is the cluster of v. Unless said otherwise, H is pair-maximal and $R = V \setminus V(H)$; every pair with an end in R is hard.

A.1 Twins

Lemma A.1 (Twins, strict form). *Let $u \neq v$ be adjacent with $N[u] = N[v]$. For every partition Π with $\Pi(u) \neq \Pi(v)$, one of the two partitions obtained by moving u into $\Pi(v)$, respectively v into $\Pi(u)$, has cost smaller than Π by at least 1; more precisely the two cost changes sum to exactly -2. Hence u, v lie in one cluster in* every *optimum, with no hypothesis on H.*

Proof. For a vertex y and a set $Y \not\ni y$ let $f(y, Y) = |Y \setminus N(y)| - |Y \cap N(y)|$. Put $C_u = \Pi(u) \setminus \{u\}$ and $C_v = \Pi(v) \setminus \{v\}$ (possibly empty); $C_u \not\ni v$, $C_v \not\ni u$, and neither contains u or v. Moving u out of $\Pi(u)$ changes the status of the pairs uz, $z \in C_u$: before, uz is an edit iff it is a non-edge; after, iff it is an edge; the change is $-f(u, C_u)$. Moving u into $\Pi(v) = C_v \cup \{v\}$ changes, for $z \in C_v \cup \{v\}$, the status from "edit iff edge" to "edit iff non-edge"; the change is $f(u, C_v \cup \{v\}) = f(u, C_v) - 1$ because $uv \in E$. All other pairs keep their status. So

$$\delta_1 = -f(u, C_u) + f(u, C_v) - 1, \qquad \delta_2 = -f(v, C_v) + f(v, C_u) - 1.$$

Since $N(u) \setminus \{v\} = N(v) \setminus \{u\}$ we have $f(u, Y) = f(v, Y)$ for all $Y \subseteq V \setminus \{u, v\}$, hence $\delta_1 + \delta_2 = -2$. Thus $\min(\delta_1, \delta_2) \leq -1$. □

This is the critical-clique property used in kernelizations of Cluster Editing [7, 10] (it is also recalled in [17] when motivating Reduction Rule 6.2); the argument above is self-contained and shows that it is strict.

A.2 Viability of a hard edit

Lemma A.2 (Viability of a hard edit). *Let S be any solution, U its hard part and $uv \in U$. Let $\mathrm{Bd}(uv)$ be the set of $z \notin Q(u) \cup Q(v) \cup \{u, v\}$ such that $\{u, v, z\}$ is an induced P_3 of $G \triangle \{uv\}$. Then for every $z \in \mathrm{Bd}(uv)$ one of uz, vz lies in U, and $|\mathrm{Bd}(uv)| \leq |U| - 1$.*

Proof. All three pairs of $\{u, v, z\}$ are hard ($z \notin Q(u) \cup Q(v)$). If neither uz nor vz is in S, the triple has in $G \triangle S$ the same pattern as in $G \triangle \{uv\}$, an induced P_3. Distinct z give distinct pairs, all different from uv. □

This gives a kernel-style condition (deleting uv needs fewer than ℓ common neighbours outside $Q(u) \cup Q(v)$) but no bound on the number of candidate pairs: all pendant edges are viable.

A.3 The vertices outside $V(H)$

Lemma A.3 (Structure of R). *Let H be pair-maximal. For $r, s \in R$: if $rs \in E$ then $N[r] = N[s]$; if $rs \notin E$ then $N(r) \cap N(s) = \emptyset$. Consequently:*

1. *every component X of $G[R]$ (a* class*) is a clique all of whose vertices have the same closed neighbourhood $N[X]$ (they are true twins);*
2. *$A_X := N(X) \setminus X$ is contained in $V(H)$;*
3. *the sets A_X, over the classes X, are pairwise disjoint.*

Proof. Every pair with an end in R is hard. Let $rs \in E$ and $z \neq r, s$ be adjacent to exactly one of r, s. Then $\{r, s, z\}$ has exactly two edges, i.e. is an induced P_3 with all three pairs hard, contradicting pair-maximality; so $N[r] = N[s]$. If $rs \notin E$ and z is a common neighbour, then $r\,z\,s$ is an induced P_3 whose pairs rz, zs, rs all have an end in R, again a contradiction.

(1) Adjacent vertices of R have equal closed neighbourhoods; along a connected component all closed neighbourhoods coincide, and then any two vertices r, t of the component satisfy $r \in N[r] = N[t]$, so they are adjacent. (2) A neighbour of $x \in X$ lying in R is in the component of x, hence in X. (3) If $a \in A_X \cap A_Y$ for classes $X \neq Y$, then a is a common neighbour of some $x \in X$ and $y \in Y$, which are non-adjacent (different components of $G[R]$); this contradicts the second clause. □

For $x \in X$ and $a \in V(H)$ we thus have $xa \in E$ iff $a \in A_X$; in particular vertices of one class are twins towards $V(H)$ as well. (Lemma A.3 fails for vertex-maximal packings, see Proposition 4.5, whose instance also shows that Theorem A.5 fails without pair-maximality.)

Lemma A.4 (Exchange formula)**.** *Let Π be any partition, X a class with $X \subseteq C$ for a cluster C, $x = |X|$ and $a = |A_X \cap C|$. Let Π' be obtained by moving X out of C into a cluster of its own. Then $\mathrm{cost}(\Pi') - \mathrm{cost}(\Pi) = x\,(x + 2a - |C|)$.*

Proof. Only the pairs between X and $C \setminus X$ change status. By Lemma A.3, a vertex of X has exactly a neighbours in $C \setminus X$, namely $A_X \cap C$. Before the move such a pair is an edit iff it is a non-edge: $x(|C| - x - a)$ edits. After the move it is an edit iff it is an edge: xa edits. The difference is $xa - x(|C| - x - a) = x(x + 2a - |C|)$. □

Theorem A.5 (C1: no edit inside R)**.** *Let H be pair-maximal. In* every *optimum Π, every cluster meets R in the empty set or in exactly one whole class. Hence no edit of Π has both endpoints in R, and for every class X the cluster $\Pi(X)$ equals $X \cup T_X$ with $T_X := \Pi(X) \cap V(H)$.*

Proof. By Lemma A.3(1) and Lemma A.1, every class lies inside one cluster (a class of size 1 trivially). Suppose a cluster C contains two distinct classes X_1, X_2. Put $x_i = |X_i| \geq 1$ and $a_i = |A_{X_i} \cap C|$. The sets $X_1, X_2 \subseteq R$ and $A_{X_1} \cap C, A_{X_2} \cap C \subseteq V(H)$ are pairwise disjoint (Lemma A.3(2),(3)), so $|C| \geq x_1 + x_2 + a_1 + a_2$. Let Δ_i be the cost change of moving X_i out of C. Since Π is an optimum, $\Delta_1 \geq 0$ and $\Delta_2 \geq 0$, i.e. by Lemma A.4 $|C| \leq x_i + 2a_i$ for $i = 1, 2$. With the lower bound on $|C|$: $x_2 + a_2 \leq a_1$ and $x_1 + a_1 \leq a_2$. Adding gives $x_1 + x_2 \leq 0$, absurd. So no cluster contains two classes.

An edit with both ends in R joins two vertices r, s. If they lie in one class, they are adjacent and (by the above) in one cluster, so rs is no edit. If they lie in different classes they are non-adjacent, and rs is an edit iff they share a cluster, which was excluded. □

The strictness of the inequality is what makes the statement hold for every optimum (no ties, and no iteration or termination argument).

A.4 Consequences

Let (G, H, ℓ) be a *YES instance* with H pair-maximal, Π any optimum, S its edit set, so $|S| = \mathrm{OPT} \leq |H| + \ell$. Let U be the hard edits. By Lemma 2.2, $|S| - |H| = \sum_i (|x_i| - 1) + |U|$, so $|U| \leq \ell$. Write $q = \max_u |Q(u)| \leq 2c$.

Corollary A.6 (Normal form)**.** *For every class X let $T_X = \Pi(X) \cap V(H)$.*

(N1) Types of hard edits. *Every hard edit is either an* HH-edit *(both ends in $V(H)$) or an* RH-edit $\{x, a\}$, *$x \in X$ a class, $a \in V(H)$; the latter is a deletion if $a \in A_X \setminus T_X$ and an insertion if $a \in T_X \setminus A_X$, and $\{x, a\}$ is edited iff $\{x', a\}$ is, for all $x, x' \in X$. With h the number of HH-edits, $|U| = h + \sum_X |X| \cdot |A_X \triangle T_X| \leq \ell - \sum_i (|x_i| - 1) \leq \ell$.*

(N2) Frozen classes. *If $|X| > \ell$, then $\Pi(X) = X \cup A_X$.*

(N3) Attachment. *Every $a \in V(H)$ lies in A_X for at most one class X. If $z, z' \in A_X$ are non-adjacent then zz' is soft. If $a \in V(H) \setminus A_X$, $z \in A_X$ and $az \in E$ then az is soft; hence $|N(a) \cap A_X| \leq |Q(a)| \leq q$.*

(N4) Shrink rule. *If $|X| \geq \ell + 2$ and $x \in X$, then (G, H, ℓ) is a YES instance iff $(G - x, H, \ell)$ is. This holds for H pair-maximal without the YES hypothesis. If $A_X = \emptyset$ then X is a component of G and may be deleted without changing* OPT.

(N5) Size. *For a pair-maximal H and after exhausting the two rules in (N4), every class has at most $\ell + 1$ vertices and at most $|V(H)|$ classes have $A_X \neq \emptyset$, so $|R| \leq (\ell + 1)|V(H)|$ and $n \leq (\ell+2)|V(H)| \leq 3(\ell+2)|H|$. ($|V(H)|$ is not bounded in terms of ℓ, so this is not a kernel; see Remark A.8.)*

Proof. (N1) Pairs with an end in R are hard; by Theorem A.5 none has both ends in R. The vertices of X are twins towards $V(H)$ and lie in one cluster, which gives the all-or-none statement; xa is an edit iff exactly one of $a \in T_X$, $a \in A_X$ holds. Counting gives the identity; the bound is the first paragraph of this subsection.

(N2) If $\Pi(X) \neq X \cup A_X$ then, since $\Pi(X) = X \cup T_X$ (Theorem A.5), $A_X \triangle T_X \neq \emptyset$ and (N1) gives $|U| \geq |X| > \ell$, contradicting $|U| \leq \ell$.

(N3) The first claim is Lemma A.3(3). Let $x \in X$. If $z, z' \in A_X$ are non-adjacent, $z\,x\,z'$ is an induced P_3 with pairs zx, xz' hard, so zz' must be soft (pair-maximality). If $a \in V(H) \setminus A_X$ and $z \in A_X$ with $az \in E$, then $xa \notin E$ and $x\,z\,a$ is an induced P_3 with xz, xa hard, so az is soft, i.e. $z \in Q(a)$.

(N4) $x \in R$, so H is still a packing of $G - x$, still pair-maximal (an induced P_3 of $G - x$ is one of G), and $X \setminus \{x\}$ is a class of $G - x$ with the same A_X. ($\Rightarrow$) Restricting a partition of G to $V \setminus x$ does not increase its cost, so $\mathrm{OPT}(G-x) \leq \mathrm{OPT}(G) \leq |H|+\ell$. ($\Leftarrow$) Let Π be an optimum of $G-x$ with cost $\leq |H| + \ell$. Now $|X \setminus x| \geq \ell + 1 > \ell$, so (N2) in $G - x$ gives $\Pi(X \setminus x) = (X \setminus x) \cup A_X$. Add x to this cluster: $N[x] = X \cup A_X$, so every pair at x is satisfied, and the cost does not change. Hence $\mathrm{OPT}(G) \leq |H| + \ell$. If $A_X = \emptyset$, X is a clique component of G, which costs nothing, and H is disjoint from it.

(N5) After the rules each class has at most $\ell + 1$ vertices; the non-empty sets $A_X \subseteq V(H)$ are pairwise disjoint (Lemma A.3), and $|V(H)| \leq 3|H|$. Classes with $A_X = \emptyset$ were deleted. □

Remark A.7 (Tightness of the threshold in (N4)). For $\ell = 0$ the threshold $|X| \geq 2$ cannot be lowered to $|X| \geq 1$: take the 4-cycle of Proposition 4.6, $H = \{(0, 1, 3)\}$, $X = \{2\}$. Then $(G, H, 0)$ is a NO instance, but $G - 2$ is the induced P_3 0 1 3 itself, so $(G - 2, H, 0)$ is a YES instance. For $\ell = 1$ the threshold $|X| \geq \ell + 2 = 3$ cannot be lowered to $|X| \geq \ell + 1 = 2$: replace vertex 2 of the 4-cycle by two adjacent twins x, x', i.e. G has vertices $0, 1, 3, x, x'$ and edges $01, 13, 0x, 0x', 3x, 3x', xx'$, and $H = \{(0, 1, 3)\}$ (pair-maximal, $R = X = \{x, x'\}$). Then $\mathrm{OPT}(G) = 3$, so $(G, H, 1)$ is a NO instance, whereas $\mathrm{OPT}(G - x) = 2 = |H| + 1$, so $(G - x, H, 1)$ is a YES instance (both by enumerating all partitions). For $\ell = 2$ the same family with $|X| = 3$ does not separate ($\mathrm{OPT} = 3$ in both graphs). So the threshold is tight for $\ell \in \{0, 1\}$; tightness for $\ell \geq 2$ is open.

Remark A.8 (The rules and the input packing). The input packing H of CEaMP need not be pair-maximal. Extend it greedily (polynomial time) to a pair-maximal $H' \supseteq H$ with $t = |H'| - |H|$ new members, set $\ell' = \ell - t$ (answer NO if $\ell' < 0$, Lemma 2.4) and apply this section to (G, H', ℓ'). Then (N4), (N5) hold with ℓ' and H'; in particular $n \le 3(\ell+2)(|H|+\ell)$ after reduction, since $|H'| \le |H| + \ell$. But $\mathrm{tw}(I(H'))$ may exceed $\mathrm{tw}(I(H))$, so these rules are a preprocessing for the parameters $(\ell, |H'|)$ or $(\ell, \mathrm{tw}(I(H')))$, not for $(\ell, \mathrm{tw}(I(H)))$ with the given H; and $|V(H')|$ is not bounded by ℓ, so (N5) is not a kernel.

Proposition A.9 (Partners of edits touching R)**.** *In the setting of Corollary A.6, let X be a class, $T = T_X$. Then:*

(P1) Deletion. *If $\{x, a\}$ is an RH-deletion, $a \in A_X$, then X is determined by a (the unique class with $a \in A_X$): it is a unary flag at a of weight $|X|$. Moreover a is then cut off from every $z \in A_X \cap T$ with $z \notin Q(a)$, each such edge az being a hard HH-deletion; so at least $|A_X \cap T| - q$ HH-edits accompany it.*

(P2) No detachment. *If $T \cap A_X = \emptyset$ then $T = \emptyset$.*

(P3) Insertion. *If $a \in T \setminus A_X$ then $T \cap A_X \neq \emptyset$, and for every $z \in T \cap A_X$ either az is soft ($a \in Q(z)$, at most q choices of a per z) or az is a hard non-edge, hence an inserted HH-edit.*

(P4) Wide classes. *If $|A_X| \ge \ell + q$ then $T \subseteq A_X$.*

Consequently, every candidate partner $a \in V(H)$ of a class in an RH-edit is unary (P1), local to some z (P3, soft case) or tied to an HH-edit (P3, second case; P1). This classifies the partners; it does not assert that the HH-edits are independent of the RH-edits.

Proof. (P1) Uniqueness is (N3). Let $a \in A_X \setminus T$ and $z \in A_X \cap T$, $z \notin Q(a)$. If $az \notin E$ then az would be soft by (N3), contradicting $z \notin Q(a)$; so $az \in E$, hard, and $a \notin \Pi(X) \ni z$, so az is deleted. At most q vertices z lie in $Q(a)$.

(P2) Suppose $T \neq \emptyset = T \cap A_X$. The cluster is $C = X \cup T$ (Theorem A.5), $a_X = |A_X \cap C| = 0$, and by Lemma A.4 moving X out changes the cost by $|X|(|X| - |X| - |T|) = -|X||T| < 0$, contradicting optimality.

(P3) $T \neq \emptyset$, so $T \cap A_X \neq \emptyset$ by (P2). Let $z \in T \cap A_X$. If $az \in E$ then az is soft by (N3) ($a \in V(H) \setminus A_X$), i.e. $a \in Q(z)$. Otherwise either az is soft (again $a \in Q(z)$) or it is a hard non-edge inside the cluster $\Pi(X)$, an insertion.

(P4) Suppose $T \not\subseteq A_X$. Put $\alpha = |A_X \cap T|$, $\delta = |A_X \setminus T|$, $\beta = |T \setminus A_X| \ge 1$. By (N1) there are $|X|\beta$ RH-insertions and $|X|\delta$ RH-deletions. For $b \in T \setminus A_X$ and $z \in A_X \cap T$ with $z \notin Q(b)$, the pair bz is hard, lies in a cluster, and is a non-edge by (N3); so it is a hard insertion, an HH-edit. There are at least $\beta \max(0, \alpha - q)$ of them. Hence $\ell \ge |U| \ge \beta(|X| + \max(0, \alpha - q)) + |X|\delta$. If $\alpha \ge q$ this gives $\ell \ge 1 + (\alpha - q) + \delta$, i.e. $|A_X| = \alpha + \delta \le \ell + q - 1$. If $\alpha < q$ it gives $\delta \le \ell - 1$ and $|A_X| \le q - 1 + \ell - 1$. Either way $|A_X| < \ell + q$. □

B Counting lemmas for arbitrary packings

These lemmas hold for every packing H (no maximality unless stated) and describe what a solution of size at most $|H| + \ell$ can look like. All proofs use only pair-disjointness, a budget, and counting. They give neither a fixed-parameter algorithm in $\ell + w$ nor hardness; they are not used in the main text.

B.1 Hubs

In this subsection H is an arbitrary packing, $c = c(H)$, and (G, H, ℓ) is a YES instance where stated.

Lemma B.1 (HUB$^+$). *Let H be any packing, (G, H, ℓ) a YES instance, u a vertex and $N_h(u) = \{x : ux \in E,\ ux \text{ not soft}\}$. Let Λ_u be the graph on $N_h(u)$ whose edges are the pairs $\{x, y\}$ with $xy \notin E$ and xy not soft (hard non-edges). Then:*

(a) $\nu(\Lambda_u) \le \ell$ (matching number);

(b) Λ_u has a vertex cover T with $|T| \le 2\ell$, and every pair of $N_h(u) \setminus T$ is an edge of G or is soft;

(c) $\alpha(G[N_h(u)]) \le 2\ell + c + 1$ where $c = c(H)$.

Proof. (a) Let $x_1y_1, \ldots, x_ty_t$ be a matching of Λ_u and $P_i = (x_i, u, y_i)$. Since $ux_i, uy_i \in E$ and $x_iy_i \notin E$, P_i is an induced P_3. Its three pairs ux_i, uy_i, x_iy_i are hard. The $2t$ vertices x_i, y_i are distinct and different from u, so the pairs ux_i, uy_i ($i \le t$) are pairwise distinct, and the pairs x_iy_i are pairwise distinct and do not contain u. So $\{P_1, \ldots, P_t\}$ is hard over H and $t \le \ell$ by Lemma 2.3.

(b) The vertex set of a maximum matching of Λ_u is a vertex cover (by maximality) of size $2\nu \le 2\ell$. A pair of $N_h(u) \setminus T$ is not an edge of Λ_u, i.e. it is an edge of G or soft.

(c) Let J be independent in $G[N_h(u)]$. By (b), every pair of $J \setminus T$ is a non-edge of G and not an edge of Λ_u, hence soft. Put $J' = J \setminus T$. A soft pair of J' lies in a member P of H and is a non-edge, so it is the pair of the two ends of P (the other two pairs of P are edges); in particular $|V(P) \cap J'| \le 2$. If $J' \ne \emptyset$ fix $x \in J'$; every $y \in J' \setminus \{x\}$ gives a member $P_y \ni x, y$, and $P_y \ne P_{y'}$ for $y \ne y'$ because $|V(P_y) \cap J'| \le 2$. So $|J'| - 1 \le c$. Hence $|J| \le |T| + |J'| \le 2\ell + c + 1$. □

Corollary B.2 (HUB, hard-independent sets). *In the setting of Lemma B.1, if $Y \subseteq N_h(u)$ and every pair of Y is a hard non-edge, then $|Y| \le 2\ell + 1$.*

Proof. Y is a clique of Λ_u, so it contains a matching of size $\lfloor |Y|/2 \rfloor$, and $\lfloor |Y|/2 \rfloor \le \ell$ by Lemma B.1(a). □

Remark B.3 (HUB$^+$ versus an independence-number bound). (1) Bounding $\alpha(G[N_h(u)])$ through the corollary (greedy choice of a subset of an independent set that is also free of soft pairs) gives only $(c + 1)(2\ell + 1) = (D/2 + 1)(2\ell + 1)$, which is weaker than (c). (2) The matching form (a) is the relevant statement for Vertex Cover on the non-adjacency graph of $N(u)$: let $N_h(u)$ be two disjoint cliques of G of size m with no soft pair between them. Then $\alpha = 2$, but (a_i, u, b_i), a_i, b_i from different cliques, are m pair-disjoint induced P_3's, and $m \le \ell$ in a YES instance. Large independent sets are not the obstruction; large matchings of Λ_u are. (3) For a general H the lemma is applied to H' of Lemma 2.4 with c', ℓ'.

B.2 Pieces and small merges

Fix an arbitrary packing H and a solution S with $|S| \le |H| + \ell$ (neither optimality nor pair-maximality is used). Let $U = U^- \cup U^+$ be its hard part and $u = |U| \le |S| - |H| \le \ell$ (Lemma 2.2). The *pieces of S* are the connected components of the graph of kept hard edges, as in Section 9; every piece lies inside one cluster of $G \triangle S$.

Lemma B.4 (SM: small merge). *Let S, u and the pieces be as above, $D \ge \max_v |Q(v)|$ (e.g. $D = D(H)$), and let C be a cluster of $G \triangle S$.*

(i) If $A \ne B$ are pieces contained in C, then every pair ab ($a \in A, b \in B$) is soft or inserted.

(ii) For such A, B with $|A| \geq |B|$: $|B|\,(|A|-D) \leq |U^+ \cap (A \times B)| \leq u$, hence $|A| \leq D+u \leq D+\ell$.

(iii) If C is the union of at least two pieces, then $|C| \leq \max\{D+u+1,\ 2D,\ 2u\}$.

Proof. (i) is Lemma 9.1.

(ii) Each vertex of B has at most D soft partners, so at most $D|B|$ pairs of $A \times B$ are soft. By (i) all other pairs of $A \times B$, at least $|A||B| - D|B|$, are inserted, and $|U^+| \leq |U| = u$. If $|A| \geq D+u+1$ then $|B|(|A| - D) \geq u+1$ because $|B| \geq 1$, a contradiction; so $|A| \leq D+u$.

(iii) Let A be a largest piece in C, $a = |A|$ and $r = |C| - a \geq 1$. By (i), every pair $\{x, y\}$ with $x, y \in C$ in different pieces is soft or inserted.

Case $a > D$. Each $x \in C \setminus A$ has at most D soft partners, so at least $a - D \geq 1$ vertices $y \in A$ give an inserted pair xy. These pairs are distinct for distinct (x, y), so $r(a-D) \leq |U^+| \leq u$. Then $r \leq u$ and $a \leq D + u/r$, so $|C| = a + r \leq D + u/r + r$. The function $r \mapsto r + u/r$ is convex on $[1, u]$ and takes the value $u+1$ at both ends, so $|C| \leq D+u+1$. (The case does not occur if $u = 0$.)

Case $a \leq D$. All pieces of C have at most D vertices. Each $x \in C$ has at least $|C| - D$ vertices of C outside its piece, at most D of which are soft partners; so x lies in at least $|C| - 2D$ inserted pairs $\{x, y\}$ with $y \in C$. Summing over x counts each pair twice: $2|U^+| \geq |C|(|C| - 2D)$. If $|C| > 2D$, then $|C| - 2D \geq 1$ and $|C| \leq 2|U^+| \leq 2u$. So $|C| \leq \max\{2D, 2u\}$. □

Remark B.5 (Qualifiers and relation to the original bound). (1) The qualifier "pieces sharing a cluster" is essential: a single piece may be an entire cluster of unbounded size, and a statement "pieces have size at most $D+\ell$" is false. (2) The bound (iii) implies the weaker bound $|C| \leq \max\{2D+\ell, 2\ell\}$: $2u \leq 2\ell$ and $2D \leq 2D + \ell$ are clear; $D + u + 1 \leq 2D + \ell$ if $D \geq 1$; if $D = 0$ and $u \geq 1$ then $u + 1 \leq 2u \leq 2\ell$; and if $D = u = 0$ no cluster with two pieces exists (the two cases of the proof are both impossible). (3) The bound (iii) is attained for instance by two singleton pieces and one inserted pair ($D = 0$, $u = 1$, $|C| = 2$); we do not claim it is tight in general. (4) Lemma B.4 says nothing about the *degrees* in a graph of pieces; it bounds the sizes of pieces that share a cluster and the number of cross pairs that are not soft.

B.3 The soft graph has bounded treewidth

Recall the soft graph G_{soft} (Section 2).

Proposition B.6 (Treewidth of the soft graph)**.** $\operatorname{tw}(G_{\text{soft}}) \leq 3(w+1) - 1 = 3w + 2$ *if $H \neq \emptyset$; G_{soft} is edgeless if $H = \emptyset$. The bound is attained for $w = 0$ (H a single P_3, G_{soft} a triangle).*

Proof. Let $(\mathcal{T}, \{\beta_t\})$ be a tree decomposition of $I(H)$ of width w and $\mathcal{T}_P = \{t : P \in \beta_t\}$, a subtree of $\mathcal{T}$ for each $P \in H$. Put $\beta'_t = \bigcup_{P \in \beta_t} V(P)$, so $|\beta'_t| \leq 3(w+1)$. For every vertex $x \notin V(H)$ add a new leaf bag $\{x\}$ to an arbitrary node. *Edges:* a soft pair lies in $V(P)$ for some $P \in H$, and $P \in \beta_t$ for some t, so the pair lies in β'_t. *Connectivity:* for $x \in V(H)$ the nodes t with $x \in \beta'_t$ form $\bigcup_{P \ni x} \mathcal{T}_P$. The members through x pairwise share x, so they are pairwise adjacent in $I(H)$ and, the edge being covered by a bag, their subtrees $\mathcal{T}_P$ pairwise intersect. A union of connected subtrees that pairwise intersect is connected (two points of the union lie in subtrees $\mathcal{T}_P, T_{P'}$ that meet; join them through a common node). The new leaf bags are single bags of size 1. So this is a tree decomposition of G_{soft} of width at most $3(w+1) - 1$. □

B.4 Completing a fixed soft assignment

Proposition B.7. *Let x assign a subset $x_i \subseteq \operatorname{pairs}(P_i)$ to each P_i and let $b = |H| + \ell - \sum |x_i|$ be the excess budget left. Whether there is a set U of at most b hard pairs with $G \triangle (x \cup U)$ a cluster graph is decidable in $3^b n^{O(1)}$ time.*

Proof. Search tree on $G_x = G \triangle x$: if the current graph has an induced P_3, some pair of it must be flipped; soft pairs are frozen by x, so branch on its at most three hard pairs, reducing b by one (if all its pairs are soft, reject). Depth is at most b; flipping a pair twice is useless, so the search is exhaustive. □

The proposition is a reformulation, not progress towards fixed-parameter tractability in (ℓ, w): finding x is the whole problem, and the property "x is completable" is not local in x.

C The dynamic program for ω and the structure of Γ'

This appendix proves Theorem 8.3, using the notation $f(C) = \binom{|C|}{2} - 2e(C)$ of Lemma 7.1, and records two further statements on the graph Γ'. Throughout, H is pair-maximal.

C.1 Proof of Theorem 8.3

Theorem C.1 (Theorem 8.3 with bounds on m^*)**.** *Let H be pair-maximal and let ℓ, D, B, Z, L_B,* BIG*, V', Γ', ω be as in Section 8, and assume a tree decomposition of Γ' of width ω is given. Then one can compute in time $(Z(\omega+1))^{O(\omega)}n + n^{O(1)}$ a number m^* such that*

$$\mathrm{OPT}(G) \le |H| + \ell \iff |E| + \sum_K f(K) + m^* \le |H| + \ell,$$

where K ranges over the hard components of BIG *and f is as in Lemma 7.1. Moreover $m_{\mathrm{all}} \le m^* \le m_{\mathrm{conn}}$, where m_{all} (m_{conn}) is the minimum of $\sum_{C\in\pi'} f(C)$ over all partitions π' of V' with clusters of at most Z vertices (and, for m_{conn}, connected in Γ').*

Proof. *($\Leftarrow$)* Let π' be the partition of V' realising m^* (the dynamic program below produces a real partition) and $\pi = \pi' \cup \{K : K \text{ a component of BIG}\}$. By Lemma 7.1, $\mathrm{cost}(\pi) = |E| + \sum_K f(K) + m^* \ge \mathrm{OPT}$, so the right-hand side gives $\mathrm{OPT} \le |H| + \ell$. No structure of G is needed for this direction.

($\Rightarrow$) Let S be optimal with $|S| \le |H| + \ell$. By Lemma 7.6(γ) every component of BIG is a whole cluster; let π' be the restriction of S's partition to V'. Its clusters are Γ'-connected and have at most Z vertices:

- *Exceptional C.* $C \cap \mathrm{BIG} = \emptyset$ by (γ), so $C \subseteq V'$; $G[C]$ is connected by (α) and its edges are edges of Γ'. $|C| \le 2D + \ell + 1 \le Z$.
- *Non-exceptional C not in* BIG. By (β), C is a union of whole hard components, none of which is in BIG (a component of BIG is a cluster by itself), so $C \subseteq V'$. Each component is connected by hard edges, which are edges of Γ'. If there are at least two components, all pairs between them are soft, hence edges of Γ', so C is Γ'-connected, and $|C| \le 2D$. If there is exactly one component $K = C$, it is clean, so every vertex y of K has all other vertices of K as hard neighbours or soft partners, $|K| \le d_h(y) + D + 1 \le B + D + 1 = Z$ because $y \notin \mathrm{BIG}$ has $d_h(y) \le B$.

So π' is feasible for m_{conn} and $\mathrm{cost}(S) = |E| + \sum_K f(K) + \sum_{C\in\pi'} f(C) \ge |E| + \sum_K f(K) + m_{\mathrm{conn}} \ge |E| + \sum_K f(K) + m^*$.

The dynamic program. Take a nice tree decomposition of Γ' of width ω with empty root bag. A state at node t is a partition $\mathcal{P}$ of the bag X_t into blocks together with a size $s(\mathsf{b}) \in \{|\mathsf{b}|, \dots, Z\}$ for each block b (the number of vertices introduced so far, in the subtree of t, of the cluster containing

b). Its value is the minimum, over partitions of the introduced vertices into clusters that realise $(\mathcal{P}, s)$ (every cluster either contains a block of $\mathcal{P}$ or has been *closed*, i.e. has no vertex left in the bag), of $\sum_C g(C)$, with $g(C) = \binom{|C|}{2} - 2e(C)$ evaluated on the introduced vertices of C. *Introduce* v into the block b of size s (or into a new block of size 1): the value increases by $s - 2k$, where k is the number of G-neighbours of v in b; here we use that an introduced vertex y that has been forgotten cannot be a G-neighbour of v, because yv would be an edge of Γ' and y, v would share a bag. *Forget* v: if its block has other bag vertices the block shrinks; otherwise the cluster is closed and is never extended. *Join* of two children with the same partition $\mathcal{P}$: for a block with bag part b and sizes s_1, s_2 in the two children, put $x_1 = s_1 - |\mathsf{b}|$, $x_2 = s_2 - |\mathsf{b}|$; the new size is $s_1 + s_2 - |\mathsf{b}|$ and the value is $g_1 + g_2 - g(\mathsf{b}) + x_1 x_2$ summed over blocks, where $g(\mathsf{b}) = \binom{|\mathsf{b}|}{2} - 2e(\mathsf{b})$ is read off the bag. This is correct because pairs inside b and edges inside b are counted in both children, pairs between the two exclusive parts number $x_1 x_2$, and there is no G-edge between the exclusive parts (such an edge would be an edge of Γ' between vertices forgotten in different subtrees, which share no bag). *The join term $x_1 x_2$ and the subtraction of $g(\mathsf{b})$ are essential*: omitting them gives a different value on some inputs. At the root the value is $\sum_C f(C)$ over a partition of V' with clusters of size $\leq Z$. Call it m^* (the minimum over states; we treat the closed clusters as final).

Every state sequence yields a real partition of V' with cluster sizes $\leq Z$ and correct cost, so $m^* \geq m_{\text{all}}$. A Γ'-connected partition is represented: if the last bag vertex of a cluster C is forgotten at t, the vertex subtrees of C have a connected union (as C is Γ'-connected), which cannot contain the parent of t, so it lies below t and all of C has been introduced; hence no later vertex needs to join a closed cluster. So $m^* \leq m_{\text{conn}}$. There are at most $(\omega+1)^{\omega+1} Z^{\omega+1}$ states per node, a join combines at most $Z^{2(\omega+1)}$ pairs of states with equal partition, and the tree has $O(\omega n)$ nodes; the total time is $(Z(\omega+1))^{O(\omega)} n$. Computing d_h, BIG and Γ' takes $n^{O(1)}$.

Combining: OPT $\leq |H| + \ell$ implies $|E| + \sum_K f(K) + m^* \leq \text{cost}(S) \leq |H| + \ell$ by ($\Rightarrow$), and the converse is ($\Leftarrow$). □

Remark C.2. (1) Cleanliness of the components of BIG need not be tested: it is implied when the instance is a YES instance (Lemma 7.6(γ)), and the inequality itself accounts for the true cost $f(K)$; rejecting an unclean component is a harmless optimisation. (2) The proof uses that an exceptional cluster is disjoint from BIG and hence lies in V'. It is *not* true that $L_B \subseteq V'$: a component of BIG can contain vertices of small hard degree (pendant vertices of a hub component). (3) Here no pointer is guessed: far merges are carried by Γ'-connectivity. As for Theorem 8.2, D, B, Z, ℓ and ω refer to the pair-maximal instance (G, H', ℓ') of Lemma 2.4. Without a tree decomposition of Γ', one is computed in $2^{O(\omega)} n$ time with width $2\omega + 1$ [14].

C.2 Further statements

Lemma C.3 (Cost decomposition along a decomposition of the soft graph)**.** *Let π be a partition of V, let $(\mathcal{T}, \beta)$ be a tree decomposition of G_{soft} (Proposition B.6) and call a hard pair* local *if it lies in a common bag. Then* $\text{cost}(\pi) = \sum_t \text{loc}_t + \text{far}$*, where each edited soft pair and each edited local hard pair is charged to one bag containing it,* loc_t *depends only on* $\pi|_{\beta(t)}$*, and* far *is the number of edited non-local hard pairs, so* $\text{far} \leq |U|$.

Proof. Bookkeeping: all edited pairs are soft, or hard and local, or hard and non-local; soft pairs always lie in a bag. □

(The $\ell = 0$ CSP of Theorem 3.5 is the local part; all non-locality is in far.)

Proposition C.4 (Structure of Γ')**.** *Let (G, H, ℓ) be a YES instance with H pair-maximal.*

(a) $\Gamma' = G_{\text{soft}}[V'] \cup G_{\text{hard}}[V']$, *where* G_{hard} *is the graph of hard edges.*

(b) *Put* $\widehat{Z} = \max\{Z, (\ell+1)(2D+\ell+1)\}$. *Every hard component* K *not in* BIG *has* $|K| \le \widehat{Z}$. *So "reject if a non-*BIG *hard component has more than* $\widehat{Z}$ *vertices" is a safe polynomial rule.*

(c) *Let* $\Gamma'_{/K}$ *be* Γ' *with every hard component contracted. Then* $\operatorname{tw}(\Gamma'_{/K}) \le \operatorname{tw}(\Gamma') \le \widehat{Z}(\operatorname{tw}(\Gamma'_{/K})+1)-1$.

Proof. (a) Every edge of G is hard or soft, and soft pairs are edges of G_{soft}. (b) Take an optimal S with $|S| \le |H| + \ell$ and let U be its edited hard pairs. If no edge of K is deleted, K lies in one cluster C. If C is exceptional, $|C| \le 2D+\ell+1 \le Z$ (Lemma 7.6(α)). If not, C is a union of whole hard components (Lemma 7.6(β)); with several components $|C| \le 2D$, and a single component $K = C$ is clean, so each $y \in K$ has all other vertices of K as hard neighbours or soft partners and $|K| \le d_h(y)+D+1 \le B+D+1 = Z$ ($K \not\subseteq$ BIG). If edges of K are deleted, there are at most $\ell+1$ pieces of K (components of the kept hard edges); K is connected, so each piece contains an endpoint of a deleted edge and lies in an exceptional cluster of size $\le 2D+\ell+1$; thus $|K| \le (\ell+1)(2D+\ell+1)$. (c) Hard components are connected in Γ' by hard edges, so $\Gamma'_{/K}$ is a minor of Γ'. Conversely replace every node of a tree decomposition of $\Gamma'_{/K}$ by the vertices of its component (of size $\le \widehat{Z}$ by (b)). □

By Proposition B.6 the soft graph G_{soft} has treewidth $\le 3w+2$, so the unbounded treewidth of Γ' in Proposition 8.4 comes from hard edges between far vertices, which are free.

D Conflict weights, pruning and far pointers

This appendix concerns the two kinds of hard edit that the locality results of Section 9 leave open: inserted pairs of type T2 and deleted hard edges. It proves a *conflict lemma* that gives every soft pair a static weight, a defect bound and a polynomial NO-test that follow from it, bounded candidate lists, a pruned version of Theorem 8.3, and the statement that putting "locally bad" members into every bag does not repair the far pointers of Proposition 9.8 for distances 2 and 3. Nothing here proves or refutes fixed-parameter tractability in $\ell + w$.

Setting. H is pair-maximal (otherwise apply Lemma 2.4 and read the primed instance), $Q(v)$ is the set of soft partners, $N_h(v)$ the set of hard neighbours, $D \ge \max_v |Q(v)|$. A solution is identified with a partition π of V; write $x(p,q) = 1$ if p, q lie in one block and 0 otherwise; U is the set of edited hard pairs (so $|U| \le |S| - |H|$ by Lemma 2.2). For a soft pair $\sigma = (z, b)$ put

$$\Delta_{\neq}(z,b) = \{y \notin \{z,b\} : \ zy, by \text{ hard, exactly one an edge of } G\},$$

$$\Delta_{=}(z,b) = N_h(z) \cap N_h(b).$$

Both sets are computed from (G, H) alone. For a solution, the *weight* of the soft pair $\sigma = (z, b)$ is $\operatorname{wt}(\sigma) = |\Delta_{\neq}(z,b)|$ if z, b lie in one cluster (*merged*) and $\operatorname{wt}(\sigma) = |\Delta_{=}(z,b)|$ otherwise. A *conflict* is a pair (σ, y) with y in the relevant set.

D.1 The conflict lemma and its consequences

Lemma D.1 (W1: conflicts)**.** *Let* π *be* any *solution and* $\sigma = (z, b)$ *a soft pair.*

(a) *If* z, b *are in one cluster, every* $y \in \Delta_{\neq}(z,b)$ *has* $zy \in U$ *or* $by \in U$.

(b) *If* z, b *are in different clusters, every* $y \in \Delta_{=}(z,b)$ *has* $zy \in U$ *or* $by \in U$.

Distinct y give distinct edited pairs, hence $|U| \geq \mathrm{wt}(\sigma)$ for every soft pair σ.

Proof. (a) In a cluster graph, z, b in one cluster implies $x(z,y) = x(b,y)$ for all y. For $y \in \Delta_{\neq}(z,b)$ the G-values of the two hard pairs zy, by differ, so at least one of them is changed. (b) If neither zy nor by is edited, then $x(z,y) = x(b,y) = 1$ (both are edges of G), so z, b, y share a cluster, a contradiction. The edited pair chosen for y contains y and one of z, b, so different y give different pairs. □

Neither maximality nor optimality is used. Lemma 9.6 is the merged case of (a): for z, b in one cluster $\Theta(z,b) \subseteq \Delta_{\neq}(z,b)$, so $|\Theta(z,b)| \leq |U|$. (The hypothesis that z, b lie in one cluster is essential: for unmerged z, b the set $\Theta(z,b)$ can be large.)

Lemma D.2 (W2: defect bound)**.** *For every solution, $\sum_{\sigma \text{ soft}} \mathrm{wt}(\sigma) \leq 2D\,|U|$.*

Proof. Map each conflict $((z,b),y)$ to an edited pair of $\{zy, by\}$ (Lemma D.1). An edited pair $e = \{p, y\}$ is chosen by conflicts (σ, y) with $\sigma \ni p$, at most $|Q(p)| \leq D$ of them, and by conflicts (σ', p) with $\sigma' \ni y$, at most D of them. □

The constant $2D$ is attained for $D = 2$ ($c = 1$) by instances with $n = 6, 7$ found by computer search; for $D = 4$ and $D = 6$ the same search reached only $1/2$ and $1/3$ of the bound, so attainment for $D \geq 4$ is open.

Corollary D.3 (LB: a lower bound from the weights)**.** *Let $\kappa_i \geq 1$ be the number of edited pairs among* pairs(P_i) *and let* $\mathrm{Wt} = \sum_\sigma \mathrm{wt}(\sigma)$ *be the total weight of a solution π. Then* $\mathrm{cost}(\pi) \geq \sum_i \kappa_i + \lceil \mathrm{Wt}/2D \rceil$.

Proof. $\mathrm{cost}(\pi) = \sum_i \kappa_i + |U|$ (Lemma 2.2) and $|U| \geq \mathrm{Wt}/2D$ by Lemma D.2. □

The right-hand side is not exact: minimised over assignments of the soft pairs that are transitive on all-soft triples, it can equal $|H|$ although $\mathrm{OPT} > |H|$ (found by computer search on small random instances; the triples with two soft pairs and one hard pair are not checked). We only use it as a lower bound.

Lemma D.4 (Static pruning)**.** *Let $|S| \leq |H| + \ell$. If $|\Delta_{\neq}(u,v)| > \ell$, the soft pair (u,v) is split in S; if $|\Delta_{=}(u,v)| > \ell$ it is merged in S.*

Proof. Lemma D.1 and $|U| \leq |S| - |H| \leq \ell$. □

Lemma D.5 (DEL: rigid hard edges)**.** *If zy is a deleted hard edge of a solution, then $|U| \geq 1 + |N_h(z) \cap N_h(y)|$. Hence, if $|S| \leq |H| + \ell$, a hard edge with at least ℓ common hard neighbours is never deleted (it is* rigid*), and the hard components of rigid edges lie inside single clusters.*

Proof. z and y are in different clusters. A common hard neighbour w cannot be kept with both, so wz or wy is a deleted hard edge; distinct w give distinct pairs (each contains w), all different from zy. □

A member $P = (x, y, z)$ with centre y has three states of member-cost one: delete xy, delete yz, insert xz. A state fixes whether each of the three pairs of P is merged, hence a weight for each. Call P *locally bad* if in each of its three states some pair of P has positive weight. This is a static property.

Lemma D.6 (BAD: few locally bad members)**.** *For every solution of size $|H| + \varepsilon$ the number of locally bad members is at most $2c\varepsilon$. Hence more than $2c\ell$ locally bad members is a NO-instance of CEaMP with budget ℓ (a polynomial test).*

Proof. By Lemma 2.2, $\varepsilon = \sum_i(\kappa_i - 1) + |U|$. A bad member with $\kappa_i \geq 2$ contributes at least 1 to the first sum. A bad member with $\kappa_i = 1$ is in one of its three cost-one states, which has a conflict (σ, y) with σ a pair of P_i; by Lemma D.1 an edit of U is incident to $V(P_i)$. An edit has two ends and each vertex lies in at most c members, so there are at most $2c|U|$ such members. In total at most $\sum_i(\kappa_i - 1) + 2c|U| \leq 2c\varepsilon$. □

Proposition D.7 (Candidate lists for (T2) pairs and deleted edges)**.** *Let H be pair-maximal, S an* optimal *solution with $|S| \leq |H| + \ell$, and $ab \in U$. Define, from (G, H) alone,*

$$L_1(a) = \begin{cases} N_h(a) \cup \bigcup_{z \in N_h(a)} Q(z) & \text{if } d_h(a) \leq D + \ell, \\ \emptyset & \text{otherwise,} \end{cases} \qquad L_2(b) = \bigcup_{z \in Q(b),\ |\Delta_{\neq}(z,b)| \leq \ell} \Delta_{\neq}(z, b).$$

Then $|L_1(a)| \leq (D + \ell)(D + 1)$ and $|L_2(b)| \leq D\ell$, and:

(i) if ab is a deleted hard edge, then $b \in L_1(a)$ and $a \in L_1(b)$;

(ii) if ab is an inserted pair of type (T2) with middle vertex z, az hard (kept) and zb soft (Theorem 9.2), then $b \in L_1(a)$ and $a \in L_2(b)$.

Pairs of type (T1) are not covered; they are bag-local by Proposition 9.5.

Proof. Every endpoint x of a pair of U has $d_h(x) \leq D + |U| \leq D + \ell$ (Lemma 7.4(a)), so the guard in L_1 holds for the ends. (i) The deleted pair is a hard edge, so $b \in N_h(a)$, and symmetrically. (ii) $z \in N_h(a)$ and $b \in Q(z)$, so $b \in L_1(a)$, and $|L_1(a)| \leq (D + \ell) + (D + \ell)D$. For the soft end: $z \in Q(b)$ ($\leq D$ choices); since az is kept and ab is inserted, a, z, b lie in one cluster, so zb is merged, and az is a hard edge while ab is a hard non-edge, so $a \in \Delta_{\neq}(z, b)$, and $|\Delta_{\neq}(z, b)| \leq |U| \leq \ell$ by Lemma D.1(a). Hence $a \in L_2(b)$ and $|L_2(b)| \leq D\ell$. □

Optimality is used only through Theorem 9.2. The lists say where the partner of a given end can be; they do not say where the first end is (Corollary 7.7).

D.2 A pruned form of Theorem C.1

Let BIG, V', Z be as in Theorem C.1 and let Γ^ℓ be the graph on V' whose edges are all hard edges inside V' and all soft pairs (u, v) inside V' with $|\Delta_{\neq}(u, v)| \leq \ell$. It is a spanning subgraph of Γ', so $\mathrm{tw}(\Gamma^\ell) \leq \mathrm{tw}(\Gamma')$, and it is computable from (G, H) alone.

Theorem D.8 (P: pruned ω-theorem)**.** *Let H be pair-maximal and let $\omega_\ell = \mathrm{tw}(\Gamma^\ell)$, with a tree decomposition of that width given. Then one can compute in time $(Z(\omega_\ell+1))^{O(\omega_\ell)}n+n^{O(1)}$ a number m^* with*

$$\mathrm{OPT}(G) \leq |H| + \ell \iff |E| + \sum_K f(K) + m^* \leq |H| + \ell,$$

K ranging over the hard components of BIG*. So CEaMP is FPT in $(\ell, D, \mathrm{tw}(\Gamma^\ell))$.*

Proof. Let $e^+(C)$ be the number of edges of G inside C that are edges of Γ^ℓ, and $\tilde{f}(C) = \binom{|C|}{2} - 2e^+(C)$. Run the dynamic program of the proof of Theorem C.1 on Γ^ℓ, with e^+ in place of e everywhere (in the introduce step k counts the Γ^ℓ-neighbours of v in the block, and in the join step there is no Γ^ℓ-edge between the exclusive parts). Exactly as there, it computes $m^* =$ the minimum of $\sum_C \tilde{f}(C)$ over Γ^ℓ-connected partitions of V' with clusters of at most Z vertices, in the stated time.

($\Leftarrow$) $e^+ \le e$, so $f(C) \le \tilde{f}(C)$ and the partition realising m^*, completed by the components of BIG, has true cost at most $|E| + \sum_K f(K) + m^*$ (Lemma 7.1); no structure is needed.

($\Rightarrow$) Let S be optimal, $|S| \le |H| + \ell$. By the proof of Theorem C.1, the clusters of S inside V' have at most Z vertices and are connected through edges of G inside the cluster and (for a cluster with several hard components) soft pairs inside the cluster. A soft pair inside a cluster is merged, so $|\Delta_{\neq}| \le \ell$ by Lemma D.4 and it is an edge of Γ^ℓ; every hard edge is an edge of Γ^ℓ. Hence these clusters are Γ^ℓ-connected and $e^+(C) = e(C)$, $\tilde{f}(C) = f(C)$. So $\mathrm{cost}(S) = |E| + \sum_K f(K) + \sum_C f(C) \ge |E| + \sum_K f(K) + m^*$. □

Theorem D.8 does not use optimality of the merges, only their cost, and it therefore cannot decide the open question.

Proposition D.9 (Theorem D.8 is strictly stronger than Theorem 8.3)**.** *Let F be a cubic graph with an orientation in which every in-degree is 1 or 2 (so every vertex has in-degree ≥ 1; for a 2-edge-connected F such an orientation exists by Robbins' theorem [16]), and let (G_F, H_F) be the instance of Proposition 8.4. For every budget $\ell \le 4$, every component of Γ^ℓ has at most 5 vertices (so $\mathrm{tw}(\Gamma^\ell) \le 4$), whereas $\mathrm{tw}(\Gamma') \ge \mathrm{tw}(F)$.*

Proof. For the oriented edge $e = (p \to q)$ the soft pairs $a_e b_e$ and $a_e c_e$ join $\mathcal{K}_p$ and $\mathcal{K}_q$. The soft partners of a_e are b_e, c_e and those of b_e are a_e, c_e (the three pairs of P_e). Hence $\Delta_{\neq}(a_e, b_e)$ consists of the vertices $y \in \mathcal{K}_p \setminus \{a_e\}$ (the pair $a_e y$ is a hard edge and $b_e y$ a hard non-edge), of which there are $2 + \mathrm{in}(p)$, and of the vertices $y \in \mathcal{K}_q \setminus \{b_e, c_e\}$ (here $b_e y$ is a hard edge and $a_e y$ a hard non-edge), of which there are $1 + \mathrm{in}(q)$. So $|\Delta_{\neq}(a_e, b_e)| = 3 + \mathrm{in}(p) + \mathrm{in}(q) \ge 5$, and the same count holds for $a_e c_e$. For $\ell \le 4$ both cross pairs are pruned, the only edges of Γ^ℓ are those inside the cliques $\mathcal{K}_v$, and $|\mathcal{K}_v| = 3 + \mathrm{in}(v) \le 5$. The bound on $\mathrm{tw}(\Gamma')$ is Proposition 8.4. □

The in-degree condition matters: with a source p (in-degree 0) the weight can be 4.

D.3 Far pointers and locally bad members

An edited pair $e = (a, b)$ of a solution S is *two-sided* if there are vertices t, t' with (t, a) soft, (t', b) soft, and $\{a, b, t\}$, $\{a, b, t'\}$ induced P_3's of $G \triangle (S \setminus \{e\})$. It is *far* (at distance d) if the members containing a and b are at distance $\ge d$ in $I(H)$.

Proposition D.10 (NF-bad fails for distance 2 and 3)**.** *The following statement is false for "far" meaning distance ≥ 2 and for "far" meaning distance ≥ 3 in $I(H)$:* there is an optimum in which every far two-sided edit touches a locally bad member.

Witnesses. Both instances have $n = 10$, a pair-maximal modification-disjoint packing, a unique optimum, $\mathrm{OPT} = |H| + 1 = 6$, no locally bad member, and exactly one edit, a hard deleted edge, which is two-sided (found by enumerating all set partitions). *Distance 2:* $E = \{01, 06, 12, 23, 29, 34, 45, 56, 67, 78, 79\}$, $H = \{(0,1,2), (2,3,4), (4,5,6), (6,7,8), (2,9,7)\}$; the optimum $\{0,1\}, \{2,9\}, \{3,4\}, \{5,6\}, \{7,8\}$ deletes $\{0,6\}$ (witnesses 1 and 5). *Distance 3:* $E = \{01, 08, 12, 14, 23, 34, 45, 49, 56, 67, 78\}$, $H = \{(0,1,2), (2,3,4), (4,5,6), (6,7,8), (1,4,9)\}$; the optimum $\{0,1\}, \{2,3\}, \{4,9\}, \{5,6\}, \{7,8\}$ deletes $\{0,8\}$ (witnesses 1 and 7). In both, $I(H)$ has $c = 3$. □

Remark D.11 (Scope). Pointers at distance $\le K$ are bag-local in a decomposition of $I(H)^K$, whose width is a function of (w, c, K), so the version of the statement that matters is the one with "far" larger than any function of (ℓ, w). Proposition D.10 does not refute it, and Proposition 9.8 does not either: the far ends of its edit are exactly the two locally bad members (for $m = 3, 4, 5$ these are the only locally bad members, by enumeration). So the idea "put the locally bad members into every bag" is excluded for distances 2 and 3 only; for unbounded distance it is open.

E Reduction on unions of members fails: the spine family

Theorem 6.10 applies Rule 6.2 of [17] to components of $I(H)$. This appendix shows that the natural generalisation to pair-closed unions of *members* (Section 6) cannot prove the reduction lemma (Open Problem 5), already for $w = 1$ and $\ell = 1$. Soundness of the rule is cited from [17]; the proposition itself is proved here, and does not use the rule. Throughout, Γ', BIG, B, Z are as in Section 8; a set $\mathcal{F}$ of members is pair-closed, reducible and irreducible as in Section 6.

Proposition E.1 (The spine family $\mathrm{Sp}(F)$)**.** *Let F be a cubic graph with an orientation of in-degree at most 2 at every vertex. Let F' be F plus a new vertex $*$ and an edge $e_0 = (t \to *)$, $t \in V(F)$. Order the edges of F' as $e_0, e_1, \ldots, e_N$ so that e_0 comes first,* consecutive edges have different heads and $\mathrm{head}(e_1) \neq \mathrm{tail}(e_0) = t$ *(possible, since every head occurs at most twice and t has in-degree at most 2). Write $h(i)$ and $\mathrm{t}(i)$ for the head and tail of e_i. The instance $\mathrm{Sp}(F)$ has vertices a_i, b_i, c_i, u_i $(0 \le i \le N)$, z and y; for $v \in V(F')$ the sets $\mathcal{K}^+_v = \{a_i : \mathrm{t}(i) = v\} \cup \{b_i, c_i, u_i : h(i) = v\}$ (so $\mathcal{K}^+_* = \{b_0, c_0, u_0\}$). Edges: all pairs inside each $\mathcal{K}^+_v$; the cross edges $a_i b_i$; the spine edges $b_i u_{i+1}$ $(i < N)$ and $b_N z$; and $y u_0$, $y c_0$ (and no other edge at y). Packing: $P_i = (a_i, b_i, c_i)$ and $X_i = (u_i, b_i, u_{i+1})$, both with centre b_i $(u_{N+1} := z)$; $\ell = 1$. Then:*

(S1) the members are induced P_3's with pairwise disjoint pair sets, H is pair-maximal, $I(H)$ is a caterpillar (spine $X_0, \ldots, X_N$, leaf P_i at X_i), so $w = 1$, $c = 2$, $D = 4$;

(S2) $(G, H, 1)$ is a YES instance and $\mathrm{OPT} = |H| + 1$;

(S3) all hard degrees are at most $7 \le B = 11$, so $\mathrm{BIG} = \emptyset$;

(S4) F is a minor of Γ', so $\mathrm{tw}(\Gamma') \ge \mathrm{tw}(F)$, which is $\Omega(|V(F)|)$ for cubic expanders [9] (cited);

(S5) $I(H)$ is connected and tight: $\tau = |H|$;

(S6) every nonempty pair-closed set $\mathcal{F}$ of members (connected or not) is irreducible.

Proof. (S1) Every induced P_3 of G uses a cross edge or a spine edge or a y-edge, because two edges inside cliques at a common vertex lie in one clique. P_i is induced: $a_i b_i$ is a cross edge, $b_i c_i$ lies in $\mathcal{K}^+_{h(i)}$, $a_i c_i \notin E$ because $\mathrm{t}(i) \neq h(i)$. X_i is induced: $u_i b_i$ lies in $\mathcal{K}^+_{h(i)}$, $b_i u_{i+1}$ is a spine edge, and $u_i u_{i+1} \notin E$ since consecutive heads differ (z is adjacent to b_N only). The pair sets $\{a_i b_i, b_i c_i, a_i c_i\}$, $\{u_i b_i, b_i u_{i+1}, u_i u_{i+1}\}$ are pairwise disjoint. Pair-maximality: no two hard edges form an induced P_3. The soft edges are $a_i b_i$, $b_i c_i$, $u_i b_i$, $b_i u_{i+1}$, so the hard edges are the other clique edges and $y u_0$, $y c_0$. Two hard clique edges at a vertex lie in one clique, so their other ends are adjacent; the hard edges at y go to u_0, c_0, which are adjacent; and a hard edge at u_0 or c_0 inside $\mathcal{K}^+_* = \{b_0, c_0, u_0\}$ can only be $u_0 c_0$, whose ends are both adjacent to y.

(S2) The partition $\{\mathcal{K}^+_v\}$ together with $\{z\}$, with y put into $\mathcal{K}^+_*$, deletes exactly the $2(N + 1) = |H|$ cross and spine edges (all soft) and inserts the single pair $y b_0$; its cost is $|H| + 1$. For $\mathrm{OPT} \ge |H| + 1$ see (S6) below.

(S3) A vertex of $\mathcal{K}^+_v$ has at most $|\mathcal{K}^+_v| - 1 = 2 + 2\,\mathrm{in}(v) \le 6$ hard neighbours inside its clique, and u_0, c_0 have one more hard neighbour, y (which has hard degree 2).

(S4) Contract every $\mathcal{K}^+_v$ (a clique). The cross edges $a_i b_i$ join $\mathcal{K}^+_{\mathrm{t}(i)}$ and $\mathcal{K}^+_{h(i)}$, so the quotient contains F.

(S5) In the partition of (S2), the pair $a_i b_i$ is deleted, $b_i c_i$ kept, $a_i c_i$ a non-edge in different clusters; $u_i b_i$ kept, $b_i u_{i+1}$ deleted, $u_i u_{i+1}$ a non-edge in different clusters. So its restriction to $V^- = V \setminus \{y\}$ edits exactly one pair per member: $\tau \le |H|$, hence $\tau = |H|$.

(S6) For pair-closed $\mathcal{F}$ the restriction of the solution of (S2) to $W_{\mathcal{F}}$ has at most $|\mathcal{F}|$ edits (no member outside $\mathcal{F}$ has a pair inside $W_{\mathcal{F}}$, and yb_0 leaves $W_{\mathcal{F}}$), so $\tau(G[W_{\mathcal{F}}]) = |\mathcal{F}|$ and the optimal $S_{\mathcal{F}}$ are the *tight* ones: exactly one edited pair per member of $\mathcal{F}$ and no hard pair is edited. Suppose such an $S_{\mathcal{F}}$ satisfied the boundary condition. We use repeatedly: two vertices $p, q \in W_{\mathcal{F}}$ joined by a hard non-edge are non-adjacent in $G[W_{\mathcal{F}}] \triangle S_{\mathcal{F}}$, and a hard edge is kept.

Case 1: $\mathcal{F}$ contains a spine member; let X_i be the one of smallest index.

$i \geq 1$. Then $X_{i-1} \notin \mathcal{F}$, and $P_{i-1} \notin \mathcal{F}$: otherwise $b_{i-1}, u_i \in W_{\mathcal{F}}$ and the pair $b_{i-1}u_i$ of the outside member X_{i-1} would lie inside $W_{\mathcal{F}}$. So $a_{i-1}, b_{i-1}, u_{i-1} \notin W_{\mathcal{F}}$ and $N(b_{i-1}) \cap W_{\mathcal{F}} = \{u_i\} \cup Y$ with $Y = \mathcal{K}^+_{h(i-1)} \cap W_{\mathcal{F}}$. For $m \in Y$ the pair $u_i m$ is a non-edge ($h(i) \neq h(i-1)$) and is soft only if $m = u_{i+1}$. If Y contains some $m \neq u_{i+1}$, then u_i, m are two non-adjacent vertices of $N(b_{i-1}) \cap W_{\mathcal{F}}$: the boundary condition fails. If $Y = \{u_{i+1}\}$, the set $\{u_i, u_{i+1}\}$ must be a cluster, so X_i inserts $u_i u_{i+1}$ and keeps $u_i b_i$, $b_i u_{i+1}$; then b_i lies in this cluster, but $b_i \notin N(b_{i-1})$: contradiction. So $Y = \emptyset$ and $\{u_i\}$ is a cluster, i.e. X_i deletes $u_i b_i$. If $P_i \notin \mathcal{F}$, the vertex $c_i \notin W_{\mathcal{F}}$ is adjacent to u_i and b_i, which lie in different clusters: the boundary condition fails at c_i. If $P_i \in \mathcal{F}$, the hard edge $u_i c_i$ is kept, contradicting that u_i is a singleton.

$i = 0$. The vertex $y \notin W_{\mathcal{F}}$ has $N(y) \cap W_{\mathcal{F}} = \{u_0\}$ if $P_0 \notin \mathcal{F}$ and $\{u_0, c_0\}$ if $P_0 \in \mathcal{F}$. First case: $\{u_0\}$ is a cluster, so X_0 deletes $u_0 b_0$, and $c_0 \notin W_{\mathcal{F}}$, adjacent to $u_0, b_0 \in W_{\mathcal{F}}$ in different clusters, violates the condition. Second case: $\{u_0, c_0\}$ must be a cluster, so b_0 is separated from both: X_0 deletes $u_0 b_0$ and P_0 deletes $b_0 c_0$. Then $a_0 b_0$ and $b_0 u_1$ are kept, so a_0, u_1 are adjacent to b_0. Now $a_0 \in \mathcal{K}^+_t$ and $u_1 \in \mathcal{K}^+_{h(1)}$ with $h(1) \neq t$ *by the ordering condition*, so $a_0 u_1$ is a hard non-edge (the soft partners of a_0 are b_0, c_0), and $a_0 b_0 u_1$ is an induced P_3 of $G[W_{\mathcal{F}}] \triangle S_{\mathcal{F}}$: contradiction.

Case 2: $\mathcal{F}$ contains only leaves. Take $P_i \in \mathcal{F}$; $u_i, u_{i+1} \notin W_{\mathcal{F}}$ (for $i = N$ read z for u_{i+1}). If $i = N$ then $N(z) \cap W_{\mathcal{F}} = \{b_N\}$ must be a cluster, i.e. b_N is a singleton; but each of the three single edits of P_N (delete $a_N b_N$, delete $b_N c_N$, insert $a_N c_N$) keeps b_N with a_N or c_N. So let $i < N$. Then $N(u_{i+1}) \cap W_{\mathcal{F}} = \{b_i\} \cup Y$ with $Y = \mathcal{K}^+_{h(i+1)} \cap W_{\mathcal{F}}$. For $m \in Y$ the pair $b_i m$ is a non-edge ($h(i+1) \neq h(i)$) unless $m = a_i$ (possible when $\mathrm{t}(i) = h(i+1)$), and it is soft only if $m = a_i$ (the soft partners of b_i are a_i, c_i, u_i, u_{i+1}). *(2a)* If Y contains some $m \neq a_i$, then b_i, m are non-adjacent in $N(u_{i+1}) \cap W_{\mathcal{F}}$: failure. *(2b)* If $Y = \emptyset$, then $\{b_i\}$ is a cluster, impossible as above. *(2c)* If $Y = \{a_i\}$, then $P_{i+1} \notin \mathcal{F}$ and $b_{i+1} \in \mathcal{K}^+_{h(i+1)}$ is outside $W_{\mathcal{F}}$ with $N(b_{i+1}) \cap W_{\mathcal{F}} = \{a_i\}$; this must be a cluster, so a_i is a singleton and $a_i b_i$ is deleted. But then $\{a_i, b_i\} = N(u_{i+1}) \cap W_{\mathcal{F}}$ is not a cluster ($a_i b_i$ is not an edge of $G \triangle S_{\mathcal{F}}$): contradiction.

So $\mathcal{F}$ is irreducible. For the lower bound of (S2): if OPT $= |H|$, an optimum S has $|S| = \tau$ for the whole component $\mathcal{C} = I(H)$, so $S = S_{\mathcal{C}}$ is tight and optimal on $V^- = V \setminus \{y\}$; by (S6) it violates the boundary condition at y, and Lemma 6.5 gives an edited pair $\{y, u\}$, which contradicts $S = S_{\mathcal{C}}$. So OPT $\geq |H| + 1$. □

Remark E.2 (Finite checks). Claims (S1)–(S5) and OPT $= |H| + 1$ were verified by computer for F the triangle ($n = 18$, OPT by an exact subset dynamic program) and for K_4 ($n = 30$). Claim (S6) was checked, in agreement with the proof, for the triangle on all 90 pair-closed subfamilies with $|\mathcal{F}| \leq 6$ and on the whole component, and for K_4 on all 632 pair-closed subfamilies with $|\mathcal{F}| \leq 4$ for 8 random admissible orderings. The proof above is the justification.

Consequence: an obstruction, not a refutation of the reduction lemma. No reduction obtained by iterating Rule 6.2 on pair-closed unions of members (the form used for components in Section 6) can prove Open Problem 5 by itself: on $\mathrm{Sp}(F)$ it removes nothing, although $\ell = 1$, $w = 1$, BIG $= \emptyset$ and $\mathrm{tw}(\Gamma') = \Omega(|V(F)|)$. In particular the counting argument of Lemma 6.6 does not extend from components to members: all $2(N+1)$ members are irreducible although $\ell = 1$. The

mechanism is Lemma 6.5: along the spine each member is made irreducible by an edit at a shared vertex of its neighbour, down to the trigger y. Other generalisations of the rule (to vertex sets that are not unions of members) are not excluded. Repacking, by contrast, removes the obstruction: Proposition 11.11 gives a packing H' of size $|H|+1 = \mathrm{OPT}$ with $\mathrm{tw}(I(H')) \le 2$, and also a two-line proof of $\mathrm{OPT} \ge |H|+1$.

F Tunnels: locks, memory and a restricted model

The natural route to W[1]-hardness in $\ell + w$ is a reduction from Grid Tiling in which hard edges between far-apart vertices (*tunnels*) carry equalities between distant parts of $I(H)$ of bounded width. This appendix proves what a tunnel can and cannot do: the lock lemmas used by the reduction of Section 5 (Lemmas F.1, F.2, F.3), the cores and the splice lemmas, and the *restricted* tunnel-release model, in which, without soft triangles, tunnels cannot carry the reduction (Theorem 10.28). The reduction of Section 5 shows that with soft triangles they can; the results of this appendix about the restricted model are not used in its proof, which does not assume the restricted model.

F.1 Zero-excess solutions and tunnels

Soft and hard pairs and $Q(v)$ are as in Section 2. A solution S is *zero-excess* (ZE) if $|S| = |H|$. A member (a, b, c) (centre b) is in state 1, 2 or 3 if its edited pair is ab, bc or ac. By Lemma 2.2 a ZE solution has exactly one edit in each member and no other edit, so it is determined by its vector of states; let $\Sigma = \Sigma(G, H) \subseteq \{1, 2, 3\}^H$ be the set of these vectors. None of the lemmas of this section except Lemma F.7 needs pair-maximality.

Lemma F.1 (Zero-excess structure)**.** *Let S be a ZE solution. (a) Every hard edge lies inside a cluster and every hard non-edge joins two clusters. (b) Every member is in one of the states $1, 2, 3$. (c) Every hard component lies in one cluster, and two vertices of one cluster in different hard components form a soft pair.*

Proof. (a),(b) are Lemma 2.2 with $|S| = |H|$: no hard pair is edited and each member has exactly one edit. A member (a, b, c) with togetherness pattern (t_{ab}, t_{bc}, t_{ac}) has the cluster patterns $000, 100, 010, 111$ of costs $2, 1, 1, 1$, so the one-edit patterns are the states $2, 1, 3$. (c) A hard edge is unedited, so its ends are together; two vertices of one cluster in different hard components are not joined by a hard edge, and a hard non-edge inside a cluster would be edited, so the pair is soft. □

A *tunnel* is a hard edge xx' with $N_h(x) = \{x'\}$, $N_h(x') = \{x\}$ and $Q(x) \cap Q(x') = \emptyset$; tunnels are always taken pairwise vertex-disjoint. A *stub* at x is a new vertex x' adjacent to x only (then $Q(x') = \emptyset$). A vertex is an *anchor* if it lies in a member and has no hard edge. *Cutting* a tunnel means editing xx' (one hard edit).

Lemma F.2 (A tunnel is a lock)**.** *Let xx' be a tunnel and S a solution in which all hard pairs at x and at x' are unedited. Then the cluster of x is $\{x, x'\}$. If S is ZE, then no member has x as centre, every member through x has x as an end with the pair at x that is an edge deleted, and the other two vertices of that member are together.*

Proof. Let $t \ne x, x'$ be in the cluster of x. If $t \notin Q(x)$ then xt is hard, hence a non-edge as $N_h(x) = \{x'\}$, hence edited, a contradiction; so $t \in Q(x)$, and likewise $t \in Q(x')$, contradicting $Q(x) \cap Q(x') = \emptyset$. In a ZE solution a member (y, x, z) with centre x would have both pairs at x apart, hence two edits. If x is an end, the pair x–centre is apart (an edit), the non-edge at x is apart (no edit), so the remaining pair is together. □

Lemma F.3 (Anchors: singleton = forced states; inactive anchors cost nothing)**.** *Let x be an anchor and σ a ZE solution of (G, H). Then x is a singleton of σ iff every member through x has x as an end and is in the state that deletes the edge at x. Moreover: (a) if G' is G with a stub at x, then G' has a ZE solution iff G has a ZE solution in which x is a singleton; (b) if x_1, x_2 are anchors that lie in no common member, are non-adjacent and have $Q(x_1) \cap Q(x_2) = \emptyset$, and G' is G plus the edge x_1x_2, then G' has a ZE solution iff G has a ZE solution in which x_1 and x_2 are both singletons.*

Proof. The first statement: ($\Rightarrow$) as in Lemma F.2; ($\Leftarrow$) the soft partners of x are its mates in members, and these are apart from x; x has no hard edge, and its hard non-edges are unedited. (a),(b) ($\Rightarrow$): by Lemma F.2 the cluster is $\{x, x'\}$ (resp. $\{x_1, x_2\}$); split it into singletons. No other pair changes status or togetherness. ($\Leftarrow$) join the singleton to its stub (resp. join the two singletons). □

Thus an anchor costs nothing when it is a singleton in some ZE solution, and the point of a tunnel is precisely that this singleton requirement is a constraint on the member states through x: a *lock*. (A lock fixes the states of the members through x to unit values; it never forces more.)

Lemma F.4 (Rigidity at a lock)**.** *Let S be ZE. (a) If y, z are together then for every $q \notin \{y, z\}$, q is together with y iff with z. Hence if qz is hard and qy soft, the soft pair qy is forced to be together iff $qz \in E$; if both are soft, the owners of yz, qy, qz form a clique of $I(H)$ (a triangle if distinct). (b) If u, w and w, v are together, then uv is an edge or a soft pair.*

Proof. Transitivity of "together"; for (b) a hard pair inside a cluster is an unedited edge (Lemma F.1(a)). □

Consequently the two vertices that a lock bridges (the mates y, z of a locked member) are *occupied*: $y \sim z$ is a cluster that cannot also be bridged to a third vertex unless the pair is an edge or soft.

F.2 Memory across a tree decomposition

Let $(\mathcal{T}, \beta)$ be a tree decomposition of $I(H)$, $e = tt'$ an edge of $\mathcal{T}$, $\mathcal{B}_e$ the members in both bags of e, and $\mathcal{M}_1$, $\mathcal{M}_2$ the members that occur only on the side of t, resp. t'. A set S of anchors is a *face* if some ZE solution of (G, H) makes every $x \in S$ a singleton; equivalently (Lemma F.3) the partial vector of forced states is the restriction of an element of Σ. A *minimal non-face* (a *core*) is a non-face all of whose proper subsets are faces.

Lemma F.5 (Splice)**.** *(a) If $\sigma, \sigma' \in \Sigma$ agree on $\mathcal{B}_e$, the vector equal to σ on $\mathcal{M}_1 \cup \mathcal{B}_e$ and to σ' on $\mathcal{M}_2$ is in Σ. (b) Let $S = S_1 \cup S_2$ be a core with all members through anchors of S_i in $\mathcal{M}_i$ ($i = 1, 2$, both parts non-empty). For $b \in S$ let $\sigma^b \in \Sigma$ make all anchors of $S \setminus \{b\}$ singletons. Then $\{\sigma^b|_{\mathcal{B}_e} : b \in S_1\}$ and $\{\sigma^a|_{\mathcal{B}_e} : a \in S_2\}$ are disjoint.*

Proof. (a) $\sigma \in \Sigma$ iff no vertex triple is an induced P_3 after editing; the status of a triple's pairs depends on the owners of its soft pairs, a clique of $I(H)$ (Observation 3.2), so contained in one bag, hence in $\mathcal{M}_1 \cup \mathcal{B}_e$ or in $\mathcal{M}_2 \cup \mathcal{B}_e$; all constraints are satisfied by the combined vector. (b) If $\sigma^b|_{\mathcal{B}_e} = \sigma^a|_{\mathcal{B}_e}$ with $b \in S_1$, $a \in S_2$, splice σ^a on $\mathcal{M}_1 \cup \mathcal{B}_e$ with σ^b on $\mathcal{M}_2$: the anchors of S_1 are singletons (they are in $S \setminus \{a\}$, and their members are in M_1), those of S_2 likewise, so S is a face, a contradiction. □

Lemma F.6 (Majority closure bounds cores)**.** *Suppose Σ is closed under* some *majority operation $m : \{1, 2, 3\}^3 \to \{1, 2, 3\}$ applied coordinatewise ($m(a, a, b) = m(a, b, a) = m(b, a, a) = a$). Then every core has at most two anchors.*

Proof. Let S be a core with $|S| \geq 3$; we derive a contradiction.

(1) Each member contains at most one anchor of S, and as an end. If an anchor $x \in S$ is the centre of a member, x is never a singleton of a ZE solution (first statement of Lemma F.3: x a singleton and the centre of (y, x, z) would mean both edges at x are edited, two edits in one member), so $\{x\}$ is a non-face, contradicting minimality of S. If both ends a, c of a member (a, b, c) are anchors of S, then making a a singleton forces state 1 and making c a singleton forces state 2 (Lemma F.3), so $\{a, c\}$ is a non-face, again contradicting minimality, since $|S| \geq 3$. (For cores of size at most 2 a member can carry two anchors.)

(2) The forced vector. Let T be the set of members through anchors of S. By Lemma F.3 and (1), the requirement that every anchor of S be a singleton fixes the state of each member of T to a single value, giving a vector $t \in \{1, 2, 3\}^T$ without conflicts, and S is a face iff $t \in \Sigma_T := \mathrm{proj}_T(\Sigma)$, which is closed under m. The same holds for every $S' \subseteq S$ with its forced vector t' on its members T'.

(3) Binary projections determine t. Claim: if for every two members $P, P' \in T$ the pair $(t_P, t_{P'})$ lies in $\mathrm{proj}_{P,P'}(\Sigma)$, then $t \in \Sigma_T$. (This is the Baker–Pixley argument [1].) Induction on $|T|$; the base case $|T| = 2$ is the hypothesis. For $|T| \geq 3$ choose three members i_1, i_2, i_3 and, by induction applied to $T \setminus \{i_j\}$ and the m-closed relation $\mathrm{proj}_{T\setminus\{i_j\}}(\Sigma)$, choose $u_j \in \Sigma_T$ that agrees with t on $T \setminus \{i_j\}$. Then $m(u_1, u_2, u_3) = t$: at i_1 the vectors u_2, u_3 agree with t, similarly at i_2, i_3, and elsewhere all three agree.

(4) Since S is a non-face, $t \notin \Sigma_T$, so by (3) some pair of members $P, P' \in T$ has $(t_P, t_{P'}) \notin \mathrm{proj}_{P,P'}(\Sigma)$. Let S' be the set of anchors of S lying in P or P'; by (1), $|S'| \leq 2 < |S|$. The vector forced by S' agrees with t on P and P'. If S' were a face, some $\sigma \in \Sigma$ would extend it and $(\sigma_P, \sigma_{P'}) = (t_P, t_{P'})$, a contradiction. So S' is a non-face and a proper subset of S, contradicting that S is a core. □

F.3 Fan-out of released tunnels, and the tunnel-release model

Lemma F.7 (Released ends have small hard degree)**.** *Let H be pair-maximal, $D \geq \max_v |Q(v)|$, and S an optimal solution with $|S| \leq |H| + \ell$. (a) Every endpoint of an edited hard pair (in particular of a cut hard edge) has hard degree at most $D + \ell$. (b) Every hard component containing a vertex of hard degree larger than $2D + 3\ell$ is a whole cluster of S in which no* hard *pair is edited.*

Proof. (a) Lemma 7.4(a) gives $d_h(x) \leq D + u$ and $u \leq |S| - |H| \leq \ell$. (b) Lemma 7.6($\gamma$) with $B = 2D + 3\ell$; hard pairs inside the cluster are unedited by Lemma 7.6(β). Soft pairs inside the component may be edited (take K_9 minus one edge ac with $H = \{(a, b, c)\}$). □

Proposition F.8 (Bounded-degree Grid Tiling is easy)**.** *Grid Tiling [6] (find $a_1, \ldots, a_k, b_1, \ldots, b_k \in [n]$ with $(a_i, b_j) \in S_{ij}$ for all i, j) is solvable in time $n\, d^{2k}$* poly *on instances where every value has at most d partners in every S_{ij}.*

Proof. Guess a_1 (n ways); each b_j has at most d candidates, being a partner of a_1 in S_{1j}; then each a_i has at most d candidates, being a partner of b_1 in S_{i1}. □

F.4 The tunnel-release model

Fix a packing H and a set $\mathcal{J}$ of tunnels (stubs allowed). A *TR-solution* is a solution with exactly one edit in every member whose hard edits are cuts of tunnels; its excess is the number of cuts. For $Y \subseteq \mathcal{J}$ let $f(Y) = 1$ iff G minus the tunnel edges of Y has a ZE solution; call $S \subseteq \mathcal{J}$ *lockable* if $f(\mathcal{J} \setminus S) = 1$, and a minimal non-lockable set a *core*.

Proposition F.9 (Bounded cores give FPT in the model)**.** *(i) f is monotone. (ii) $f(Y) = 1$ iff Y meets every core. (iii) The minimum number of cuts of a TR-solution is the minimum size of a hitting set of the cores. (iv) If every core has at most d tunnels, this minimum is computable in time $d^\ell 3^{w+1} n^{O(1)}$ (or decides that it exceeds ℓ), given a tree decomposition of $I(H)$ of width w.*

Proof. (i) Given a ZE solution of G minus Y' and $Y' \subseteq Y$, each tunnel of $Y \setminus Y'$ is a cluster $\{x, x'\}$ (Lemma F.2); splitting it into singletons changes no edit. (ii),(iii): $f(Y) = 1$ iff $\mathcal{J} \setminus Y$ is lockable iff it contains no core, as non-lockability is upward closed; a TR-solution with cut set Y exists iff $f(Y) = 1$. (iv) Test f with the $\ell = 0$ dynamic program of Theorem 3.5; find a core inside $\mathcal{J} \setminus Y$ by deleting tunnels one at a time while the rest stays non-lockable; branch on which of its at most d tunnels is cut, to depth ℓ. □

This is a statement about a restricted model, not about CEaMP: solutions in which a member receives two edits, or a hard pair other than a tunnel edge is edited, are excluded, and they can be strictly cheaper. Inside the model, a W[1]-hardness proof therefore needs cores of unbounded size; such cores exist with soft triangles (Remark I.13), and the hardness of Section 5 is proved for the unrestricted problem.

F.5 Examples

Example F.10 (A local flip gadget)**.** Let G^* have vertices $0, 1, 3, 4, 5, 6, 7$ and edges $01, 04, 05, 14, 15, 45, 03, 13, 36, 37, 06, 56$, with members $(6, 3, 7), (4, 1, 3), (5, 0, 3), (4, 5, 6), (6, 0, 1)$ and anchor $x = 7$ ($c = 3$; the only hard edges are 04 and 15, and 7 has none). It is valid, pair-maximal and tight (OPT $= |H| = 5$), and has exactly five ZE solutions (found by enumerating all partitions): with x a singleton, $\{0, 1, 3, 4, 5, 6\}$ and $\{0, 1, 4, 5\}, \{3, 6\}$; with x not a singleton, $\{0, 1, 4, 5, 6\}, \{3, 7\}$, $\{0, 1, 4, 5\}, \{3, 6, 7\}$ and $\{0, 1, 4, 5\}, \{3, 7\}, \{6\}$. On the pair of members consisting of the fourth member $(4, 5, 6)$ and the second member $(4, 1, 3)$ in the order listed the locked relation (anchor a singleton) is the equality on $\{2, 3\}$, while releasing the anchor admits $(3, 2)$: the stored value flips only when the tunnel at x is cut. The relation of G^* is not determined by its binary projections (22233 is pairwise consistent but not a solution).

Example F.11 (The uniform majority operation need not close Σ)**.** Let $n = 6$, $E = \{03, 05, 12, 14, 23, 24, 25, 34, 35, 45\}$ and $H = \{(0, 5, 4), (1, 2, 5), (0, 3, 2), (1, 4, 3)\}$ ($c = 2$, tight). Then $\Sigma = \{1111, 1313, 3131, 2222\}$ (by enumeration of all partitions). The uniform majority m_3 ($m_3(a, b, c) = 3$ for pairwise distinct a, b, c) maps $(1313, 3131, 2222)$ to $3333 \notin \Sigma$, so Σ is not closed under m_3. It is, however, closed under other majority operations, for instance the one with value 1 on all pairwise distinct triples, so Lemma F.6 still applies to this instance. In particular the example refutes closure under the specific operation m_3 for $c \leq 2$, not closure under every majority operation. (The packing of this example has four soft triangles; without soft triangles Σ is closed under all majority operations, Theorem 10.4.)

The object that a hardness proof through tunnels needs is a family of tight instances with bounded c and $\mathrm{tw}(I(H))$ and a core of unbounded size. Lemma F.4(b) explains why the obvious compositions of copies of G^* fail: the two vertices bridged by a lock are occupied and cannot be shared with the next gadget. By Lemma F.6, such a family needs a set Σ that is closed under no majority operation, and by Theorem 10.4 it must contain a soft triangle (Section 10). The thick selection chain of Section 5 is such a family (Remark I.13); it needs $c = 4p + 2 \geq 26$.

G Proofs for Section 10

Notation and the labels "proved", "by computer" and "evidence" are as in Section 10. Computer checks were carried out by two independent implementations, the second of which built every instance and every set Σ directly from the graph and the definition (all set partitions, or an exact subset dynamic program), without using the structure lemmas.

G.1 Proof of Theorem 10.4

By Lemma 10.3, Σ is empty or the intersection of unary lists and of binary relations R_{AB}; it suffices to show that each of these is closed under every majority operation m. A static violation makes Σ empty, which is closed.

Lists. Let $S \subseteq \{1,2,3\}$ and $u, v, w \in S$. If two of them coincide then $m(u,v,w)$ is their common value, which lies in S. If all three are distinct then $S = \{1,2,3\}$.

Binary relations (classification by computer). R_{AB} is determined by the roles of the shared vertex q in A and in B (three each) and the four cross bits, 144 configurations. Computing the relation from the formula of Lemma 10.3(iii) gives exactly 40 distinct relations, with the following shapes, where p, q range over $\{1,2\}$ (the states that edit a pair at the shared vertex):

shape	number
$\emptyset$	1
singletons	9
two-element sets	18
rows $\{p\} \times \{1,2,3\}$, columns $\{1,2,3\} \times \{q\}$ (4) and the 3-element subsets of $\{1,2\}^2$ (4)	8
crosses $\big(\{p\} \times \{1,2,3\}\big) \cup \big(\{1,2,3\} \times \{q\}\big)$	4

The same list results from an independent construction: all 576 graphs on five vertices consisting of two members sharing exactly one vertex (all 6×6 orientations, 16 patterns of the four cross pairs), with each triple $\{p,q,r\}$ tested for being an induced P_3 after editing, without the formula.

Closure of each shape. Let m be a majority and $t_1, t_2, t_3 \in R$, $u = m(t_1,t_2,t_3)$. (1) $|R| \le 2$: at every coordinate at most two values occur among t_1, t_2, t_3, so u is the coordinatewise majority; if, say, t_1 occurs at least twice among the three tuples, then $u = t_1 \in R$. (2) A row $\{p\} \times \{1,2,3\}$: the first coordinate is constantly p, so $u_1 = p$ and $u \in R$; columns are symmetric. (3) A 3-element subset of $\{1,2\}^2$: all values lie in $\{1,2\}$, so m acts as Boolean majority, and every binary relation on $\{1,2\}$ is closed under it (it is definable by clauses with at most two literals). (4) A cross $R = (\{p\} \times \{1,2,3\}) \cup (\{1,2,3\} \times \{q\})$: suppose $u_1 \ne p$; we show $u_2 = q$. If two of the three first coordinates coincide, u_1 is their value, which differs from p, so these two tuples are not in the row and are of the form (x,q); two second coordinates equal q, so $u_2 = q$. If the three first coordinates are distinct, at most one of them equals p, so at least two tuples are of the form (x,q) and again $u_2 = q$. The case $u_2 \ne q$ is symmetric. So $u_1 = p$ or $u_2 = q$, i.e. $u \in R$. □

Orientation. The state names $1, 2$ depend on the orientation (a,b,c) versus (c,b,a) of a member (they are exchanged). The set of all majority operations is invariant under relabelling of the states by any permutation, so closure under *all* majorities does not depend on this convention (closure under one fixed operation would).

G.2 Proof of Theorem 10.14

Call a pair of vertices a *cross pair* or *far* as in Section 10.2. For a graph $G' = G \triangle W$ and a member $A = (a,b,c)$ and vertex $t \notin V(A)$ whose three pairs $\{x,t\}$, $x \in V(A)$, are hard, the *list of* (A,t) is

the set of states of Observation 10.13. It is *liftable* if at least one of the three pairs is far. A far pair lies in no binary triple (the third pair of a binary triple is a cross pair) and in no triple with two soft pairs. A triple $\{x, y, t\}$ with one soft pair xy of $A = (x, y, z)$ (up to naming) and a far pair xt also has yt and zt hard: if yt (or zt) were soft, with owner B, then B shares the vertex y (or z) with A and xt would be a cross pair. So the triple belongs to the list of (A, t), which is liftable. By Lemma 10.10 a far pair therefore changes only static triples and liftable lists. Let $\Sigma_{\text{perm}}(G')$ be the set of vectors satisfying all constraints of G' except the liftable lists; it is closed under every majority operation by Theorem 10.4 (only unary lists and the relations R_{AB} remain), and $\Sigma_{\text{perm}}(G \triangle W) = \Sigma_{\text{perm}}(G)$ for every set W of far pairs.

Algorithm solve(G', b), deciding whether some set W of at most b far pairs has $\Sigma(G' \triangle W) \neq \emptyset$. (1) If G' has an induced P_3 with three hard pairs, branch on its far pairs, $b \to b - 1$ (if there is none, or $b = 0$, answer NO). (2) Else if $\Sigma(G') \neq \emptyset$, answer YES; if $b = 0$, NO. (3) If $\Sigma_{\text{perm}}(G') = \emptyset$, NO. (4) Else choose a minimal family $\mathcal{F}$ of liftable lists with $\Sigma_{\text{perm}}(G') \cap \mathcal{F} = \emptyset$ (greedy deletion). By Lemma 10.12, $|\mathcal{F}| \leq 2$ (a list is a unit list, or empty; empty is a family of one). For each list of $\mathcal{F}$ and each of its far pairs e (at most 3) call solve$(G' \triangle e, b - 1)$.

Correctness. Soundness is clear. Completeness by induction on b: let W be a smallest set of at most b far pairs with $\Sigma(G' \triangle W) \neq \emptyset$. If $W = \emptyset$ step (2) answers YES (static violations are absent by (1) and then $\Sigma \neq \emptyset$ is tested). Otherwise $b \geq 1$. A static all-hard induced P_3 of G' has to be hit by a far pair of W (only far pairs are edited), so (1) branches correctly. In (4), W changes no relation and no permanent list, so $\Sigma(G' \triangle W) \subseteq \Sigma_{\text{perm}}(G') \cap \{\text{unchanged liftable lists}\}$; as $\Sigma(G' \triangle W) \neq \emptyset$ and $\Sigma_{\text{perm}}(G') \cap \mathcal{F} = \emptyset$, W changes some list of $\mathcal{F}$, i.e. contains a far pair $\{x, t\}$ with $x \in V(A)$ of it. Recursing on $G' \triangle e$ with $W \setminus \{e\}$ completes the induction (the pair e stays far in $G' \triangle e$, and a far pair is a far pair of H whichever graph is considered).

Time. Each node takes polynomial time (Corollary 10.5), at most 6 children, depth at most ℓ. □

G.3 Algorithm $\mathcal{A}$ and the proof of Theorem 10.18

Let $I(H)$ be a forest. Write $\text{Ex}_C(G')$ as in Section 10.3. For a graph G' (obtained from the input by flipping hard pairs) without an all-hard induced P_3, every triple whose three pairs are far is not an induced P_3, so it can be deleted without changing Ex; hence $\text{Ex}_C(G') = \text{Ex}(G')$ if C is the set of all sites.

Algorithm $\mathcal{A}(G', r)$ (excess budget r; accept iff $\text{OPT}(G') \leq |H| + r$). (1) If G' has an induced P_3 with three hard pairs: if $r = 0$ reject; else call $\mathcal{A}(G' \triangle e, r - 1)$ for each of its three pairs e (cross pairs included; the recursion is on graphs, so this stays sound) and accept iff one call accepts. (2) Now H is pair-maximal in G'. If $\text{Ex}(G') \leq r$, accept; if $r = 0$, reject. (3) Compute an inclusion-minimal set C of sites with $\text{Ex}_C(G') > r - 1$ by greedy deletion (at most $2|H|$ oracle calls). (4) With $B = 2D + 3r$ let L_B be the set of Section 7, computed from G', H, and $P(x)$ the set of $y \in \text{Ball}_{G'[L_B]}(x, 2)$ with xy far. For every $x \in V(C)$ and $y \in P(x)$ call $\mathcal{A}(G' \triangle xy, r - 1)$; accept iff one accepts.

Soundness. Acceptance in (2) gives a solution of size $\leq |H| + r$ by Lemma 10.15 (take $W = \emptyset$); in (1) and (4) a solution of $G' \triangle e$ of size $\leq |H| + r - 1$ gives one of G' of size $\leq |H| + r$.

Completeness. Suppose $\text{OPT}(G') \leq |H| + r$, there is no static violation and $\text{Ex}(G') > r$ (otherwise the algorithm accepts). In step (1), a solution contains one of the three pairs of the static P_3, and removing it gives a solution of $G' \triangle e$ of size $\leq |H| + r - 1$. In (4): by Lemma 10.15 choose a set W of far pairs minimising $|W| + \text{Ex}(G' \triangle W) \leq r$, with minimising F, x; (W, F, x) is an optimum of G'. As $\text{Ex}(G') > r$, $W \neq \emptyset$ and $\text{Ex}(G' \triangle W) \leq r - 1$. Let $Z(W)$ be the set of members through an end of a pair of W together with all edges of $I(H)$ at these members. Every triple that contains a pair of W has all its sites in $Z(W)$ (an owner of a soft pair of it is a member through an end

of the pair; a cross pair $\{x, z\}$ of it lies at an edge incident to a member through x or z). Let C' be the set of sites outside $Z(W)$. The triples kept in $\mathrm{Ex}_{C'}(G')$ contain no pair of W, so they have the same status in G' and in $G' \triangle W$, and (F, x) is feasible for $\mathrm{Ex}_{C'}(G')$ with the same cost: $\mathrm{Ex}_{C'}(G') \le \mathrm{Ex}(G' \triangle W) \le r - 1$. The set C of step (3) has $\mathrm{Ex}_C(G') > r - 1$, so by monotonicity $C \not\subseteq C'$: C contains a site of $Z(W)$, a member through an end x of some $e = xy \in W$, or an edge incident to such a member, and in both cases $x \in V(C)$. Since (W, F, x) is an optimum of G' of size $\le |H| + r$ and H is pair-maximal in G', Corollary 7.7 gives $y \in \mathrm{Ball}(x, 2)$ of $G'[L_B]$, so $y \in P(x)$. The recursion on $G' \triangle e$ with $W \setminus \{e\}$ satisfies $|W \setminus \{e\}| + \mathrm{Ex} \le r - 1$.

Time. The depth is at most $r \le \ell$; a node has at most $\max\{3, |V(C)|Z^2\} \le 6|C|Z^2$ children, $Z = 3D + 3r + 1$; each node makes at most $2|H| + 1$ oracle calls. Correctness did not use BC; only $|C| \le K$ does. □

G.4 Proof of Theorem 10.21

Variables of the CSP: the state of each member (seven non-empty edit sets, cost = size) and a bit for each local hard pair (cost 1 if edited); remote pairs are constants (in T) or absent. A vertex triple τ whose remote pairs all lie in T gives the constraint "τ is not an induced P_3"; its scope consists of the owners of its soft pairs and the bits of its local hard pairs. For a triple τ let $P(\tau)$ be its non-remote pairs and $W(\tau)$ the union of $\mathcal{M}(u)$ over the ends u of pairs of $P(\tau)$ (a set of members).

Lemma G.1 (Triple scopes)**.** *If $P(\tau) \neq \emptyset$, then $W(\tau)$ lies in a ball of radius 3 of $I(H)$ around one member.*

Proof. One pair uv: there are $A \in \mathcal{M}(u)$ and $B \in \mathcal{M}(v)$ equal or adjacent; every member of $\mathcal{M}(u)$ is within distance 1 of A (it shares u with A), every member of $\mathcal{M}(v)$ within 1 of B, so $W(\tau)$ lies within distance 2 of A. Two pairs uv, vw (they share v): there are $A \in \mathcal{M}(u)$, $B, B' \in \mathcal{M}(v)$, $C \in \mathcal{M}(w)$ with $d(A, B) \le 1$, $d(B', C) \le 1$, and $d(B, B') \le 1$ as both contain v. Around B: $\mathcal{M}(v)$ is within 1, $\mathcal{M}(u)$ within 2, C within 2 and $\mathcal{M}(w)$ within 3. Three pairs: two of them share a vertex and already involve all three vertices of τ, so the argument for two pairs applies. □

Radius 3 is attained: take six members forming a path A'–A–B–B'–C–C' in $I(H)$ with $u \in A \cap A'$, $v \in B \cap B'$, $w \in C \cap C'$, so that uv and vw are local and uw is remote; the minimal radius covering $W(\tau)$ is 3.

Proof of Theorem 10.21. Let $(\mathcal{T}, \{\beta_t\})$ be the given decomposition and B_t the set of members within distance 3 of β_t in $I(H)$; $|B_t| \le (w + 1)(1 + \Delta + \Delta^2 + \Delta^3)$ with $\Delta \le 3(c - 1) \le 3w$. For a member m the set $\{t : m \in B_t\}$ is the union of the subtrees $\{t : y \in \beta_t\}$ over the members y within distance 3 of m; these subtrees are chained along the connected ball, so the union is a subtree. A bit for a local pair uv is placed in the bags t with $\mathcal{M}(u) \cup \mathcal{M}(v) \subseteq B_t$, an intersection of subtrees, non-empty: take t with $A \in \beta_t$ for $A \in \mathcal{M}(u)$ adjacent to some $B \in \mathcal{M}(v)$; then $\mathcal{M}(u)$ and B are within 1, $\mathcal{M}(v)$ within 2. By Lemma G.1, all variables of a triple are in B_t for any t whose bag contains the centre member of the lemma. Hence this is a tree decomposition of the primal graph, with at most $|B_t|$ members and $\binom{3|B_t|}{2}$ bits per bag, and min-sum dynamic programming over it (at most $7^{|B_t|} 2^{\binom{3|B_t|}{2}}$ states per bag) solves the CSP in time $f(w) n^{O(1)}$. Dropping triples with a remote pair outside T is the definition of $F'(T)$. □

(Checks: against an independent solver over pair variables on 280 instances, no disagreement; the algorithm of Theorem 10.20 against exact OPT on 449 (instance, budget) pairs, and, with an independent solver, on 237 pairs, no mismatch; random tree-like instances needed radius 3 and never 4.)

G.5 Proof of Theorem 10.22

Let $\ell = 0$. Then $F'(T) > |H|$ iff there is no zero-excess assignment (states in $\{1,2,3\}$, no local hard edit) satisfying all triples whose remote pairs lie in T.

Lemma G.2 (Guards). *Let τ be a triple with a remote pair, $\ell = 0$. (i) If all three pairs are remote, or the non-remote pairs of τ are hard, then τ is a* static test*: violated iff τ is an induced P_3 of G; it is active iff its (at most three) remote pairs lie in T. (ii) If exactly one pair xz of τ is non-remote and it is soft with owner M, with remote pairs xy, zy, the triple is a* guard*, a list on M active iff $xy, zy \in T$: $t_{xz}(s_M) = 1$ if both are edges, $= 0$ if exactly one is an edge, no condition if none. For the three pairs of M the lists are $\{2,3\},\{1,3\},\{3\}$ (for $t = 1$) and $\{1\},\{2\},\{1,2\}$ (for $t = 0$). (iii) If two pairs uv, vw are non-remote they are not both soft (their owners would share v and uw would be a cross pair, hence local); one soft and one local hard gives a guard on the soft owner, active iff the remote pair $uw \in T$, of size 1 or 2; two local hard pairs give a static test.*

Proof. Read off the definitions: a local hard pair is unedited at $\ell = 0$ and a soft pair of a member in state s is together iff $t_e(s) = 1$; the triple is not a P_3 iff the number of edges is not 2. □

Let Σ_0 be the set of zero-excess vectors that satisfy all triples with no remote pair (and are empty if one of these triples is a static violation). If H has no soft triangle, Σ_0 is an intersection of unary lists and of the relations R_{AB} (all four cross pairs are local), so it is closed under every majority operation (proof of Theorem 10.4).

Lemma G.3 (Helly lemma for lists of size at most two). *Let $|D| = 3$ and $\Sigma \subseteq D^V$ closed under all majority operations. Let $L_i \subseteq D$, $|L_i| \leq 2$, for i in a finite index set I, such that $\Sigma \cap \prod_i L_i = \emptyset$ but for every i there is $x^i \in \Sigma$ with $x^i_l \in L_l$ for all $l \neq i$. Then $|I| \leq 3$.*

Proof. Suppose $|I| \geq 4$; then $x^i_i \notin L_i$. For $i,j,k \in I$ distinct and a majority operation m, $u = m(x^i, x^j, x^k) \in \Sigma$. For $l \notin \{i,j,k\}$ the three values lie in L_l, $|L_l| \leq 2$, so two coincide and $u_l \in L_l$. At i: if $x^j_i = x^k_i$ then $u_i = x^j_i \in L_i$. Otherwise $L_i = \{x^j_i, x^k_i\}$ and (x^i_i, x^j_i, x^k_i) is a permutation π_i of D; $u_i \in L_i$ iff $m(\pi_i)$ differs from the *first* entry of π_i. Likewise at j ($m(\pi_j) \neq$ second entry) and at k (third entry). The values of m on the six permutations are free, so a suitable m exists unless $\pi_i = \pi_j = \pi_k$ (then $m(\pi)$ would have to avoid all three entries), i.e. unless L_i, L_j, L_k are 2-lists whose missing values are pairwise distinct. Among four coordinates two missing values coincide in every triple that contains them, so some triple has a suitable m; then $u \in \Sigma \cap \prod_i L_i$, a contradiction. □

Proof of Theorem 10.22. Let T be a minimal certificate. If one of its active static tests is violated, the violated triple has at most three remote pairs, all in T by minimality, so $|T| \leq 3$. Otherwise the active guards of T have empty intersection with Σ_0. Let $\mathcal{G}$ be a minimal family of active guards with $\Sigma_0 \cap \bigcap \mathcal{G} = \emptyset$ and, for each member i, let L_i be the intersection of the guards of $\mathcal{G}$ at i ($|L_i| \leq 2$). Take a minimal set I of coordinates with $\Sigma_0 \cap \prod_{i \in I} L_i = \emptyset$; then for each $i \in I$ some x^i exists as in Lemma G.3, so $|I| \leq 3$, and by minimality of $\mathcal{G}$ all guards of $\mathcal{G}$ are at coordinates of I. At one coordinate, the partial intersections of its guards (non-redundant by minimality) strictly shrink: $\leq 2 \to \leq 1 \to 0$. If some $L_i = \emptyset$, then $\mathcal{G}$ consists of the at most three guards at i; otherwise there are at most two guards at each coordinate. Hence $|\mathcal{G}| \leq 6$. Each guard has at most two remote pairs, and the remote pairs of $\mathcal{G}$ form a certificate contained in T, so equal to T: $|T| \leq 12$. □

G.6 The family G_N and the proof of Theorem 10.23

Construction. For $j = 1, \dots, N$ take vertices p_j, q_j, r_j, c_j, b_j; add a vertex a_1 and put $a_j := b_{j-1}$ for $j \geq 2$. Members (centre in the middle)

$$A_j = (q_j, p_j, a_j), \qquad B_j = (q_j, r_j, b_j), \qquad C_j = (p_j, c_j, r_j).$$

The edges of G_N are the pairs $q_jp_j, p_ja_j, q_jr_j, r_jb_j, p_jc_j, c_jr_j$ of the members and the hard edges c_jq_j $(1 \leq j \leq N)$; all other pairs among these vertices are hard non-edges. Pendant vertices (in no member): w_j adjacent to p_j only $(1 \leq j \leq N)$, w_{N+1} adjacent to a_1 only, w_{N+2} adjacent to b_N only; all their pairs are remote. Then $|V| = 6N + 3$ and H_N has $3N$ members. Every vertex lies in at most two members (p_j in A_j, C_j; q_j in A_j, B_j; r_j in B_j, C_j; b_j in B_j, A_{j+1}), $I(H_N)$ is a chain of N triangles joined by single edges (tw $= 2$), each triple $\{p_j, q_j, r_j\}$ is a soft triangle, and the hard edges form a matching, so no induced P_3 has three hard pairs: H_N is pair-maximal (all by direct inspection; also checked by computer for $N \leq 3$).

$$T_N = \{w_jp_j,\, w_jr_j : 1 \leq j \leq N\} \cup \{w_{N+1}a_1,\, w_{N+1}p_1,\, w_{N+2}b_N,\, w_{N+2}r_N\}, \qquad |T_N| = 2N + 4.$$

Lemma G.4 (Structure)**.** *Write $s_j = (A_j, B_j, C_j)$ for the states of block j. Then $\Sigma_0(G_N)$ (all zero-excess vectors satisfying every triple that has no remote pair) is the set of vectors with $s_j \in \{121, 212, 223\}$ for all j and no $j < N$ with $s_j = 212$ and $s_{j+1} = 121$.*

Proof (by computer, for the first blocks, and by isomorphism of the interior blocks). A triple without remote pair lies in $W_j \cup W_{j+1}$ for some j, where W_j is the vertex set of block j (with a_j, b_j), by the definition of local. Inside a block the triples force the three states $121, 212, 223$ (the relation of a triangle block, by enumeration). The only triples with vertices of two blocks that carry soft pairs have both soft pairs at the glue vertex $b_j = a_{j+1}$ (the pairs r_jb_j of B_j and $p_{j+1}a_{j+1}$ of A_{j+1}) and the hard non-edge r_jp_{j+1} as third pair: it is a P_3 iff both soft pairs stay edges, i.e. B_j and A_{j+1} are both in $\{1, 3\}$, which within the allowed block states means $B_j = 1$ ($s_j = 212$) and $A_{j+1} = 1$ ($s_{j+1} = 121$). The other mixed triples have one soft pair and two non-edges. Enumeration over all 3^{3N} vectors for $N \leq 4$ gives $|\Sigma_0| = 3, 8, 21, 55$, equal to the transfer-matrix count, and contains every interior configuration (a block with both neighbours); interior blocks are isomorphic. □

The active guards of T_N. A triple is active iff its remote pairs lie in T_N. A triple containing w_j $(j \leq N)$ has two remote pairs at w_j, so it is $\{w_j, p_j, r_j\}$; this is the guard of Lemma G.2(ii) with $xz = p_jr_j$, the pair ac of C_j, w_jp_j an edge and w_jr_j a non-edge: $t_{ac}(C_j) = 0$, i.e. $C_j \in \{1, 2\}$. Likewise $\{w_{N+1}, a_1, p_1\}$ gives $t_{bc}(A_1) = 0$, i.e. $A_1 = 2$, and $\{w_{N+2}, b_N, r_N\}$ gives $B_N = 2$. No other triple is active, and each guard has exactly two remote pairs.

Proof of Theorem 10.23. Lower bound. $F'(T_N) \geq |H|+1$: otherwise there would be a zero-excess vector in Σ_0 with $C_j \neq 3$ for all j, $A_1 = 2$ and $B_N = 2$. By Lemma G.4 and $C_1 \neq 3$, block 1 is 121 or 212; $A_1 = 2$ excludes 121, so $s_1 = 212$, $B_1 = 1$. The link forbids $s_2 = 121$, and $C_2 \neq 3$ excludes 223, so $s_2 = 212$; inductively $s_N = 212$, $B_N = 1$, contradicting $B_N = 2$. *Minimality.* Removing a pair $e \in T_N$ deactivates exactly one guard; for each guard a zero-excess vector exists: without the A_1-guard take all blocks 121 (the link needs $s_j = 212$, never); without the B_N-guard all blocks 212 ($A_{j+1} = 2$); without the C_k-guard, blocks $j < k$ equal to 212, block k equal to 223 and blocks $j > k$ equal to 121 (the link applies only after a block 212, and the next block is then 212 or 223). These satisfy all remaining guards, and $F'(T_N \setminus \{e\}) = |H|$. Also $F'(\emptyset) = |H|$ by $\Sigma_0 \neq \emptyset$. □

(The values for $N \leq 4$ were also recomputed by enumeration; the two further statements, $F'(T_N) = |H| + 1$ for $N \leq 10$ and the connected family at $\ell = 1$, are computer evidence, as stated in Section 10.4. $\mathrm{OPT}(G_N) = |H| + N$ was computed exactly for $N \leq 3$ by an independent subset dynamic program and for $N \leq 5$ by CP-SAT.)

G.7 Proof of Lemma 10.25

Identify $\{1,2\}$ with $\{0,1\}$ and put $R_5 = \{(x,y,z) \in \{0,1\}^3 : x+y+z \neq 1\}$; then $R_k \cap \{1,2\}^3 = R_5$ (the words $111, 122, 212, 221, 222$). Let f be a conservative near-unanimity operation of arity $r \geq 3$ preserving R_k and all unary relations. Conservativity gives $f(\{1,2\}^r) \subseteq \{1,2\}$, so f preserves R_5 on $\{0,1\}$. Let $\mathcal{N}_0 = \{B \subseteq [r] : f(\mathbf{1}_B) = 0\}$ ($\mathbf{1}_B$ the indicator vector). For $B, C \in \mathcal{N}_0$ apply f to the three rows $\mathbf{1}_{B\cup C}, \mathbf{1}_B, \mathbf{1}_C$; every column is $(1,1,1), (1,1,0), (1,0,1)$ or $(0,0,0)$, hence in R_5, so $(f(\mathbf{1}_{B\cup C}), 0, 0) \in R_5$, i.e. $f(\mathbf{1}_{B\cup C}) = 0$ and $B \cup C \in \mathcal{N}_0$. Near-unanimity of arity $r \geq 3$ gives $f(\mathbf{1}_{\{i\}}) = 0$ (all but one argument equal 0), so every singleton is in $\mathcal{N}_0$ and by union-closure $[r] \in \mathcal{N}_0$, i.e. $f(1, \ldots, 1) = 0$, contradicting idempotence ($f(x, \ldots, x) = x$). □

The relation R_k itself is realised as follows (by computer): six vertices $0, \ldots, 5$, members $A = (2,1,0)$, $B = (4,3,1)$, $C = (5,0,3)$ (centres $1, 3, 0$), hard non-edges $25, 45$, and four further members that own the pairs $24, 23, 40, 51$ (using fresh vertices; each of the vertices $0, \ldots, 4$ lies in three members, so $c = 3$). The conjunction of the triples of the six vertices that contain none of these four pairs is R_k, with the states written $1, 2, 3$ for ab, bc, ac.

G.8 Proof of Theorem 10.27

Grid Tiling with $\leq$ [6]: given sets $S_{ij} \subseteq [n]^2$ ($1 \leq i, j \leq k$), choose $s_{ij} = (a_{ij}, b_{ij}) \in S_{ij}$ with $a_{ij} \leq a_{i,j+1}$ and $b_{ij} \leq b_{i+1,j}$; it is W[1]-hard in k (and has no $f(k)n^{o(k)}$ algorithm under ETH). (This is the version of the cited chapter after exchanging the two coordinates.)

Construction. For each cell (i,j) create two *arms*: A_{ij} with one node per tile of S_{ij} ordered by (a,b), and B_{ij} with one node per tile ordered by (b,a). Row i is the path obtained by concatenating $A_{i,k}, A_{i,k-1}, \ldots, A_{i,1}$ (each in increasing order), column j the path $B_{k,j}, \ldots, B_{1,j}$. The forest consists of these $2k$ paths. Obstruction paths: (1) every arm; (2) for every row i, $j < k$ and $\alpha \in [1, n-1]$ the set ρ of nodes of $A_{i,j+1}$ with $a \geq \alpha + 1$ together with the nodes of A_{ij} with $a \leq \alpha$ (a subpath of the row, as $A_{i,j+1}$ immediately precedes A_{ij}); (3) the same for columns with the b-coordinate and the arms B. Links: the node of the tile e in A_{ij} with the node of e in B_{ij} (a perfect matching). $L = k^2$.

Correctness. ($\Rightarrow$) For a tiling use, for every cell, the link of its chosen tile: all arms are hit; a path ρ of type (2) is hit iff $a_{ij} \leq \alpha$ or $a_{i,j+1} \geq \alpha + 1$, and $a_{ij} \leq a_{i,j+1}$ gives this for every α; columns likewise. ($\Leftarrow$) The $2k^2$ arms are pairwise disjoint and an operation meets at most two of them (a link only the two arms of its cell), so with $L = k^2$ every operation is a link, one per cell, hence a tile in every S_{ij}. If $a_{ij} > a_{i,j+1}$, the path ρ for $\alpha = a_{i,j+1}$ contains neither chosen node (the chosen nodes of other cells are on other arms), a contradiction; columns likewise. An empty S_{ij} gives an unhittable empty arm (NO, correctly), and an empty ρ means $a_{ij} > a_{i,j+1}$ in every choice (NO). The size is polynomial and $L = k^2$, so an $f(L)n^{O(1)}$ algorithm would give an $f(k^2)n^{O(1)}$ algorithm for Grid Tiling with $\leq$. □

(Checks: own brute force and an independent solver, generic hitting-set branching on a smallest unhit path and naive enumeration of all L-subsets, on 1,710 instances in total, including $k = 2, 3$ and $n = 2, 3, 4$: no mismatch.)

H Proofs for Section 11

Notation and conventions are those of Section 11.

H.1 The exchange calculus

Proof of Lemma 11.1. The pairs of T are not soft pairs of $H-Q$, and the members of $H-Q$ have pairwise disjoint pair sets; hence $H' = (H-Q)+T$ is modification-disjoint, and $T \notin H-Q$ (a member of $H-Q$ has soft pairs), so $|H'| = |H|$ (if $T = Q$ there is nothing to prove). As graphs on the node set $H-Q$, $I(H')-T = I(H)-Q$, an induced subgraph of $I(H)$. Adding one node to a graph raises the treewidth by at most one (put it in every bag), so $\mathrm{tw}(I(H')) \le \mathrm{tw}(I(H)-Q)+1 \le \mathrm{tw}(I(H))+1$. For j exchanges the same argument applies with $H - \{Q_1, \ldots, Q_j\}$, the T_i being pairwise pair-disjoint and disjoint from its soft pairs, and j added nodes.

Now let (G, H, ℓ) be a YES instance. Since $|H'| = |H|$, $\mathrm{OPT} \le |H'| + \ell$, so (G, H', ℓ) is a YES instance. Let M be a maximal family hard over H'. By Lemma 2.3, $|M| \le \ell$, by Lemma 2.4 $H'' = H' \cup M$ is pair-maximal, $\mathrm{tw}(I(H'')) \le \mathrm{tw}(I(H')) + |M| \le \mathrm{tw}(I(H)) + 1 + |M|$, and $(G, H'', \ell - |M|)$ is equivalent to (G, H', ℓ), hence to (G, H, ℓ). That H' need not be pair-maximal is shown by the computer test of Section 11. □

Proof of Lemma 11.2. Suppose $a \in C_1$, $b \in C_2$ with $C_1 \neq C_2$. Since $N[a] = N[b]$, $ab \in E$, and ab is an edge across clusters, so it is edited. Let c_a be the number of pairs $\{a, x\}$, $x \notin \{a, b\}$, that disagree with S (that is, are an edge across clusters or a non-edge inside a cluster), and c_b likewise. These pairs together with ab contribute $c_a + c_b + 1$ to the cost; all other pairs do not involve a or b. Move a into C_2. For $x \notin \{a, b\}$, a and x have the same adjacency as b and x, and a is now in the cluster of b, so $\{a, x\}$ disagrees exactly when $\{b, x\}$ did before: c_b pairs; the pairs at b are unchanged, c_b, and ab is now an edge inside a cluster and agrees. The new cost of these pairs is $2c_b$, all other pairs are unchanged. Symmetrically, moving b into C_1 gives $2c_a$. Since $\min\{2c_a, 2c_b\} \le c_a + c_b < c_a + c_b + 1$, S is not optimal. □

Proof of Corollary 11.3. Let $ab \in S$ be inserted: a, b lie in one cluster C and $ab \notin E$. By Lemma 7.2(b) there is $z \in C$ with $az, zb \in E$, so (a, z, b) is an induced P_3 containing ab. Let $ab \in S$ be deleted: $ab \in E$ and a, b lie in different clusters. By Lemma 11.2, a, b are not closed twins, so some $x \notin \{a, b\}$ is adjacent to exactly one of them, say to a; then $x\,a\,b$ is an induced P_3 containing ab (adjacent to b instead: $x\,b\,a$). □

Proof of Lemma 11.5. By Corollary 11.3 witnesses exist. Let T be a witness; the pair e is hard, so T has at most two soft pairs. As H is pair-maximal, T does not have all three pairs hard, so it has at least one soft pair. If e is inserted, a non-edge of G, then in the induced P_3 T the pair e joins the two ends and the opposite vertex z is the centre. If e is deleted, an edge of G, then e is an edge of T and z is an end.

(a) Let T have two soft pairs. They are the two pairs other than e, namely $\{a, z\}$ and $\{b, z\}$; let P, P' own them, so $a, z \in V(P)$ and $b, z \in V(P')$. If $P = P'$ then $a, b \in V(P)$ and $e \in \mathrm{pairs}(P)$ would be soft, a contradiction. So $P \neq P'$, $z \in V(P) \cap V(P')$, and z is a common soft neighbour of a and b ($z \neq a, b$): e is bridged.

(b) Otherwise every witness has exactly one soft pair s, owned by Q, and its other two pairs (one of them e) are hard. So the pairs of T are not soft pairs of $H-Q$, and Lemma 11.1 shows that $(H-Q)+T$ is a packing of size $|H|$; $e \in \mathrm{pairs}(T)$ is soft in it. □

Proof of Corollary 11.6. If e is not bridged, no witness has two soft pairs by Lemma 11.5(a) (whose conclusion needs both ends covered), so case (b) applies to any witness, and one exists. The width bound is Lemma 11.1. □

Proof of Lemma 11.7. Let $e = \{r, r'\} \in U$ with r, r' uncovered. By Corollary 11.3 it has a witness $T = \{r, r', x\}$. The pairs $\{r, x\}$ and $\{r', x\}$ are hard, since r and r' lie in no member, and e is hard. So T is an induced P_3 with all pairs hard, contradicting pair-maximality. □

Proof of Lemma 11.8. The pair az joins two vertices of the cluster C of a, b and is an edge of G (a step of the witness), so it is not in S. Q is an induced P_3 of G on $\{a, z, r\}$ with the edge az, so exactly one of ar, zr is an edge of G and the other a non-edge. If $r \in C$, all three pairs of Q are edges of $G \triangle S$: the edge among ar, zr is kept, and the non-edge is inserted. If $r \notin C$, then ar, zr join different clusters: the edge is deleted and the non-edge is untouched. In both cases $S \cap \text{pairs}(Q) = \{f\}$ with $f \in \{ar, zr\}$. Also $r \neq b$: otherwise $\text{pairs}(Q) = \text{pairs}(T) \ni e$ would be soft. The pairs of T in S are e only ($az \notin S$, and zb is an edge inside C). So relative to $H' = (H - Q) + T$ the hard edits are $(U \setminus \{e\}) \cup \{f\}$: e is soft, $f \notin \text{pairs}(T)$ is hard. The ends of f are r and a or z. As $r \notin V(T)$, r is covered after the exchange iff it lies in a member of $H - Q$. □

Proof of Lemma 11.9. All pairs at r are hard, so $N_h(r) = N(r)$. Let $z \in N(r)$ and $w \in N_h(z)$, $w \neq r$. Then rz and zw are hard edges, so by Lemma 7.3(a) (centre z) $rw \in E$ or rw is soft; rw is not soft as r is uncovered, so $w \in N(r)$. Hence the hard component of r is $N[r]$. For $z, z' \in N(r)$, Lemma 7.3(a) with centre r gives: $zz' \in E$ (a hard edge or a soft pair) or zz' is soft. A soft pair lies in one member, so its ends are co-bagged; a pair of $N[r]$ that is not co-bagged is therefore not soft, and is an edge, hence a hard edge. □

H.2 Proofs of Propositions 11.10 and 11.11

Proof of Proposition 11.10. (a) An induced P_3 has exactly two edges. The members of a packing have disjoint pair sets, hence are edge-disjoint. As $|E(G_m)| = 2m + 1$, a packing has at most m members, and H_m has m. $\text{OPT} = m + 1$ is Proposition 9.8(ii). By Lemma 2.2 every packing H^* has $\ell^* = |H| + \ell - |H^*| \geq \text{OPT} - \nu_{\text{mod}} = 1$ in a YES instance of excess 1.

(b) Pairs of P_i: $\{2i, 2i+1\}, \{2i+1, 2i+2\}, \{2i, 2i+2\}$. $T = (1, 2m-1, 2m-2)$: $1 \sim 2m-1$ (chord), $2m-1 \sim 2m-2$ (path edge), and $1 \not\sim 2m-2$ because the neighbours of 1 are $0, 2, 2m-1$ and $2m - 2 \geq 4$ for $m \geq 3$. So T is an induced P_3 with centre $2m-1$ and pairs $\{1, 2m-1\}$, $\{2m-2, 2m-1\}$, $\{1, 2m-2\}$. The first is in no P_i ($2m - 2 \geq 4$ apart, both odd); the second lies only in P_{m-1}; the third is neither a consecutive pair nor an even-even pair ($2m - 3 \geq 3$). So T avoids the pairs of $P_0, \ldots, P_{m-2}$ and Lemma 11.1 applies with $Q = P_{m-1}$. Soft edges of H^*: the path edges $\{j, j+1\}$, $j \leq 2m-3$, and $\{2m-2, 2m-1\}$, and the chord. The only hard edge is $\{2m-1, 2m\}$; an induced P_3 has two edges, at most one hard, so it contains a soft pair: H^* is pair-maximal. $V(T) = \{1, 2m-2, 2m-1\}$ meets $V(P_i) = \{2i, 2i+1, 2i+2\}$, $i \leq m-2$, exactly for $i = 0$ (vertex 1) and $i = m-2$ (vertex $2m-2$); and $V(P_i) \cap V(P_{i+1}) = \{2i+2\}$. So $I(H^*)$ is the cycle $P_0, P_1, \ldots, P_{m-2}, T, P_0$ on $m \geq 3$ nodes, of treewidth 2.

(c) The partition has the edits: the path edges $\{2i+1, 2i+2\}$ for $0 \leq i \leq m-1$ and the chord, $m + 1$ pairs, and no inserted pair; so its cost is $m + 1 = \text{OPT}$. Relative to H^* the edits $\{2i+1, 2i+2\}$, $i \leq m-2$, lie in P_i, the chord lies in T; the only hard edit is $\{2m-1, 2m\}$ and $2m \notin V(H^*) = \{0, \ldots, 2m-1\}$. The vertex $2m$ has the single neighbour $2m-1$, whose neighbours are $2m-2, 2m, 1$. So $\text{Ball}(2m) \cap V(H^*) \subseteq \{2m-1, 2m-2, 1\} = V(T)$, pairwise co-bagged in every decomposition (one member), and $2m \in L_B$ (hard degree 1). So $2m$ is a local hub and $\{2m-1, 2m\}$ is eligible; this optimum lies in $\mathcal{S}$ for H^*, $\ell^* = 1$. □

Proof of Proposition 11.11. We use the description of $\text{Sp}(F)$ in Proposition E.1. *Y is an induced P_3:* $yu_0, u_0b_0 \in E$ ($b_0, u_0 \in \mathcal{K}_*^+$) and $yb_0 \notin E$ (y is adjacent only to u_0, c_0). Its pairs $\{y, u_0\}$, $\{y, b_0\}$ are hard (y is uncovered) and $\{u_0, b_0\} \in \text{pairs}(X_0)$ only. *Z is an induced P_3:* b_0u_1 is a spine edge,

u_1c_1 is an edge inside $\mathcal{K}^+_{h(1)}$ (both belong to it), and $b_0c_1 \notin E$, because c_1 has no edge outside $\mathcal{K}^+_{h(1)}$ and $b_0 \in \mathcal{K}^+_*$ with $* \neq h(1)$ (only e_0 has head $*$). The soft edges are $a_ib_i, b_ic_i, u_ib_i, b_iu_{i+1}$ and the soft non-edges are a_ic_i, u_iu_{i+1}; so $\{u_1, c_1\}$ and $\{b_0, c_1\}$ are hard, and $\{b_0, u_1\} \in \text{pairs}(X_0)$ only. The pairs of Y and Z are disjoint, and none is a soft pair of $H - X_0$. So $H' = (H - X_0) + Y + Z$ is a packing and $|H'| = |H| + 1$. By Lemma 2.2 OPT $\geq |H'| = |H| + 1$, and OPT $\leq |H| + 1$ by the partition of Proposition E.1(S2).

Width. $I(H)$ is a caterpillar with spine $X_0, \ldots, X_N$ and leaf P_i at X_i. Removing X_0 isolates P_0. Y meets only P_0 and Z (all through b_0; u_0 lies in X_0 and in no other member, and y in no member of H). Z meets P_0, Y (via b_0), X_1 (via u_1) and P_1 (via c_1). So $I(H')$ is obtained from the caterpillar $X_1, \ldots, X_N$ with leaves $P_1, \ldots, P_N$ by adding the triangle $\{P_0, Y, Z\}$ and joining Z to X_1 and P_1 (the triangle $\{Z, X_1, P_1\}$, since X_1P_1 is an edge). Its blocks are triangles and edges, so $\text{tw}(I(H')) \leq 2$.

Class. The optimum of (S2) has the single hard edit $\{y, b_0\}$ with y uncovered. All hard degrees are at most $7 \leq B = 11$ (S3), so $y, b_0 \in L_B$. $\text{Ball}(y) = \{y, u_0, c_0, b_0\}$ (the neighbours of u_0 and c_0 are $\mathcal{K}^+_*$ and y). The covered vertices $u_0, b_0 \in V(X_0)$ and $c_0, b_0 \in V(P_0)$, and X_0, P_0 are adjacent through b_0, so they are co-bagged in every decomposition: y is a local hub and the edit is eligible. □

H.3 Proof of Theorem 11.13 and Corollary 11.14

Fix H, ℓ and a decomposition; *eligibility* (Definition 11.12) is then computable from G, H and the decomposition. For a covered $v \in L_B$ let $\text{Elig}(v)$ be the set of partners p with $\{v, p\}$ eligible; $|\text{Elig}(v)| \leq |\text{Ball}(v)| \leq Z^2$. Eligibility of a covered pair is symmetric. Every eligible pair has a covered end in L_B.

Labels and status. A *labelling* assigns to every member P a state x_P among the five of Lemma 3.3, and to every covered $v \in L_B$ a set $E_v \subseteq \text{Elig}(v)$ with $|E_v| \leq \ell$. It defines $S = \bigcup_P x_P \cup U$ with $U = \{\{v, p\} : p \in E_v\}$ and is *symmetric* if for covered v, v' with $\{v, v'\}$ eligible, $v' \in E_v \iff v \in E_{v'}$. The status of a pair in $G \triangle S$ is G-status xor membership in S. At a bag t, the status of a pair is *determined by the state at* t if it is soft with owner in β_t (status from x_P), or hard and not eligible (status of G), or hard and eligible with an end $p \in X_t \cap L_B$ (status from $[q \in E_p]$).

Lemma H.1 (Visibility)**.** *Let H be pair-maximal. For every vertex triple Δ of G there is a bag t at which the statuses of all three pairs of Δ are determined by the state, or Δ has no soft pair and no eligible pair (then all its statuses are static and Δ is not an induced P_3 of G).*

Proof. *Δ has a soft pair.* The owners of the soft pairs of Δ form a clique of $I(H)$ (Observation 3.2), so some bag t contains all of them (Helly). Every vertex of Δ incident to a soft pair lies in X_t. A hard pair of Δ has an end among these vertices: if Δ has one soft pair $\{a, b\}$, the hard pairs are $\{a, c\}, \{b, c\}$; if it has two or three, all vertices of Δ lie in X_t. So each hard pair is static or has an end in X_t with a label, and is determined at t (a vertex outside L_B is in no eligible pair).

Δ has no soft pair and no eligible pair. Then Δ has three hard pairs, all static, and it is not an induced P_3 of G by pair-maximality.

$\Delta = \{p, q, c\}$ has no soft pair and the eligible pair $\{p, q\}$. (i) p, q covered, co-bagged: take $t \in \mathcal{T}_p \cap \mathcal{T}_q$; the pairs $\{p, c\}, \{q, c\}$ are static or determined by the labels of p and q. (ii) $p = r$ uncovered, $q = v$ covered, r a local hub. The pair $\{r, v\}$ is determined by the label of v. If c is uncovered then $\{r, c\}$ is not eligible, hence static, and $\{v, c\}$ is static or determined by the label of v: any bag of $\mathcal{T}_v$ works. If c is covered and $c \in \text{Ball}(r)$, then $v, c \in V(H) \cap \text{Ball}(r)$ are co-bagged; take $t \in \mathcal{T}_v \cap \mathcal{T}_c$: $\{r, c\}$ is determined by the label of c, $\{v, c\}$ by that of v. If c is covered and $c \notin \text{Ball}(r)$, then $\{r, c\}$ is static and a bag of $\mathcal{T}_v$ works. By Lemma 11.7 no eligible pair has two uncovered ends (by definition of eligibility an eligible pair has a covered end), so the cases are exhaustive. □

Proof of Theorem 11.13. Algorithm. Take a nice tree decomposition of $I(H)$ of width w. A state at a bag t is $(x_P)_{P\in\beta_t} \in [5]^{\beta_t}$ together with $E_v \subseteq \mathrm{Elig}(v)$, $|E_v| \le \ell$, for every $v \in X_t \cap V(H) \cap L_B$. A vertex v is introduced when the first member through v enters, its label is chosen there, it is carried unchanged through the subtree $\mathcal{T}_v$, and forgotten with the last member through v; a join node requires equal states of the two children. At each bag the table keeps, for each state, the minimum cost, where the cost is $\sum_P |x_P| + \sum_v |E_v \setminus V(H)| + \frac{1}{2}\sum_v |E_v \cap V(H)|$, each term charged at the introduction of its member or vertex. A state is *consistent at* t if (S) for $v, v' \in X_t \cap L_B$ covered: $v' \in E_v \iff v \in E_{v'}$; and (V) for every triple Δ whose three statuses are determined at t, Δ is not an induced P_3 of $G\triangle S$, where the statuses are those given by the state. Inconsistent states are discarded. The output m^* is the minimum cost at the root, over all states in the root table (and ∞ if none).

Number of states. $|\beta_t| \le w+1$ gives 5^{w+1} choices of (x_P); $|X_t| \le 3(w+1)$ vertices have a label, each a set of at most ℓ elements of a set of at most Z^2 elements, at most $(Z^2+1)^\ell$ choices. Together $5^{w+1}(Z^2+1)^{3\ell(w+1)}$ states. Each transition and each test (enumerating triples) takes $n^{O(1)}$ time, and $Z = 3D + 3\ell + 1 \le 6(w+1) + 3\ell + 1$ since $D = 2c \le 2(w+1)$ (Lemma 2.1).

Soundness. A consistent labelling of the whole decomposition gives labels E_v constant along $\mathcal{T}_v$ (carried through, equal at joins) and states x_P; define $S = \bigcup x_P \cup U$ with $U = \{\{v,p\} : p \in E_v\}$. Two covered vertices v, v' with $\{v,v'\}$ eligible are co-bagged, so some bag contains both, and (S) there gives $v' \in E_v \iff v \in E_{v'}$ globally: a covered pair of U is seen from both ends, so the cost term $\frac{1}{2}\sum_v |E_v \cap V(H)|$ counts it once, and $|S|$ equals the cost. By Lemma H.1, every triple is either static and not an induced P_3 of G (so unaffected by S) or has all three statuses determined at some bag, where it is verified. Statuses at different bags agree, since they come from the same global labelling (the symmetry of labels makes the status of an eligible covered pair independent of the end that is used). Hence no triple is an induced P_3 of $G\triangle S$: S is a solution, all of whose hard edits are eligible, each covered vertex in at most ℓ of them: $S \in \mathcal{S}$, and m^* is at least $\min\{|S| : S \in \mathcal{S}\} \ge \mathrm{OPT}$.

Completeness. Let $S \in \mathcal{S}$. Put $x_P = S \cap \mathrm{pairs}(P)$, which is one of the five states because $V(P)$ is a cluster graph in $G\triangle S$ (Lemma 2.2, Lemma 3.3), and $E_v = \{p : \{v,p\} \in S \text{ hard}\} \subseteq \mathrm{Elig}(v)$, $|E_v| \le \ell$. The labelling is symmetric by construction, its cost is $|S|$, and every triple is not an induced P_3 of $G\triangle S$; so all states are consistent and $m^* \le |S|$. Hence $m^* = \min\{|S| : S \in \mathcal{S}\}$. If some optimal solution of size at most $|H| + \ell$ is in $\mathcal{S}$ then $m^* = \mathrm{OPT}$. The last sentence of the theorem follows. □

The restriction of the labels to $\mathrm{Elig}(v)$ is used twice: it makes (S) checkable in a bag common to both ends, and it gives the local-hub condition that makes the triples at an uncovered vertex visible; an uncovered r lies in no bag, so two edits at r with partners that are not co-bagged could not be compared.

Proof of Corollary 11.14. Let a decomposition co-bag every pair of $\Lambda(H)$. Let S be an optimal solution of size at most $|H| + \ell$ and $\{x,y\} \in U$ with x covered. By Corollary 7.7, $x, y \in L_B$ and $y \in \mathrm{Ball}(x)$; so if y is covered, $\{x,y\} \in \Lambda(H)$ (take $r = x$), hence co-bagged and eligible. If y is uncovered, $y \in L_B$ is a hub: any two covered $a, b \in \mathrm{Ball}(y)$ lie in $\Lambda(H)$ (take $r = y$), so they are co-bagged, and $\{y,x\}$ is eligible. Lemma 11.7 excludes two uncovered ends, and $|U| \le \ell$. So $S \in \mathcal{S}$ and Theorem 11.13 is exact. The equivalence with the treewidth of $I(H)$ plus links: a decomposition of $I(H)$ that puts a member $P \ni a$ and a member $P' \ni b$ in a common bag is a decomposition of $I(H)$ plus the link PP', and conversely a decomposition of $I(H)$ plus links is one of $I(H)$ in which the linked members share a bag. □

H.4 Proofs for the limits of repacking

Proof of Theorem 11.15. (a) At $\ell = 0$ CEaMP is NP-hard [15, Thm. 1], even with every vertex in at most 23 members (Theorem 4.1). Suppose the algorithm $\mathcal{R}$ exists with constants $s = s(0)$, $f = f(0)$. Given $(G, H, 0)$, run $\mathcal{R}$ and let H^* be its output. The guarantee holds only on YES instances, so validate: (i) test that H^* is a modification-disjoint packing of induced P_3's of G, else reject; (ii) compute, in time $f^{O(f^3)}|H^*|$ [3], a tree decomposition of $I(H^*)$ of width at most f, else reject; (iii) put $\ell^* = |H| - |H^*|$ and reject unless $0 \le \ell^* \le s$. Finally run the CSP algorithm (Theorem 3.6) on (G, H^*, ℓ^*) with this decomposition, in time $5^{f+1}n^{2s+O(1)}$, polynomial, and answer accordingly. Correctness: for a valid packing H^*, (G, H^*, ℓ^*) is a YES instance iff $\mathrm{OPT} \le |H^*| + \ell^* = |H|$, i.e. iff $(G, H, 0)$ is. So whenever a decision is made it is correct; the rejections in (i)–(iii) are unreachable on YES instances (where the guarantees hold, and $\ell^* \ge 0$ because $|H^*| \le \mathrm{OPT} \le |H|$), and are correct on NO instances. Hence NP is solved in polynomial time. The theorem used is the algorithm of Section 3, valid for every packing; the majority-closure theorem (Theorem 10.4) is not used, and the soft triangles in the gadgets of [15] play no role.

(b) Suppose $\mathcal{R}$ maps every YES instance $(G, H, 0)$ to H^* with $|H^*| = |H|$ and $\mathrm{tw}(I(H^*)) \le |H|^{1-\varepsilon}$. Take the instances of the proof of Theorem 4.1 from a sparsified 3-SAT formula on N variables, with $|H| = O(N)$. Run $\mathcal{R}$, validate that H^* is a packing of induced P_3's with $|H^*| = |H|$ (else reject), compute a tree decomposition of $I(H^*)$ of width at most $2\mathrm{tw} + 1$ in time $2^{O(|H|^{1-\varepsilon})}|H|$ [14], rejecting if its width exceeds $2|H|^{1-\varepsilon} + 1$, and apply Theorem 3.5 with it, in time $3^{2|H|^{1-\varepsilon}+3}n^{O(1)}$. Since $(G, H^*, 0)$ is equivalent to $(G, H, 0)$ for a valid packing of the same size, this decides 3-SAT in time $2^{O(N^{1-\varepsilon})}$ (plus the $2^{o(N)}$ of sparsification and the polynomial running time of $\mathcal{R}$), contradicting ETH.

The two remarks follow: a tight instance ($\ell = 0$, $\nu_{\mathrm{mod}} = \mathrm{OPT} = |H|$) has no packing larger than H, so "large treewidth implies a better packing" fails on it, and tight instances of unbounded treewidth exist under ETH by Theorem 4.1 (they are the instances of its proof). □

Proof of Proposition 11.16. The induced P_3's of G are (x, h, y) with x, y leaves and h a hub (a leaf and both hubs form a triangle). Every member has a hub as its centre. Two members with the same centre h and leaf pairs $\{x, y\}, \{x', y'\}$ have disjoint pair sets only if $\{x, y\} \cap \{x', y'\} = \emptyset$, impossible among three leaves. So each hub is the centre of at most one member, and $\nu_{\mathrm{mod}} \le 2$. A packing of size 2 has one member with centre 3 and leaf pair $\{x, y\}$ and one with centre 4 and a different leaf pair $\{x', y'\}$; two different 2-subsets of three leaves cover all leaves, so every maximum packing covers all five vertices ($\nu_{\mathrm{mod}} = 2$, six maximum packings). Enumerating all 52 partitions gives $\mathrm{OPT} = 3$ and the four optima $\{0, 1, 2, 3, 4\}$, $\{0\}, \{1, 2, 3, 4\}$, $\{1\}, \{0, 2, 3, 4\}$, $\{2\}, \{0, 1, 3, 4\}$; for each of the six maximum packings each optimum has a hard edit (all vertices are covered, so it is covered-covered). For example, for $H^* = \{(0, 3, 1), (1, 4, 2)\}$ and the optimum $\{0\}, \{1, 2, 3, 4\}$ the hard edit is $\{0, 4\}$ and $0, 4$ have the common soft neighbour 1: it is bridged. The counts 3/20, 2/111, 3/852, 305/11,116 are from exhaustive enumeration of all connected graphs with at least one induced P_3, over all maximum packings and all optima (computer). □

Proof of Proposition 11.17. The hard edges of H_0 are $\{0, 1\}$ and $\{2m + 1, 2m + 2\}$, which do not share a vertex; an induced P_3 of the path is three consecutive vertices, with two edges, at most one hard, so it contains a soft pair: H_0 is pair-maximal. A packing has at most $\lfloor (2m + 2)/2 \rfloor = m + 1$ members (two edges each, edge-disjoint), and $\{(2i, 2i+1, 2i+2) : 0 \le i \le m\}$ attains it; $\mathrm{OPT} \le m+1$ by the partition $\{0, 1\}, \ldots, \{2m, 2m + 1\}, \{2m + 2\}$, so $\nu_{\mathrm{mod}} = \mathrm{OPT} = m + 1$ and $\ell = 1$ for H_0 ($|H_0| = m$), $w = 1$. A packing of size $m + 1$ must use all $2m + 2$ edges. The edge $\{0, 1\}$ lies only in $(0, 1, 2)$, so this is a member; then $\{2, 3\}$ lies only in $(2, 3, 4)$ among the remaining ones, and so on: the packing is unique, $\{(2i, 2i + 1, 2i + 2)\}$, and it shares no member with H_0 (whose members

start at odd vertices). An improving exchange removes r members of H_0 and adds $r+1$ members pair-disjoint from the rest; the result is a packing of size $m+1$, hence the unique one, so all m members of H_0 are removed: $r = m$. □

Proof of Proposition 11.19. Run the algorithm of Open Problem 1, validate that H^* is a pair-maximal packing of induced P_3's with $\ell^* \geq 0$ (else reject, correct because then $\mathrm{OPT} \geq |H^*| > |H| + \ell$ or the output is invalid on a NO instance), and run Theorem 11.13 on (G, H^*, ℓ^*) with the decomposition of width at most $g(\ell, w)$; the time is $5^{g+1}(Z^2+1)^{3\ell^*(g+1)}n^{O(1)}$, a function of (ℓ, w) times a polynomial. Accept iff $m^* \leq |H^*| + \ell^*$. Soundness holds for every output by Theorem 11.13, and on a YES instance some optimum lies in the class, so $m^* = \mathrm{OPT} \leq |H^*| + \ell^* = |H| + \ell$. □

H.5 Computational checks for Section 11

All checks are finite and support, but do not prove, the statements; the structural claims of the section are proved above. Reference solutions were obtained by enumerating all set partitions ($n \leq 10$) or by an exact MILP.

- *Exchange calculus.* Random graphs with $6 \leq n \leq 10$ and random greedy pair-maximal packings, all optimal partitions: 5,867 instances, 25,899 optima, 26,387 hard edits. Every hard edit lay in an induced P_3 (0 exceptions; 1,634 closed-twin pairs were never separated). Every witness had one or two soft pairs; with two soft pairs the owners were distinct and contained the shared vertex z (0 violations); 11,433 two-soft witnesses had z not at the centre, all of them for deleted edits; of 6,595 hard edits 2,699 had witnesses of both kinds, 832 only two-soft and 3,064 only one-soft. No optimum had a hard edit with both ends uncovered (0 of 26,387). Of 27,177 exchanges $H - Q + T$ all gave packings of size $|H|$ with e soft and $\mathrm{tw} \leq \mathrm{tw}(I(H)) + 1$ (exact treewidth); 16,079 (59%) were pair-maximal. Displacement: 5,027 inserted edits, Q always contained exactly one edit, new hard set $U - e + f$, r uncovered in 1,650 and covered in 3,377. Outside vertices: 6,200, no violation of Lemma 11.9.

- *FAR.* $m = 3, \ldots, 9$ ($n = 7, \ldots, 19$): H_m valid and pair-maximal, $\nu_{\mathrm{mod}} = m$ and $\mathrm{OPT} = m + 1$ by MILP, H^* valid, pair-maximal, $I(H^*)$ a cycle; for $m = 3, 4, 5$ all optima $(7, 9, 11)$ enumerated, the chord is deleted in all of them and every optimum has at most one hard edit relative to H^*, touching the uncovered vertex $2m$.

- *Spine family.* F the triangle ($n = 18$), K_4 ($n = 30$), $K_{3,3}$ ($n = 42$): all claims of Proposition E.1(S1)–(S5), H' a packing of size $|H| + 1$ avoiding the pairs of $H - X_0$, $\mathrm{tw}(I(H')) = 2$ (exactly, triangle and K_4); $\mathrm{OPT} = |H| + 1$ by MILP for the triangle (9) and K_4 (15).

- *Theorem 11.13.* A full implementation of the algorithm (random elimination orders for the decompositions) on 984 instances ($n = 6, \ldots, 9$, random pair-maximal H with $|H| \leq 5$, width at most 2; 807 random and 177 sparse): value $\geq \mathrm{OPT}$ always, and value $= \min\{|S| : S \in \mathcal{S}\}$ (computed by enumerating partitions) in all 984; among 948 instances with an optimum in $\mathcal{S}$ the value was OPT in all, and in the 36 others it exceeded OPT; every triple was verified at some bag. FAR_m, $m = 3, 4, 5$: with the path tiling the value was $m + 2 = \mathrm{OPT} + 1$ and the YES instance was rejected, with H^* it was $m + 1 = \mathrm{OPT}$. Sp(triangle) with the original H, $\ell = 1$: value $9 = \mathrm{OPT}$. Negative control: with covered partners not restricted to co-bagged ones and all uncovered vertices allowed as hubs, 17 of 177 instances gave a value below OPT. (A control without the symmetry check (S) was not completed.)

- *Conjecture 11.18 and Proposition 11.16.* All connected graphs with $n \le 8$ (11,116 with an induced P_3 at $n = 8$), 1,500 random connected graphs with $n = 9$ and 300 with $n = 10$: for each graph a maximum packing and an optimum all of whose hard edits have an uncovered end or are bridged were found. For the naive form (uncovered end only) the failing graphs numbered $3/20, 2/111, 3/852, 305/11{,}116$ for $n = 5, \dots, 8$ and, in the random samples, 23/1,500 and 15/300.
- *Proposition 11.17.* $m = 2, \dots, 6$: exhaustive search for the smallest improving exchange gave $r = m$.

I Proof of Theorem 5.1: the reduction from Grid Tiling

This appendix contains the construction and all proofs for Section 5. Steps that use a computer are marked **(CA1)**–**(CA4)**; they are listed in Section 5.3 and collected in a status table at the end of the appendix. The appendix is organised so that each lemma can be corrected locally: Lemmas I.1–I.2 are statements about one chain; Lemma I.4 and Proposition I.5 give the ZE semantics; Lemmas I.7, I.8 and Theorem I.9 give soundness; Lemmas I.6–I.12 the parameters.

I.1 Conventions

Soft and hard pairs, members, states and zero-excess (ZE) solutions are as in Section 2 and Appendix F. A *ZE partition* of an instance (G, H) is a partition of $V(G)$ whose cost is $|H|$; by Lemma 2.2 and Lemma F.1 it is a partition in which no hard pair is edited (hard edges lie inside clusters, hard non-edges join different clusters) and every member has exactly one edited pair. For a member (a, b, c) with centre b the state is L if the single edit deletes ab ($b \sim c$, a apart), R if it deletes bc, and M if it inserts ac; these are the states $1, 2, 3$ of Appendix F. Write $x \sim y$ for "x, y are in one cluster". The *excess* of a solution S is $\mathrm{ex}(S) = |S| - |H|$. Grid Tiling instances are assumed to have all $S_{ij} \neq \emptyset$ and $n \ge 2$ (an instance with an empty S_{ij} is mapped to a fixed NO instance; $n = 1$ is trivial).

I.2 The thin chain

Definition. Fix $m \ge 7$. The *thin chain* OH_m has the vertices $s_0, \dots, s_{2m}$ and $e_0, \dots, e_m$ and the following members, listed as (a, b, c) with centre b:

- spine members $P_j = (s_{2j}, s_{2j+1}, s_{2j+2})$, $0 \le j \le m - 1$;
- $G_j = (e_j, s_{2j+1}, e_{j+2})$ for $2 \le j \le m - 4$ and $F_j = (e_j, s_{2j+2}, e_{j+1})$ for $2 \le j \le m - 3$;
- left cap: (e_0, s_2, e_1), (s_1, e_1, s_3), (e_0, s_4, e_2);
- right cap: $(s_{2m-5}, e_{m-3}, s_{2m-1})$, (e_{m-2}, s_{2m-2}, e_m), (s_{2m-4}, e_{m-1}, e_m), $(s_{2m-3}, e_{m-1}, s_{2m})$, $(e_{m-2}, e_{m-1}, s_{2m-1})$, $(s_{2m-3}, e_m, s_{2m-1})$.

The edges are the pairs ab, bc of the members and the four *hard edges* s_0e_1, e_0s_3, $s_{2m-3}e_{m-2}$, $s_{2m}e_m$; every other pair of vertices is a hard non-edge. The members have pairwise disjoint pair sets and are induced P_3's (checked by computer, and by the pair-disjointness argument of Lemma I.12 for the thick version). The *cell* of a vertex is $\lfloor t/2 \rfloor$ for s_t and t for e_t; every member has its vertices in at most three consecutive cells, and every soft pair joins vertices in cells at distance at most 2. A *position* is an integer $a \in \{0, \dots, m - 1\}$; a *group* is a set $\{2u, 2u + 1\}$, $0 \le u \le U - 1$, where $m = 2U$.

The partitions P_a. Put $\varepsilon(0)=e_1$, $\varepsilon(1)=e_0$, $\varepsilon(j)=e_j$ for $2\le j\le m-2$ and $\varepsilon(m-1)=e_m$. For $0\le a\le m-1$ let P_a be the partition of $V(\mathrm{OH}_m)$ whose clusters $C_0,\dots,C_{m-1}$ are

$$C_j=\begin{cases}\{s_{2j},s_{2j+1}\} & j<a\\ \{s_{2j},s_{2j+1},s_{2j+2}\} & j=a\ \cup\ \{\varepsilon(j)\}\\ \{s_{2j+1},s_{2j+2}\} & j>a\end{cases}\quad(\text{for } j\le m-2),$$

together with the following changes at the right end. If $a\le m-2$ then

$$C_{m-1}=\{s_{2m-1},s_{2m},e_{m-1},e_m\};$$

if $a=m-1$ then C_{m-2} contains in addition e_{m-1} and $C_{m-1}=\{s_{2m-2},s_{2m-1},s_{2m},e_m\}$. In P_a the spine members $P_0,\dots,P_{m-1}$ are in the states R^aML^{m-1-a}.

Lemma I.1 (Chain lemma)**.** *For every $m\ge 7$ the ZE partitions of OH_m are exactly $P_0,\dots,P_{m-1}$; they are pairwise distinct.*

Proof. (a) Each P_a is a ZE partition. This is a conjunction of local conditions: every member, and every hard pair that is an edge, involves vertices of at most three consecutive cells, and the pairs of vertices that must be apart are only constrained through clusters, which have vertices in at most three consecutive cells. A condition at a cell therefore depends only on the relation of the position a to this cell (smaller by at least 3, or $a-j\in\{2,1,0,-1,-2\}$, or larger), on the distance to the left end up to a constant, and on the distance to the right end up to a constant. Windows at the left end do not depend on m, windows at the right end depend only on the distance to the right end, and interior windows are isomorphic. All window types occur for $m\le 20$. **(CA1)**: for every $7\le m\le 60$ and every a the partition P_a was checked by computer to have cost exactly $|H|$ with one edit per member and no hard pair edited (no failure). By the window argument this covers all $m\ge 7$.

(b) There are no other ZE partitions. **(CA2)**. Process the cells $0,1,\dots,m$ from left to right. After cell c, a *partial ZE partition* is a partition of the vertices of the cells $\le c$ in which every hard pair and every member with all its vertices in these cells is respected. A vertex of cell c' can be together with a vertex of a cell $c''\ge c'+3$ only if the pair is soft or a hard edge, which never happens (soft pairs join cells at distance at most 2 and the hard edges join vertices in cells at distance at most 1). So only the partition induced on the vertices of the last two cells matters for the future, together with the information which blocks already contain an older vertex (such a block is *closed*). The state after cell c is the set partition of the vertices of the cells $c-1,c$ with its set of closed blocks, with blocks renamed canonically and vertices described by (type, cell offset). At every interior cell there are exactly six reachable states, and the transition matrix between consecutive interior cells is the same matrix

$$M=\begin{pmatrix}0&0&0&0&1&1\\0&0&0&0&0&0\\1&1&1&1&0&0\\0&0&0&0&0&0\\0&0&0&0&1&1\\0&0&0&0&0&0\end{pmatrix},\qquad \chi_M(x)=x^4(x-1)^2,$$

with a Jordan block of size two at the eigenvalue 1 (computed with exact integer arithmetic; the equality of the matrices was checked at the cells $7,\dots,19$ of $m=26$, and the transition rule at a cell involves only members and hard pairs within three consecutive cells, while the member list is

translation invariant in the interior, so the matrix is the same at every interior cell of every m). The number N_m of ZE partitions is $N_m = u^\top M^{\,m-c_0} v_m$ where u is the vector after the left cap, which does not depend on m, and v_m is determined by the right cap and depends only on the distance to the right end, hence is the same for all m after the shift by the exponent. Since x^4 divides χ_M, the generalised eigenspace of the eigenvalue 0 is annihilated by M^4, and M^t is for $t \geq 4$ an affine function of t (the Jordan block at 1). Hence N_m is an affine function of m for $m \geq m_1$. A direct computation gives $N_m = m$ for $7 \leq m \leq 40$, so $N_m = m$ for all $m \geq 7$. Together with (a), which provides m distinct ZE partitions (their spine words $R^a M L^{m-1-a}$ differ), these are all of them. □

Further evidence (not used in the proof): an independent implementation (own validity checker, own SAT enumeration of all ZE partitions) found exactly the m partitions P_a for $m = 7, \ldots, 14$ and for a sample of m up to 100.

Profiles. Let $m = 2U$. The *host* of the group u is $h_u = s_{4u+2}$; it is the last vertex of P_{2u} and the first vertex of P_{2u+1}. Write $\varepsilon_u^- = \varepsilon(2u)$ and $\varepsilon_u^+ = \varepsilon(2u+1)$; $\varepsilon_u^- \neq \varepsilon_u^+$ for all u.

Lemma I.2 (Profile of a host)**.** *For the group u the positions fall into four classes $A_0 = \{a \leq 2u-1\}$, $A_1 = \{2u\}$, $A_2 = \{2u+1\}$, $A_3 = \{a \geq 2u+2\}$, and for $a \in A_i$ the cluster of h_u in P_a is $\{h_u\} \cup \Pi_i$ with*

$$\Pi_0 = \{s_{4u+1}, \varepsilon_u^-\}, \quad \Pi_1 = \{s_{4u}, s_{4u+1}, \varepsilon_u^-\}, \quad \Pi_2 = \{s_{4u+3}, s_{4u+4}, \varepsilon_u^+\}, \quad \Pi_3 = \{s_{4u+3}, \varepsilon_u^+\}.$$

($A_0 = \emptyset$ for $u = 0$ and $A_3 = \emptyset$ for $u = U-1$.) In particular $\Pi_0 \subset \Pi_1$, $\Pi_3 \subset \Pi_2$, $s_{4u} \in \Pi_1 \setminus \Pi_0$, $s_{4u+4} \in \Pi_2 \setminus \Pi_3$, and Π_0, Π_1 are disjoint from Π_2, Π_3.

Proof. Read off the definition of P_a: the vertex s_{4u+2} is s_{2j+2} for $j = 2u$ and s_{2j} for $j = 2u+1$. If $a < 2u$ the cluster C_{2u} has the form $\{s_{4u+1}, s_{4u+2}\} \cup \{\varepsilon(2u)\}$ and $C_{2u+1} \not\ni s_{4u+2}$; if $a = 2u$ the cluster is $C_{2u} = \{s_{4u}, s_{4u+1}, s_{4u+2}, \varepsilon(2u)\}$; if $a = 2u+1$ it is $C_{2u+1} = \{s_{4u+2}, s_{4u+3}, s_{4u+4}, \varepsilon(2u+1)\}$; if $a \geq 2u+2$ it is $C_{2u+1} = \{s_{4u+2}, s_{4u+3}, \varepsilon(2u+1)\}$. At the caps, the label ε takes the exceptional values stated before Lemma I.1. **(CA3)**: the statement was in addition verified by computer for all even m with $8 \leq m \leq 60$ and all u (1,836 cases), including the groups $u = 0$ and $u = U-1$. □

Allowed sets and forbidden positions. For a set W of thin vertices of one chain with $h_u \in W$ let $F_u(W) = \{a :$ the cluster of h_u in $P_a \not\subseteq W\}$ be the set of positions *forbidden* by W. Define, for the group u,

$$\mathrm{Pad}_u = \{h_u\} \cup \Pi_0 \cup \Pi_3, \qquad X_u = \{h_u\} \cup \Pi_0, \qquad Y_u = \{h_u\} \cup \Pi_3, \qquad \mathrm{All}_u = \{h_u\} \cup \Pi_0 \cup \Pi_1 \cup \Pi_2 \cup \Pi_3.$$

Corollary I.3. *$F_u(\mathrm{Pad}_u) = \{2u, 2u+1\}$, $F_u(X_u) = \{a \geq 2u\}$, $F_u(Y_u) = \{a \leq 2u+1\}$ and $F_u(\mathrm{All}_u) = \emptyset$.*

Proof. By Lemma I.2: for Pad_u the clusters at A_0 and A_3 are inside Pad_u, while at A_1 the vertex $s_{4u} \notin \mathrm{Pad}_u$ and at A_2 the vertex $s_{4u+4} \notin \mathrm{Pad}_u$. For X_u: A_0 is allowed, A_1 is forbidden by s_{4u}, and A_2, A_3 by the non-empty sets Π_2, Π_3 disjoint from Π_0 (if $A_3 = \emptyset$ there is nothing to forbid). Y_u is symmetric: A_3 is allowed, A_2 is forbidden by s_{4u+4}, and A_1, A_0 by Π_1, Π_0, which are disjoint from Π_3. All_u contains every cluster. □

I.3 Thick chains and guests

Let $p \geq 6$ be even. A fixed *one-factorisation* $\mathcal{F}_1, \ldots, \mathcal{F}_{p-1}$ of K_p on the vertex set $\mathbb{Z}_p$ (round robin) is used throughout.

Thick chain. The thick version of a thin chain with tag c has the vertices (v,i), v a thin vertex, $i \in \mathbb{Z}_p$; $K_v = \{(v,i) : i \in \mathbb{Z}_p\}$ is the *super-vertex* of v. For each thin member (a,b,c') there are the p^2 *Latin members* $((a,i),(b,j),(c',i+j))$ with centre (b,j), $i,j \in \mathbb{Z}_p$. For each thin hard pair uv all p^2 pairs between K_u and K_v are hard, edges or non-edges as uv. Inside each K_v all pairs are hard edges (the *glue*) except the pairs used as pad or host pairs below. Every pair between K_a and K_b (ab a pair of a thin member) is covered by exactly one Latin member: the pair $(a,i),(b,j)$ by the member with these two vertices, the pair $(b,j),(c',k)$ by $i = k-j$, and the pair $(a,i),(c',k)$ by $j = k-i$. So the Latin members have pairwise disjoint pair sets, and if $i+j \neq k$ the triple $(a,i),(b,j),(c',k)$ is a *soft triangle*.

Guests. A *guest* is a single vertex y together with a set W_y of thin vertices (of one or two chains) such that all pairs between y and the super-vertices K_v, $v \in W_y$, are soft and all other pairs of y are hard non-edges, apart from the single-vertex members specified below. Realisation: for every $v \in W_y$ a perfect matching $\mathcal{M}(y,v)$ of K_p is chosen, a different class $\mathcal{F}_r$ for different guests at the same v; for every $\{i,k\} \in \mathcal{M}(y,v)$ $(i<k)$ there is the *pad member* $(y,(v,i),(v,k))$ with centre (v,i) (edges $y(v,i)$ and $(v,i)(v,k)$, non-edge $y(v,k)$). A *permanent guest* z with a *host* $h \in W_z$ has instead, for the matching $\mathcal{M}(z,h)$, the *host members* $((h,i),z,(h,k))$ with centre z (edges $z(h,i)$, $z(h,k)$, non-edge $(h,i)(h,k)$), and pad members for all $v \in W_z \setminus \{h\}$. The pairs of K_v in a matching used at v are soft; the glue is the complete graph on K_v minus these pairs. All pairs between different super-vertices that are not soft and not thin hard edges are hard non-edges, and so are all pairs between guests (except those in the members (x,y,z) and (z,y,z') below).

Thin-guest model. A *thin configuration* is a partition π of the set of thin vertices, guests, anchors and stubs of the instance such that: (T1) a pair that is a hard edge in the thin picture (a thin hard edge, an uncut tunnel, a pair (z,h) of a permanent guest and its host) lies inside a cluster; every other pair of two elements is split, except the *free pairs* (y,v), $v \in W_y$, and the pairs of the members; (T2) every thin chain member and every single-vertex member (x,y,z), (z,y,z') has exactly one edited pair.

Lemma I.4 (Blow-up)**.** *Assume that the glue graph of every K_v is connected (Lemma I.6). Then the ZE partitions of a thick instance are exactly the lifts (replace each thin vertex v by K_v) of the thin configurations. In particular in a ZE partition (a) every K_v lies in one cluster; (b) a guest y is together with K_v only if $v \in W_y$; (c) a permanent guest is together with its host K_h; (d) the thick members of a thin member are all in the state of the thin member.*

Proof. Let σ be a ZE partition. No hard pair is edited, so the connected glue puts each K_v into one cluster, which is (a). A pair between K_u and K_v that is a thin hard pair is hard of the same type for all p^2 pairs, so $K_u \sim K_v$ iff the thin pair is an edge. For a thin member (a,b,c') the p^2 Latin members see the same togetherness pattern of the three super-vertices, so they are in the same state, which is a state of the thin member (one edit); this is (d). A pad member $(y,(v,i),(v,k))$ with $(v,i) \sim (v,k)$ has the togetherness patterns $(1,1,1)$ (if $y \sim K_v$) with one edit, inserting $y(v,k)$, and $(0,1,0)$ (if $y \not\sim K_v$) with one edit, deleting $y(v,i)$; so a pad member imposes no condition. If $v \notin W_y$ the pairs $y(v,i)$ are hard non-edges, so $y \not\sim K_v$, which is (b). A host member $((h,i),z,(h,k))$ with $(h,i) \sim (h,k)$ has one edit iff $z \sim (h,i)$ (pattern $(1,1,1)$: insert $(h,i)(h,k)$); if z is apart it has two edits. This is (c). Conversely, the lift of a thin configuration has no edited hard pair (hard pairs between super-vertices and glue edges are respected by (T1), pairs of guests with K_v, $v \notin W_y$, are split) and one edit in every thick member, by the same case analysis. Note that in a lift the host pair $(h,i)(h,k)$ is a soft non-edge inside a cluster and is edited, while a pad pair is a soft edge inside a cluster and is not. □

I.4 The reduction

Let $(k, n, (S_{ij}))$ be given, $n \geq 2$, and let p be the least even integer $\geq k^2 + 5$.

Chains and groups. For every cell $(i, j) \in [k]^2$ there are two chains A_{ij} and B_{ij}, each a thick copy of OH_m with $m = 2U$, $U = 1 + n(n+1)$ (so $m \geq 14$). For $\alpha \in [n]$ the *block* α has the *slot groups* $u = (\alpha-1)(n+1) + s$, $s \in [n]$, and is followed by the *separator group* $u_\alpha = \alpha(n+1)$; group 0 is the left end pad, and the last separator group $u_n = U - 1$ is the right end pad.

Tiles and tunnels. A tile $(a, b) \in S_{ij}$ has its *slot* in A_{ij} at the group $(a-1)(n+1) + b$ (block a) and in B_{ij} at the group $(b-1)(n+1) + a$ (block b). The groups that are slots of tiles are *tile groups*.

Guests. For every chain c and every group u there is a permanent guest $z_{c,u}$ with host h_u; if u is a tile group, $W_z = \mathrm{All}_u$, and in addition there are a *lock guest* $y_{c,u}$ with $W_y = \mathrm{Pad}_u$, a vertex $x_{c,u}$ (the *anchor*) and the single-vertex member $(x_{c,u}, y_{c,u}, z_{c,u})$ with centre $y_{c,u}$. If u is not a tile group, $W_z = \mathrm{Pad}_u$ (a *permanent lock*). All other pairs of x, y, z are hard non-edges.

Tunnels. For every cell and every tile $e \in S_{ij}$ the two anchors of e (in A_{ij} and in B_{ij}) are joined by a hard edge, the *tunnel* of e. The anchors have no other hard edge.

Comparators. For $\alpha \in [n-1]$, for every row i and $j < k$, let $c_1 = A_{ij}$, $c_2 = A_{i,j+1}$, and for every column j and $i < k$ let $c_1 = B_{ij}$, $c_2 = B_{i+1,j}$. In each case there is a *comparator guest* y with $W_y = X_{u_\alpha}(c_1) \cup Y_{u_\alpha}(c_2)$ (sets of thin vertices of the two chains) and the single-vertex member $(z_{c_1,u_\alpha}, y, z_{c_2,u_\alpha})$ with centre y.

The instance. G is defined by the members above (edges ab, bc of every member), the hard edges (thin hard edges, glue, tunnels) and no other edges. H is the set of all members: Latin members, host members, pad members, lock members (x, y, z) and comparator members. The excess is $\ell = k^2$. The instance is denoted I_{tile}; for a set C of tunnels, I_C is I_{tile} with the pairs of C made non-edges (they stay hard). A *stub instance* is the same with some anchors' tunnels ending at stubs (vertices of degree one).

Lemma I.6 below shows that H is a modification-disjoint packing of induced P_3's, that the glue graphs are connected and that matchings suffice, so Lemma I.4 applies to I_{tile}, to every I_C, and to the stub instances used below.

I.5 Tunnel-release semantics

For a chain c and a cell let $\beta(c) \in [n]$ denote the *block* of the tile whose tunnel is cut: $\beta(A_{ij}) = a$ and $\beta(B_{ij}) = b$ if the tile is (a, b).

Proposition I.5 (Tunnel release). *Let $t_{ij} = (a_{ij}, b_{ij}) \in S_{ij}$ be a tile of every cell, C the set of their tunnels, and σ a ZE partition of I_C. Then in σ every chain c is in a position a_c that lies in the group of its cut tunnel, and $a_{ij} \leq a_{i,j+1}$, $b_{ij} \leq b_{i+1,j}$ for all cells. Conversely, if (t_{ij}) satisfies these inequalities, I_C has a ZE partition.*

Proof. ($\Rightarrow$) By Lemma I.4 take the thin configuration. Its restriction to a chain c is a ZE partition of OH_m (all non-soft pairs inside the chain are hard), so by Lemma I.1 it is P_{a_c} for one position a_c.

Permanent guests. By Lemma I.4(c), $z = z_{c,u} \sim K_{h_u}$, so z is in the cluster of h_u in P_{a_c}; for each v in that cluster the pair zK_v is not a hard non-edge, so $v \in W_z$ by (b). Hence $a_c \notin F_u(W_z)$. If u is not a tile group, Corollary I.3 gives $a_c \notin \{2u, 2u+1\}$.

Locks. Let u be a tile group whose tunnel xx' ($x = x_{c,u}$) is not cut. It is a hard edge of I_C with $N_h(x) = \{x'\}$ and $Q(x) \cap Q(x') = \emptyset$ (the soft partners are guests of different chains), so by Lemma F.2 the cluster of x is $\{x, x'\}$; in particular $x \not\sim y, z$ for the members (x, y, z). In the member (x, y, z) with centre y this leaves the single edit "delete xy" only if $y \sim z$ (if $y \not\sim z$ also yz is deleted: two edits). So $y \sim z$ and y is in the cluster of h_u; as above $a_c \notin F_u(\mathrm{Pad}_u) = \{2u, 2u+1\}$.

Exactly one tunnel of the chain c is cut (the tunnel of t_{ij} serves both A_{ij} and B_{ij}). All groups except the group of that tunnel exclude their two positions, and the groups partition $\{0, \dots, m-1\}$; hence a_c lies in the group of the cut tunnel. With the slot indices above, this gives for the group u_c of c: $u_c = (a_{ij} - 1)(n+1) + b_{ij}$ for A_{ij} and $(b_{ij} - 1)(n+1) + a_{ij}$ for B_{ij}. Since $s \in [n]$, the group u_c lies in the block $\beta(c)$: $u_c < u_\alpha$ iff $\beta(c) \le \alpha$, and $u_c > u_\alpha$ iff $\beta(c) \ge \alpha + 1$; so $a_c \le 2u_\alpha - 1$ iff $\beta(c) \le \alpha$ and $a_c \ge 2u_\alpha + 2$ iff $\beta(c) \ge \alpha + 1$.

Comparators. Let y be the comparator guest of (c_1, c_2, α), $u = u_\alpha$. In the member (z_1, y, z_2) ($z_i = z_{c_i,u}$) the state M would give $z_1 \sim z_2$, impossible because z_1, z_2 lie in clusters of different chains (a pair of thick vertices of different chains is a hard non-edge). So the state is L ($y \sim z_1$) or R ($y \sim z_2$). In the first case the cluster of $h_u(c_1)$ is inside W_y, i.e. inside $X_u(c_1)$ (the part of W_y in c_1), so $a_{c_1} \notin F_u(X_u)$, i.e. $a_{c_1} \le 2u - 1$, i.e. $\beta(c_1) \le \alpha$. In the second case $a_{c_2} \notin F_u(Y_u)$, i.e. $a_{c_2} \ge 2u + 2$, i.e. $\beta(c_2) \ge \alpha + 1$. So for every $\alpha \in [n-1]$: $\beta(c_1) \le \alpha$ or $\beta(c_2) > \alpha$, which is equivalent to $\beta(c_1) \le \beta(c_2)$ (if $\beta(c_1) > \beta(c_2)$ take $\alpha = \beta(c_2) \le n - 1$). For rows, $\beta(A_{ij}) = a_{ij}$, for columns $\beta(B_{ij}) = b_{ij}$, which are the inequalities of Grid Tiling.

($\Leftarrow$) Put every chain c at the position $a_c = 2u_c$, u_c the group of its cut tunnel. Build the thin configuration: the base clusters are those of P_{a_c}. *Permanent guests:* $z_{c,u}$ joins the cluster of h_u; this is allowed: for a tile group the set is All_u, for the others $a_c \notin \{2u, 2u+1\}$ and Corollary I.3 applies. *Locks:* for a tile group $u \ne u_c$ (tunnel not cut) the lock guest $y_{c,u}$ joins the cluster of $z_{c,u}$ (allowed, $a_c \notin \{2u, 2u+1\}$), and the anchors x, x' at the two ends of that tunnel form the cluster $\{x, x'\}$; the member (x, y, z) has the single edit "delete xy" (state L). For $u = u_c$ the tunnel is cut, $y_{c,u}$ is not joined to anything but its anchor, and the anchor forms the cluster $\{x, y\}$ (state R: delete yz). The pair xx' is a cut tunnel, a hard non-edge, and the two anchors are in different clusters. *Comparators:* the guest y of (c_1, c_2, α) joins z_1 if $\beta(c_1) \le \alpha$ (allowed: then $a_{c_1} \le 2u_\alpha - 1$, so the cluster of h_{u_α} is inside X_{u_α}) and joins z_2 otherwise (then $\beta(c_2) > \alpha$ and $a_{c_2} \ge 2u_\alpha + 2$, inside Y_{u_α}). Every guest is in at most one cluster with a base vertex and two guests are never together, except in the pairs y, z of members: guests join base clusters only through a permanent guest; two hosts $h_u, h_{u'}$ are never in one cluster (their positions in the spine differ by at least four); a tile group carries one lock guest and no comparator, and a separator group carries up to two comparator guests on the same z (one as first chain with condition $\beta \le \alpha$, one as second chain with $\beta \ge \alpha + 1$), which cannot both join. All thin chain members have one edit since P_{a_c} is ZE, the members (x, y, z) and (z_1, y, z_2) have one edit as described, and every remaining pair is as required in (T1). By Lemma I.4 this lifts to a ZE partition of I_C. □

I.6 Soundness and completeness

Lemma I.6 (Parameters and validity)**.** *(a) H is a modification-disjoint packing of induced P_3's of G. (b) Let $u(v)$ be the number of matchings used at K_v (pad matchings and the host matching). Then $u(v) \le 3$ for all v and $u(v) \le 2$ if $k \le 2$; at a vertex lying in four thin members $u(v) \le 2$, and at a vertex in at most three thin members $u(v) \le 3$. (c) The glue graph of every K_v is connected, and $p - 1 \ge 3$ matchings are available. (d) $c(H) \le 4p+2$ and $|V(G)| = O(k^2n^2p)$, $|H| = O(k^2n^2p^2)$.*

Proof. (a) Every pair between guests and super-vertices, and every pair inside a super-vertex that is used as a pad or host pair, lies in exactly one member because different guests use different classes of the one-factorisation at the same v and a pair of a matching lies in exactly one pad (or host) member; Latin members are pair-disjoint as shown above and have no pair inside a super-vertex; the single-vertex members (x, y, z), (z, y, z') use pairs between guests and anchors only. Every member (a, b, c) has the edges ab, bc and the non-edge ac by definition, and these adjacencies are consistent because a soft pair belongs to one member.

(b) **(CA4)** A hand count from Lemma I.2: the sets W contain h_u, Π_0, Π_3 (pad), Π_0 (X), Π_3 (Y) or all Π_i (All) of a single group u, so a super-vertex is used only by guests of the group u, except $s_{4u+4} = s_{4(u+1)}$, which occurs in Π_2 of the group u and Π_1 of the group $u+1$. A group carries: a tile group the permanent guest z and a lock guest y (both $W \ni h_u$); a separator group (with $W_z = \mathrm{Pad}_u$) the permanent guest and at most two comparator guests (one with the chain as first chain, using X_u, one as second chain, using Y_u); any other group only z. At the host h_u: z (host matching), plus y, or plus up to two comparator guests: at most 3. At $s_{4u+1}, \varepsilon_u^-$: z plus y or the comparator guest of the first chain: at most 2; at $s_{4u+3}, \varepsilon_u^+$: z plus y or the comparator of the second chain: at most 2; at s_{4u}, s_{4u+4}: only in All, i.e. only the permanent guests of tile groups, at most one from each of two groups: at most 2. If $k = 1$ there are no comparators, and if $k = 2$ each chain has at most one neighbour, so the host has at most 2. The thin member counts: a vertex lying in four thin members is an e-vertex, and the hosts s_{4u+2} lie in three; this, and the above counts, were confirmed by computer for $k \le 4$ and $n \le 3$ (maximum of $u(v)$ and of the number of thin members through every vertex, with the exceptional values at the caps).

(c) A vertex of K_v has glue degree at least $p - 1 - u(v)$. For $p \ge 8$ and $u(v) \le 3$ this is at least $p - 4 \ge (p-1)/2$, hence the glue graph is connected (diameter at most 2). For $p = 6$ we have $k = 1$, so $u(v) \le 2$ and the degree is at least $3 \ge (p-1)/2$. At most $3 \le p - 1$ matchings are used at any vertex.

(d) A thick vertex (v, i) lies in p Latin members of each thin member through v, and in $u(v)$ guest members. By (b) this is at most $4p+2$ at a vertex in four thin members and at most $3p+3 \le 4p+2$ otherwise. The sizes: $2k^2$ chains with $O(n^2)$ cells each and $O(p)$ thick vertices per thin vertex; each thin member gives p^2 Latin members; guests contribute $O(p)$ members each. □

Lemma I.7 (No super-vertex is split)**.** *Let S be a solution of I_{tile} with $\mathrm{ex}(S) \le k^2$. Then every K_v lies in one cluster of $G \triangle S$. In particular no pad pair (a soft edge inside a super-vertex) is edited.*

Proof. Suppose K_v meets at least two clusters; let $X \subset K_v$ be its smallest part, $x = |X| \in [1, p/2]$, $Y = K_v \setminus X$. Every pair between X and Y that is an edge of G is deleted. A vertex of X has at most $u(v) \le 3$ soft pairs inside K_v, so at least $p - x - u(v)$ hard edges to Y; altogether at least $\varphi(x) = x(p - x - u(v))$ hard edits. Since φ is concave, $\varphi(x) \ge \min\{\varphi(1), \varphi(p/2)\}$. We have $\varphi(1) = p - 1 - u(v) \ge p - 4 \ge k^2 + 1$. For $k \ge 3$: $\varphi(p/2) \ge \frac{p}{2}(\frac{p}{2} - 3) \ge \frac{(k^2+5)(k^2-1)}{4} > k^2$. For $k \le 2$ we have $u(v) \le 2$ and $p \in \{6, 10\}$: $\varphi(p/2) \ge 3 > 1$ resp. $\ge 15 > 4$. So there are more than k^2 hard edits, but by Lemma 2.2 the number of hard edits is at most $\mathrm{ex}(S) \le k^2$, a contradiction. A pad pair is an edge inside the cluster of K_v, hence not edited. □

(The host pairs $(h, i)(h, k)$ are soft *non*-edges inside K_h; they are edited whenever the permanent guest z joins its host, which happens in every ZE partition. Only pad pairs, in particular those of comparator pad members, are unedited.)

Regions. For a chain c let R_c be the set of its thick vertices and of its guests $z_{c,u}, y_{c,u}$ and anchors $x_{c,u}$ (not the comparator guests). Let T_c be the set of tunnel partners of anchors of c (they are

anchors of the partner chain, $A_{ij} \leftrightarrow B_{ij}$). The sets R_c are pairwise disjoint, and $T_c \subseteq R_{c'}$ for the partner chain c'. Let H_c be the set of members inside R_c (Latin, host, pad members of its permanent and lock guests, lock members), H_{cmp} the set of members that contain a comparator guest (their pad members and the comparator members); H is the disjoint union of the H_c and H_{cmp}. Let Π_c be the set of pairs inside $R_c \cup T_c$ except the pairs with both ends in T_c. A *cross-anchor pair* is a pair $\{x, x''\}$ of an anchor x of c carrying a tunnel and an anchor x'' of the partner chain carrying a tunnel; cross-anchor pairs are hard (tunnels and hard non-edges). A pair lies in two sets Π_c, Π_d iff it is a cross-anchor pair (and then d is the partner of c); every other pair lies in at most one Π. Pairs of H_{cmp} members other than the pad pairs involve a comparator guest or join z_{c_1} to z_{c_2} and lie in no Π_c.

Lemma I.8 (Region bound)**.** *Let S be a solution with $\text{ex}(S) \le k^2$. For a chain c let $x_M = S \cap \text{pairs}(M)$ and h_c the number of edits in Π_c on pairs that belong to no member. Then $|S \cap \Pi_c| = \sum_{M \in H_c} |x_M| + h_c \ge |H_c| + \max\{1, h_c\}$.*

Proof. By Lemma I.7 pad pairs of comparator pad members are unedited, so the edits in Π_c are on pairs of members of H_c or on pairs of no member; this gives the equality. Each member has at least one edit, so $\sum |x_M| \ge |H_c|$, and if $h_c \ge 1$ we are done. Suppose $h_c = 0$ and $\sum |x_M| = |H_c|$: then every member of H_c has exactly one edit and S has no other edit in Π_c. In particular no tunnel at c is cut. Two stubs $x'_1, x'_2 \in T_c$ (partners of anchors x_1, x_2 of c) are not together: otherwise $x_1 \sim x'_1 \sim x'_2 \sim x_2$ (tunnels unedited), but $x_1 x_2$ is an unedited hard non-edge. So the partition restricted to $R_c \cup T_c$ has no edit on pairs with both ends in T_c either, and is a ZE partition of the instance $G_c = G[R_c \cup T_c]$ with the packing H_c (the comparator pad pairs inside R_c are plain hard edges of G_c, and the vertices of T_c are stubs: each has as only edge its tunnel). G_c is a stub instance, so Lemma I.4 applies. The proof of Proposition I.5 ($\Rightarrow$) for the single chain c, with all tunnels present, uses only Lemma I.1, Corollary I.3, the permanent guests and the locks, and no comparators; it shows that the position a_c lies outside every group (every group is excluded either by a permanent lock or by an uncut tunnel), a contradiction since the groups cover all positions. □

Theorem I.9 (Soundness)**.** *Every solution S of I_{tile} has $\text{ex}(S) \ge k^2$. If $\text{ex}(S) = k^2$, then S consists of one cut tunnel in every cell and one edit in every member, and nothing else; hence $S \setminus C$ is a ZE partition of I_C with C the set of cut tunnels, and Grid Tiling has a solution.*

Proof. Let $\text{ex}(S) \le k^2$ (otherwise there is nothing to show). Let q be the number of edited cross-anchor pairs and S_{rest} the set of edits outside $\bigcup_c \Pi_c$. Since only cross-anchor pairs lie in two sets, $|S| = \sum_c |S \cap \Pi_c| - q + |S_{\text{rest}}|$. Every member of H_{cmp} has an edit; its pad pair is unedited (Lemma I.7) and its other pairs are in no Π_c; these members are pair-disjoint, hence $|S_{\text{rest}}| \ge |H_{\text{cmp}}|$. With Lemma I.8 and $|H| = \sum_c |H_c| + |H_{\text{cmp}}|$,

$$\text{ex}(S) \;\ge\; \sum_c \max\{1, h_c\} - q.$$

Let $Z = \{c : h_c = 0\}$ and let h_{in} be the number of hard edits lying in exactly one Π_c. A cross-anchor edit counts in two chains, so $\sum_c h_c = h_{\text{in}} + 2q$, and $\sum_c \max\{1, h_c\} = |Z| + h_{\text{in}} + 2q$. Hence $\text{ex}(S) \ge |Z| + h_{\text{in}} + q$. The number of chains not in Z is at most $\sum_c h_c$, so $|Z| \ge 2k^2 - h_{\text{in}} - 2q$, and therefore $\text{ex}(S) \ge 2k^2 - q$. On the other hand every edited cross-anchor pair is a hard edit, so $q \le |U| \le \text{ex}(S)$ (Lemma 2.2). Thus $\text{ex}(S) \ge \max\{q, 2k^2 - q\} \ge k^2$.

Let $\text{ex}(S) = k^2$. Then $q = k^2$ (as $\text{ex} \ge q$ and $\text{ex} \ge 2k^2 - q$ give $q = k^2$), and from $\text{ex}(S) \ge |Z| + h_{\text{in}} + q$ we get $Z = \emptyset$ and $h_{\text{in}} = 0$. Every chain has $h_c \ge 1$ and $\sum_c h_c = 2q = 2k^2$, so $h_c = 1$ for all $2k^2$ chains: the only non-member edit at a chain is one cross-anchor pair. All hard edits

are these k^2 pairs (as $|U| \le \mathrm{ex} = k^2 = q$); equality throughout the chain of inequalities also forces $\sum_{M \in H_c} |x_M| = |H_c|$ for every c, i.e. all members of the H_c have exactly one edit, and S_{rest} has exactly one edit per member of H_{cmp}. In a cell (i,j) exactly one cross-anchor pair is edited (each chain has $h_c = 1$ and sees only cross pairs of its own cell). It is a tunnel: suppose the edited pair is a non-edge $x_A x_B$ of anchors of A_{ij} and B_{ij} whose tunnels are $x_A x_B^*$ and $x_A^* x_B$ (with $x_B^* \ne x_B$). It is inserted, so $x_A \sim x_B$; the tunnel $x_A x_B^*$ is an unedited edge, so $x_A \sim x_B^*$; hence two distinct anchors of B_{ij} are together, but their pair is an unedited hard non-edge ($h_{B_{ij}} = 1$ is used up by the cross pair), a contradiction. So the edited cross pair is a tunnel of the cell, deleted, and S is the set C of k^2 cut tunnels (one per cell) together with one edit per member. By Lemma I.7 the partition is a partition with K_v intact, and it is a ZE partition of I_C. Proposition I.5 ($\Rightarrow$) gives a tiling: the tiles t_{ij} of the cut tunnels satisfy $a_{ij} \le a_{i,j+1}$ and $b_{ij} \le b_{i+1,j}$. □

Theorem I.10 (Completeness). *If the Grid Tiling instance has a solution (s_{ij}), then* $\mathrm{OPT}(I_{\mathrm{tile}}) = |H| + k^2$.

Proof. Let C be the tunnels of the tiles s_{ij}. By Proposition I.5 ($\Leftarrow$) the instance I_C has a ZE partition, whose edit set (relative to G) is the set of its $|H|$ member edits together with the k^2 deleted tunnels. So $\mathrm{OPT} \le |H| + k^2$, and $\mathrm{OPT} \ge |H| + k^2$ by Theorem I.9. □

I.7 Pathwidth, pair-maximality and proof of Theorem 5.1

Lemma I.11 (Pathwidth). *Every component of $I(H)$ is contained in the set of members of the k chains of one row $\{A_{i,1}, \dots, A_{i,k}\}$ or one column $\{B_{1,j}, \dots, B_{k,j}\}$ together with their guests and comparator guests; so $I(H)$ has at most $2k$ components, and* $\mathrm{pw}(I(H)) < 24kp^2$, *independently of n.*

Proof. Members of different chains share a vertex only through a comparator guest or through its pad members (they share vertices with Latin members of both chains); tunnels are hard pairs and are not members. This gives the components. Define a *level* $\lambda(M)$ of a member: for a Latin member of a thin member, the smallest cell of the thin member (all three cells lie in $[\lambda, \lambda+2]$); for every member belonging to a guest of the group u (host, pad, lock, comparator members), $\lambda = 2u$. A guest of group u has pad and host vertices in cells $[2u, 2u+2]$ (the vertices $s_{4u}, \dots, s_{4u+4}, \varepsilon_u^{\pm}$), so for every thick vertex (v,i) all members through it have levels in $[\mathrm{cell}(v)-2, \mathrm{cell}(v)]$, and all members through a guest have the same level. Hence two members that share a vertex have levels differing by at most 2. For a component Q put $B_j = \{M \in Q : \ j \le \lambda(M) \le j+2\}$; every member lies in the bags $j \in [\lambda(M)-2, \lambda(M)]$, an interval, and two members sharing a vertex have a common bag; so $(B_j)_j$ is a path decomposition. At one level a chain has at most 6 thin members (at most 6 at the right cap, at most 3 elsewhere), hence at most $6p^2$ Latin members. At an even level $2u$, the guest members of one chain are at most $3.5p$ (permanent guest) $+2.5p+1$ (lock) in a tile group and $2.5p$ in a separator group, and the comparator guests of a row or column at a separator group contribute at most $k-1$ guests with at most $3p+1$ members each. So a level of a component has at most $6kp^2 + k(6p+1)$ members, and $|B_j| \le 3 \cdot 6kp^2 + 2k(6p+1) \le 24kp^2$ for $p \ge 6$. □

Lemma I.12 (Pair-maximality). *H is pair-maximal.*

Proof. An induced P_3 with three hard pairs consists of two hard edges sharing a vertex and a hard non-edge. The hard edges form the graph Γ_h whose components are: (a) for a thin vertex without a thin hard edge, the glue of K_v; (b) for each of the four thin hard edges uv of a chain, $K_u \cup K_v$ with the glue inside both and all edges between them; (c) the tunnels xx' (the anchors carry no other hard edge, guests carry none). In the components (a) and (b) the pairs that are not adjacent

are exactly the pad and host pairs inside a super-vertex, which are soft; a component (c) is a single edge. So in every component every non-adjacent pair is soft, and an induced P_3 with two hard edges has its three vertices in one component of Γ_h, hence its third pair is soft or an edge. Hence no induced P_3 has all pairs hard. (Checked by computer for $(k, n, p) = (1, 2, 8), (1, 2, 6), \ldots, (1, 5, 6)$ and $(2, 2, 10)$.) □

Proof of Theorem 5.1. The instance (G, H, k^2) is constructed in time polynomial in k and n (Lemma I.6(d)). (i) holds by definition, Lemma I.6(a),(d) and Lemma I.12; (ii) is Lemmas I.6(d) and I.11 ($\mathrm{tw} \le \mathrm{pw}$); (iii) is Theorems I.9 and I.10 (a solution of size at most $|H| + k^2$ has excess at most k^2, hence exactly k^2, hence a tiling exists by Theorem I.9). The parameters $\ell + \mathrm{tw}(I(H))$, $\ell + \mathrm{pw}(I(H))$ and $\ell + c$ are bounded by $k^2 + 24k(k^2 + 6)^2$, which is a function of k only, and the reduction is polynomial: it is a parameterized reduction from Grid Tiling with $\le$ parameterized by k, which is W[1]-hard. An algorithm of running time $f(\ell, w)n^{O(1)}$ would decide Grid Tiling in time $f'(k)\, n^{O(1)}$, so FPT = W[1]. The ETH statement is Corollary 5.2. □

Remark I.13 (The chain object). Consider one chain with N tile groups, each with a lock guest and an anchor whose tunnel ends at a stub, all other groups being permanent locks. The proof of Proposition I.5 ($\Rightarrow$) shows that with all tunnels present there is no ZE partition, and the proof of ($\Leftarrow$) shows that cutting any single tunnel gives one. In the language of Appendix F the set of all N tunnels is therefore a core, of unbounded size N, while $c \le 4p+2$ and the pathwidth do not depend on N (the bound of Lemma I.11). This is the family of tight instances with bounded c and width and cores of unbounded size that the discussion of Appendix F asks for; by Corollary 10.7 it must have soft triangles, and it does (the Latin members). The argument is the proof above applied to a single chain and was not run separately by computer. The hypotheses p even, $p \ge 8$ suffice for this statement (the tunnel semantics do not use the no-split lemma); for the full problem $p > \ell$ is needed.

I.8 Status of the verification

Statement	Status
Lemma I.1(a) (P_a is ZE, all m)	proved modulo (CA1): finite check $m \le 60$ plus window argument
Lemma I.1(b) (no other ZE partition)	proved modulo (CA2): transfer matrix $\chi = x^4(x-1)^2$, $N_m = m$ for $7 \le m \le 40$; independent SAT enumeration up to $m = 100$ (sample)
Lemma I.2, Cor. I.3 (profiles)	read off the formula; (CA3) checked for even $m \le 60$ and all groups
Lemma I.4 (blow-up)	proved by hand
Prop. I.5 (tunnel release, comparators)	proved by hand from the above; thin model tested on random instances ($k \le 3$), thick comparator on all position pairs
Lemma I.6 ($u_{\max} \le 3$, $c \le 4p + 2$)	hand count, (CA4) confirmed by computer, $k \le 4$
Lemma I.7 (no split)	proved by hand (needs $p \ge k^2 + 5$)
Lemma I.8, Thm. I.9 (soundness)	proved by hand; exact solves on small instances only ($k = 1$; comparator instance 401 vertices, $\ell = 2$); none for $k \ge 2$
Thm. I.10 (completeness)	proved by hand
Lemma I.11 (pathwidth $< 24kp^2$)	proved by hand (cell sweep); consistent with measured vertex separation
Lemma I.12 (pair-maximal)	proved by hand; checked on small instances

J Computational checks

The proofs in this paper do not depend on computation, except (i) that the statements about explicit small instances are verified by exhaustive enumeration, as indicated where they occur, (ii) the finite classification steps marked *by computer* in Section 10, and (iii) the four computer-assisted steps (CA1)–(CA4) in the proof of Theorem 5.1 (the chain lemma and the transfer-matrix certificate, the host profiles, and the counts $u_{\max} \le 3$, $c \le 4p+2$), listed in Section 5.3 and Appendix I. This appendix lists what was enumerated and what was tested.

Exhaustive enumeration of explicit instances. For each of the following instances all set partitions of the vertex set were enumerated and the cost of each partition computed: the instances of Remark 3.7 (domain size), Propositions 4.5 and 4.6, the shrink-rule example of Appendix A (for twin classes of size $1,\dots,4$), the three instances of Proposition 9.7, the family of Proposition 9.8 for $m = 3,4,5$, the two instances of Proposition D.10, the gadget G^* and the majority example of Appendix F. In each case the packing was checked to consist of induced P_3's with pairwise disjoint pair sets and to be pair-maximal, and the optima and the structural claims were read off from the enumeration.

Randomised tests of lemmas. The lemmas of Sections 7 and 9 and of Appendices A–D were tested on random small graphs ($n \le 10$; random, clustered, hub and sparse graphs; random and maximal packings) against all partitions of cost at most $|H|+3$, with code written independently of the proofs. No violation was found: the decomposition identity, Lemma 7.2, Theorem 9.2, Corollary 9.3, Lemma 9.6, Lemmas 7.4 and 7.6, Corollary 7.7 (inserted pairs had distance at most 2 in every test), Theorem A.5, Corollary A.6, Lemmas B.1, B.4, D.1 and D.2, and Lemma 6.5. Control experiments in which a constant is weakened by one (for example $\alpha \le 2\ell + c$ in Lemma B.1, or $|C| \le \max\{D+u, 2D, 2u\}$ in Lemma B.4), or in which partitions of cost $\mathrm{OPT}+1$ or $\mathrm{OPT}+2$ are tested, produce violations, so the tests are not vacuous. The dynamic program of Theorem 8.3 was implemented and agreed with brute force on 2,778 pairs of an instance ($n \le 10$) and a budget ℓ. The checks belonging to Section 10 are listed in Appendix G, those belonging to Section 11 at the end of Appendix H, and those belonging to Section 5 in Section 5.3 and at the end of Appendix I; the latter include SAT-based exact solves of small instances of the reduction, in which UNSAT answers are proofs for the instance solved. These tests are evidence for the statements; the proofs are the justification. The scripts are available from the author.

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